



STRUCTURAL MODEL 3D VIEW

	CONCRETE
	CONCRETE - EXISTING
	EARTH
	GRAVEL OR GRANULAR FILL
	GROUT OR DRYPACK OR SAND
	CMU OR MASONRY
	METAL / COLD-FORM STUD
	WOOD / STUD
	PRECAST CONCRETE
	STEEL
	OTHER/SPECIALTY

#	NUMBER OR POUNDS AT DEGREE
Ø	DIAMETER
A.B.	EXISTING
ANCHOR	ANCHOR BOLT
ARCH	ARCHITECT - URE. -URAL
B.F.	BOTTOM OF
B.F.	BEAM FLANGE WIDTH
BF	BRACE FRAME
BM	BEAM
B.N.	BUILDING
BOTT	BOTTOM
BTWN	BETWEEN
CFSE	COLD FORM STEEL FRAMING
CJS	CENTER OF GRAVITY OF THE TENDON
CJP	CORNER JOINT PENETRATION WELD
CL	CLEAR
CL	CENTERLINE
CNU	CORNER MASONRY UNIT
COL	COLUMN
CONC	CONCRETE
CONN	CONNECTION
CONST	CONSTRUCTION
CONT	CONTINUOUS
COORD	COORDINATION
CO	CORNER
CO	DIAMETER
DL	DEAD LOAD
DET	DETAIL
DWG	DRAWING
DWL	DOWEL
EACH	EACH
EA	EACH FACE
EFF	EFFECTIVE
EF	ELEVATION
ELEC	ELECTRICAL
EMBED	EMBEDMENT
E.H.	EDGE NAILING
ED	EDGE OF DECK
EOS	EDGE OF SLAB
EQ	EQUAL
EQUIP	EQUIPMENT
ETC	ETCETERA
EW	EACH WAY
EXP	EXPANSION
EXT	EXTERIOR
FC	CONCRETE COMPRESSIVE STRENGTH
FND	FOUNDATION
FN	FIELD NAILING
FT	FOOT
FT	FOOTING
FY	YIELD STRESS
GA	GAGE OR GAUGE
GALV	GALVANIZED
GL	GLULAM BEAM
GT	GIRDED TRUSS
HORIZ	HORIZONTAL
HSA	HIGHER STITCH ANCHOR
HS	HORIZONTAL BOLT
HT	JOINT
K	KIP
KIP	KIP PER SQUARE FOOT
KSI	KIPS PER SQUARE INCH
KB	LENGTH
LB	POUNDS
LL	LONG LIVE
LLV	LONG LIVE HORIZONTAL
LLV	LONG LIVE VERTICAL
LONG	LONGITUDINAL
LS	LONG LIVE HORIZONTAL
LT	LONG LIVE VERTICAL
LT WT	LENGTH WEIGHT
MECH	MECHANICAL
MANUF	MANUFACTURER
MIN	MINIMUM
NIC	NOT IN CONTRACT
NT	NOT TO SCALE
OC	ON CENTER
OP	OPPOSITE HAND
OPNG	OPENING
OSB	ORIENTED STRAND BOARD
P.C.	POUNDS PER CUBIC FOOT
P.H.	PENHOUTSE
PL	PARALLEL JOINT PENETRATION WELD
PLF	PLATE
PL	POUNDS PER LINEAR FOOT
PS	POUNDS PER SQUARE FOOT
PSI	POUNDS PER SQUARE INCH
PS	TENSION, END, -ING
R	RADIUS
REIN	REINFORCING, -MENT, -ED
REIN	REQUIRED
RTU	ROOF TOP UNIT
SC	SPLIT CRITICAL
SCHED	SCHEDULE
SCH	STEEL FORCE-RESISTING SYSTEM
SIM	SIMILAR
SN	SNOW LOAD
S.M.S.	SHEET METAL SCREW
SP	SPACE(S)
SPECS	SPECIFICATIONS(S)
SC	SQUARE
STIFF	STIFFENER
STL	STEEL
SY	SYMMETRICAL
T&B	TOP AND BOTTOM
TO	TOP OF
TO	PRE-TENSIONED BOLT
TEMP	TEMPERATURE
THK	THICK FLANGE THICKNESS
THK	THICK
TRANS	TRANSVERSE
TY	TYPICAL
UN	UNLESS OTHERWISE NOTED
VERT	VERTICAL
VIF	VELOCITY IN FIELD
W	WITH
W	WORK POINT
WT	WEIGHT
WWR	WELDED WIRE REINFORCING

S001	COVER SHEET
S002	GENERAL NOTES
S002	GENERAL NOTES
S110	FOUNDATION PLAN
S190	ROOF FRAMING PLAN
S190	CARPOT PLANS
S300	TYPICAL FOUNDATION SECTIONS AND SCHEDULES
S301	FOUNDATION SECTIONS
S302	FOUNDATION SECTIONS
S400	FIREPLACE SECTIONS
S503	STEEL SECTIONS
S504	STEEL SECTIONS
S520	TYPICAL STEEL ROOF FRAMING DETAILS
S530	STEEL BASE PLATE DETAILS
S602	WOOD SECTIONS
S700	TYPICAL COLD FORM SECTIONS
GRAND TOTAL: 16	

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architect's stamp

consultant

KEY PLAN

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[illegible]

PROJECT TITLE:

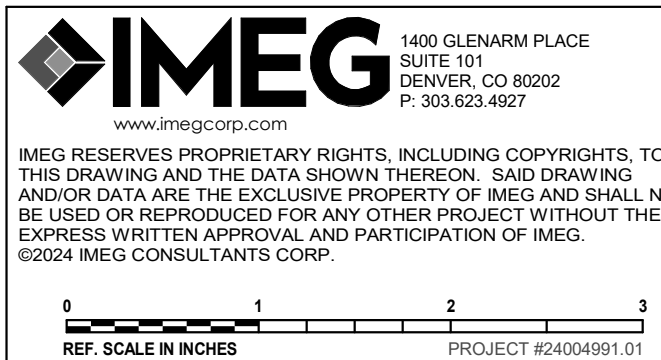
3101 EAST CAMELBACK ROAD
PHOENIX, ARIZONA 85016

PC/PM: LG
DRAWN BY: STAFF
JOB NO.: 24004991.01

DRAWING TITLE:
COVER SHEET

SHEET NO.:

\$000



DESIGN CRITERIA

1. STRUCTURE HAS BEEN DESIGNED TO COMPLY WITH 2018 IBC AND SUBSEQUENT REFERENCE STANDARDS.
2. RISK CATEGORY: II
3. SUPERIMPOSED DEAD LOADS:

ROOF	TYPICAL	20 PSF
	MFP ROOMS	50 PSF
FLOOR	TYPICAL	100 PSF

4. MECHANICAL UNITS: THE CONTRACTOR SHALL CONFIRM THE ACTUAL MECHANICAL UNITS PURCHASED ARE OF WEIGHT AND SIZE SUCH THAT:
- a. THE UNIT WEIGHT IS LESS THAN OR EQUAL TO WEIGHT LISTED ON PLAN.
- b. THE UNIT LENGTH IS LESS THAN 15 FEET. IF UNIT LENGTH IS GREATER THAN 15 FEET, SNOW DRIFT ON ROOF SHALL BE ACCOUNTED FOR.
- c. THE TOTAL SUPERIMPOSED LOAD UNDER EACH UNIT DOES NOT EXCEED THE LIVE LOAD FOR WHICH THE FLOOR WAS DESIGNED IN THE AREA AREA WHERE THE UNIT IS LOCATED AS INDICATED BY THE EQUATION BELOW:

TOTAL SUPERIMPOSED LOAD UNDER UNIT =	TOTAL EQUIPMENT WEIGHT / UNIT FOOTPRINT AREA =	INDICATED SERVICE LEVEL DESIGN LIVE LOAD
TOTAL EQUIPMENT WEIGHT =	OPERATING WEIGHT OF UNIT + CONCRETE HOUSEKEEPING PAD + INERTIA BLOCK + OTHER ANCILLARY ITEMS ASSOCIATED WITH THE EQUIPMENT	

UNIT FOOTPRINT AREA =	EFFECTIVE AREA OF UNIT OR HOUSEKEEPING PAD, WHICHEVER IS LARGER
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EFFECTIVE AREA =	(LENGTH + 6 FEET) X (WIDTH + 6 FEET)
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CONTRACTOR SHALL NOTIFY THE STRUCTURAL ENGINEER PRIOR TO CONSTRUCTION ON THE LOCATION OF ANY MECHANICAL UNITS WHERE THE SUPERIMPOSED LOAD UNDER THE UNIT EXCEEDS THE INDICATED DESIGN INERTIA AND LOAD FOR THE ROOF AND/OR LIVE LOAD FOR THE FLOOR AS INDICATED ON THE PLAN.

5. SEISMIC:
- SEISMIC DESIGN CATEGORY B
- IMPORTANCE FACTOR 1.0
- SITE CLASS D
- SS 0.187 g
- S1 0.066 g
- S0S 0.2 g
- SDS 0.106 g
- SEISMIC FORCE RESISTING SYSTEM STEEL ORDINARY CANTILEVERED COLUMNS, AT GLEATORY WINDOWS CONCENTRICALLY BRACED FRAMES BELOW CLEDESTORY
- RESPONSE MODIFICATION COEFFICIENT, R 1.25
- Cd 1.25
- O 1.25
- ANALYSIS PROCEDURE EQUIVALENT LATERAL FORCE
- SEISMIC RESPONSE COEFFICIENT, CS 0.16
- DESIGN BASE SHEAR (STRENGTH LEVEL) 26.3 KIPS

BASIC WIND SPEED	VULT = 115 MPH VASD = 90 MPH
EXPOSURE CLASS	C
INTERNAL PRESSURE COEFFICIENT, Gcpi	±0.18
MAIN WIND FORCE PER SQUARE FOOT (STRENGTH LEVEL)	21 PSF

COMPONENTS & CLADDING:	
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ROOF COMPONENTS	ZONE 1	ZONE 2	ZONE 3
SUPPORT BEAMS (A=10 SF)	16.0/-35.9 PSF	16.0/-47.7PSF	16.0/-56.8 PSF
ROOF SHEATHING (A=50 SF)	16.0/-38.9 PSF	16.0/-51.6 PSF	16.0/-64.6 PSF
DECK FASTENERS (A=10 SF)	16.0/-46.0 PSF	16.0/-60.7 PSF	16.0/-82.7 PSF
WALL COMPONENTS	ZONE 4	ZONE 5	
A=200 SF	21.4/-23.6 PSF	21.4/-25.1PSF	
A=50 SF	23.7/-25.9 PSF	23.7/-28.9 PSF	
A=20 SF	21.4/-23.6 PSF	25.2/-32.9 PSF	

- C & C NOTES:
- a. THE PRESSURES LISTED ARE IN ACCORDANCE IBC AND ASCE 7, AND THE DESIGN FORCES USED BY THE SUBCONTRACTOR FOR A SPECIFIC APPLICATION ARE THE RESPONSIBILITY OF THE SUBCONTRACTOR.
- b. WIND PRESSURES ARE ULTIMATE DESIGN LEVEL.
- c. SEE ASCE 7 FOR ZONE DEFINITIONS AND EXTENT OF ZONES.
- d. SUBMIT DESIGN CALCULATIONS ABOVE THE CONTRACTOR SHALL HAVE THE PROJECT'S JURISDICTION FOR ANY DESIRED MODIFICATION TO THE STATED PRESSURES.

7. ALL LATERAL LOAD RESISTANCE AND STABILITY OF THE BUILDING IS COMPLETED STRUCTURE IS PROVIDED BY STEEL BRACE FRAMES AND MOMENT FRAMES TO THE CLEAR STORY ELEVATION AND CANTILEVERED COLUMNS UP TO THE ROOF IN EACH ORTHOGONAL DIRECTION. SEE PLANS FOR LOCATIONS. THE SHEATHING AND METAL DECK SERVE AS HORIZONTAL DIAPHRAGMS DISTRIBUTING THE LATERAL FORCES TO THE VERTICAL LATERAL ELEMENTS WHICH IN TURN CARRY THE LOAD TO THE BUILDING FOUNDATIONS.

GENERAL

1. DURING THE CONSTRUCTION PERIOD, THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE SAFETY OF PERSONNEL AND PROPERTY ON AND AROUND THE JOBSITE. THE CONTRACTOR SHALL PROVIDE ADEQUATE SHORING, BRACING, GUYS, ETC. IN ACCORDANCE WITH ALL NATIONAL, STATE, AND LOCAL SAFETY ORDINANCES. TEMPORARY BRACING, SHORING, GUYS, ETC. SHALL AVOID EXCESSIVE STRESSES AND HOLD STRUCTURAL ELEMENTS IN PLACE DURING CONSTRUCTION. THE STRUCTURE SHOULD NOT BE CONSIDERED STABLE UNTIL ALL STRUCTURAL ELEMENTS HAVE BEEN CONSTRUCTED IN ACCORDANCE WITH THE CONTRACT DOCUMENTS.
2. ALL DRAWINGS AND SPECIFICATIONS ARE CONSIDERED TO BE A PART OF THE CONTRACT DOCUMENTS. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR THE REVIEW AND COORDINATION OF ALL DRAWINGS PRIOR TO THE START OF CONSTRUCTION. ANY DISCREPANCIES OR OMISSIONS SHALL BE BROUGHT TO THE ATTENTION OF THE ARCHITECT PRIOR TO THE START OF CONSTRUCTION SO A CLARIFICATION CAN BE ISSUED. ANY WORK THAT DEVIATES FROM OR IS PERFORMED IN CONFLICT WITH THE CONTRACT DOCUMENTS OR ANY CODE REQUIREMENTS SHALL BE CORRECTED BY THE CONTRACTOR AT THEIR OWN EXPENSE AND AT NO EXPENSE TO THE OWNER OR THE DESIGN PROFESSIONALS.
3. THE CONTRACT DOCUMENTS REPRESENT THE FINISHED STRUCTURE. THEY DO NOT INDICATE THE METHOD OF CONSTRUCTION. THE CONTRACTOR IS RESPONSIBLE FOR DETERMINING ALLOWABLE CONSTRUCTION LOADS AND FOR DETERMINING SEQUENCES OF CONSTRUCTION. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE SAFETY OF THE PROJECT STRUCTURE AND SAFETY OF WORKERS DURING CONSTRUCTION. SUCH MEASURES SHALL INCLUDE, BUT NOT BE LIMITED TO: FALSEWORK, FORMWORK, STAGING, BRACING, AND SHORING FOR LOADS DUE TO CONSTRUCTION EQUIPMENT, ETC. OBSERVATION VISITS TO THE SITE BY THE ARCHITECT SHALL NOT INCLUDE INSPECTION OR APPROVAL OF THE ABOVE ITEMS AND DO NOT IN ANY WAY RELIEVE THE CONTRACTOR OF THEIR RESPONSIBILITIES FOR THE ABOVE. THE CONTRACTOR SHALL TAKE ALL NECESSARY PRECAUTIONS TO MAINTAIN AND ENSURE THE INTEGRITY OF THE STRUCTURE AT ALL STAGES OF CONSTRUCTION.
4. ALL DIMENSIONS AND SITE CONDITIONS SHALL BE VERIFIED BY THE CONTRACTOR AT THE JOBSITE PRIOR TO BID SUBMITTAL. START OF SHOP DRAWINGS, START OF CONSTRUCTION, AND/OR FABRICATION OF MATERIALS. IF DISCREPANCIES ARE ENCOUNTERED, OR CONDITIONS DEVELOP THAT ARE NOT COVERED BY THE CONTRACT DOCUMENTS, THE ARCHITECT SHALL BE NOTIFIED FOR CLARIFICATION.
5. STRUCTURAL SUBSTITUTIONS MAY BE ALLOWED WITH THE APPROVAL OF THE STRUCTURAL ENGINEER. SUPPLIER SHALL PROVIDE SIGNED AND SEALED DESIGN CALCULATIONS OR SUITABLE PRODUCT LITERATURE FOR THE COMPONENTS. ALL PRODUCT SUBSTITUTIONS SHALL INCLUDE A CODE EVALUATION REPORT SPECIFIC TO THE BUILDING CODE LISTED IN THE DESIGN CRITERIA.
6. STRUCTURAL DRAWINGS INDICATE DIMENSIONS AND DIMENSIONS FOR STRUCTURAL INTEGRITY BUT DO NOT SHOW ALL DETAIL DIMENSIONS TO FIT INTRICATE ARCHITECTURAL AND MECHANICAL DETAILS. CONTRACTOR SHALL CONSTRUCT THE WORK SO IT WILL CONFORM TO THE CLEARANCES REQUIRED BY ARCHITECTURAL, MECHANICAL, AND ELECTRICAL DESIGN.
7. ALL SYMBOLS AND ABBREVIATIONS USED ON THE DRAWINGS ARE CONSIDERED TO BE CONSTRUCTION STANDARDS. IF CLARIFICATION IS REQUIRED, THE CONTRACTOR SHALL NOTIFY THE ARCHITECT PRIOR TO PROCEEDING WITH THE WORK.

8. DO NOT SCALE DRAWINGS. PRINTED DIMENSIONS HAVE PRECEDENCE OVER SCALED DRAWINGS AND LARGE-SCALE OVER SMALL-SCALE DRAWINGS. CONTRACTOR TO DETERMINE FINAL DIMENSION WITH ARCHITECT.
9. TYPICAL DETAILS SHALL BE PROVIDED FOR ANY OCCURRING ON THE PROJECT THAT ARE THE SAME OR SIMILAR TO THOSE SPECIFICALLY REFERENCED. WHERE NO DETAILS ARE GIVEN, CONSTRUCTION SHALL BE AS SHOWN FOR SIMILAR WORK.
10. SEE ARCHITECTURAL, ELECTRICAL, AND MECHANICAL DRAWINGS FOR DETAILS. CONDITIONS, PITS, TRENCHES, PADS, DEPRESSIONS, ROOF, FLOOR OPENINGS, TOP OF WALL ELEVATIONS, STAIRS, SLEEVES, ITEMS TO BE EMBEDDED OR ATTACHED TO STRUCTURAL ELEMENTS, ETC. NOT SHOWN ON THE STRUCTURAL DRAWINGS. FOR THESE NON-STRUCTURAL ELEMENTS SHOWN ON STRUCTURAL DRAWINGS, THEY ARE FOR GENERAL INFORMATION ONLY.
11. COORDINATE FLOOR FINISH INCLUDING, BUT NOT LIMITED TO THE "FLATNESS" AND "LEVELNESS" OF FLOORS WITH THE FLOOR FINISH CONTRACTOR, PROVIDE UNDERLAYMENT / TOPPING WHERE REQUIRED TO PROVIDE A SURFACE ACCEPTABLE FOR INSTALLATION OF FLOOR FINISHES. SEE ARCHITECTURAL DRAWINGS AND SPECIFICATIONS FOR ADDITIONAL INFORMATION AND REQUIREMENTS.
12. ESTABLISH AND VERIFY ALL OPENINGS AND INSERTS FOR MECHANICAL, ELECTRICAL, AND PLUMBING WITH APPROPRIATE TRADE CONTRACTORS. OPENING SIZES AND LOCATIONS SHOWN FOR DUCTS, PIPE, INSERTS, AND OTHER PENETRATIONS WHEN SHOWN ARE FOR GENERAL INFORMATION ONLY AND SHALL BE VERIFIED PRIOR TO FORMING.
13. THE EXACT WEIGHTS, DIMENSIONS, AND LOCATIONS OF ALL MECHANICAL UNITS AND ELECTRICAL GEAR SUPPORTED ON STRUCTURAL FRAMING SHALL BE DETERMINED AND COORDINATED BY THE CONTRACTOR PRIOR TO DETAILING THE STRUCTURAL FRAMING SUPPORTING THOSE UNITS. IF THE UNIT WEIGHTS ARE GREATER THAN THE WEIGHTS SHOWN ON THE STRUCTURAL DRAWINGS, THE STRUCTURAL ENGINEER SHALL BE NOTIFIED PRIOR TO DETAILING THE STRUCTURE. UNIT WEIGHTS, DIMENSIONS, AND LOCATIONS SHOWN ON THE STRUCTURAL DRAWINGS ARE APPROXIMATE ONLY AND SHALL NOT BE USED FOR DETAILING THE STRUCTURE.
14. PROVIDE TEMPORARY BLOCKOUTS AND TEMPORARY OPENINGS IN THE STRUCTURE AS REQUIRED TO PERMIT INSTALLATION OF WALL, BLOCKOUTS AND TEMPORARY OPENINGS SHALL BE LOCATED, CONFIGURED, DETAILED, AND FILLED IN A MANNER THAT ALTERS NEITHER THE STRENGTH OF THE STRUCTURE NOR THE STRENGTH OF CONNECTIONS. INFILL ALL BLOCKOUTS AND TEMPORARY OPENINGS USING THE MATERIALS SPECIFIED FOR THE FRAMING AT THE LOCATIONS WHERE THE BLOCKOUTS AND OPENINGS OCCUR. SUBMIT DRAWINGS AND DETAILS FOR APPROVAL. PROVIDE DETAILS AND DETAILS OF ALL PROPOSED BLOCKOUTS AND OPENINGS AND DETAILS INCLUDING THE MANNER IN WHICH THE BLOCKOUTS AND OPENINGS WILL BE FILLED.
15. NO HOLES, NOTCHES, BLOCK-OUTS, ETC. ARE ALLOWED IN STRUCTURAL ELEMENTS UNLESS SPECIFICALLY DETAILED ON THE STRUCTURAL DRAWINGS OR APPROVED BY THE STRUCTURAL ENGINEER.
16. BEFORE SUBMITTING A PROPOSAL FOR THIS WORK, CONTRACTOR SHALL VISIT THE PREMISES AND BECOME FULLY ACQUAINTED WITH ALL EXISTING CONDITIONS, TEMPERA- CONSTRUCTION REQUIRED, QUANTITIES AND TYPE OF EQUIPMENT, ETC. THE PROPOSAL SHALL INCLUDE ALL SUMS REQUIRED TO DO THE WORK.
17. ELEMENTS SUCH AS NON-BEARING PARTITIONS, ETC. ATTACHED TO AND/OR SUPPORTED BY THE STRUCTURE SHALL TAKE INTO ACCOUNT DEFLECTIONS AND OTHER STRUCTURAL MOVEMENTS. THE STRUCTURAL FRAMING WAS DESIGNED TO LIMIT DRIFT AND DEFLECTION OF THE STRUCTURAL SYSTEMS LESS THAN THE MAXIMUM PERMITTED DEFLECTIONS LISTED IN THE BUILDING CODE. THE CONTRACTOR SHALL COORDINATE THE WORK OF OTHER TRADES TO ACCOMMODATE THESE DEFLECTIONS AND TO ACCOMMODATE CONSTRUCTION TOLERANCES.

18. FIRE PROTECTION FOR ALL STRUCTURAL PARTS SHALL BE PROVIDED AS SPECIFIED BY THE ARCHITECTURAL DRAWINGS. UL FIRE RESISTANCE RATING RESTRAINT CLASSIFICATION AS FOLLOWS:
- a. "PRIMARY STRUCTURAL FRAME," AS DEFINED BY IBC SECTION 202 IS "RESTRAINED" EXCEPT AS INDICATED IN FOLLOWING NOTE.
- b. THE FOLLOWING FRAMING IS "UNRESTRAINED"
1. FRAMING SUPPORTED BY BEARING WALLS, OTHER THAN CAST-IN-PLACE CONCRETE WALLS, IN END BAYS AND OTHER LOCATIONS WHERE THE END OF THE FRAMING IS NOT ABUTTING FRAMING IN AN ADJACENT BAY.
2. HOLLOW CORE SLABS IN END BAYS AND OTHER LOCATIONS WHERE THE END OF THE SLAB IS NOT ABUTTING SLABS IN AN ADJACENT BAY.
3. STEEL JOISTS IN END BAYS ON ROOFS SUPPORTING STEEL DECK.
4. ALL JOIST CONSTRUCTION.

SUBMITTALS

1. SUBMITTALS ARE:
- a. CONCRETE MIX DESIGNS
- b. MATERIAL PRODUCT DATA FOR STRUCTURAL MATERIALS
- c. CONCRETE REINFORCING
- d. ENGINEERED LUMBER
- e. HEAVY TIMBER
- f. STEEL FABRICATION AND MISCELLANEOUS METALS
- g. STEEL DECK
- h. COLD FORMED STEEL FRAMING AND CONNECTIONS
2. SUBMITTALS SHALL BE REVIEWED AND COORDINATED PRIOR TO SUBMITTING TO THE ARCHITECT. EACH SHOP DRAWING SUBMITTED SHALL BE STAMPED INDICATING REVIEW BY THE CONSTRUCTION MANAGER/GENERAL CONTRACTOR AND REVIEWED BY THE ARCHITECT SHALL NOT BEGIN UNTIL THIS IS COMPLETE. WORK SHALL NOT BEGIN WITHOUT REVIEW BY THE DESIGN PROFESSIONALS.
3. SUBMITTALS SHALL BE REVIEWED BY THE DESIGN PROFESSIONALS FOR GENERAL CONFORMANCE WITH DESIGN CONCEPT ONLY. NOTATIONS MADE BY THE DESIGN PROFESSIONALS ON THE SHOP DRAWINGS DO NOT RELIEVE THE CONTRACTOR FROM COMPLYING WITH THE REQUIREMENTS OF THE DESIGN.
4. FOR ADDITIONAL INFORMATION ON REQUIRED SUBMITTALS, SEE INDIVIDUAL MATERIAL SECTIONS.

DEFERRED SUBMITTALS

1. DEFERRED SUBMITTALS PER SECTION 107.3.4.1 SHALL BE SUBMITTED TO THE BUILDING OFFICIAL AND THE DESIGN PROFESSIONALS AND REVIEWED PRIOR TO INSTALLATION.
2. DEFERRED SUBMITTALS ARE:
- a. EXCAVATION, SHORING
- b. ROOFTOP EQUIPMENT ANCHORAGE AND CURBS
3. ALL DEFERRED SUBMITTALS SHALL BEAR THE STAMP AND SIGNATURE OF THE QUALIFIED PROFESSIONAL ENGINEER, REGISTERED IN THE STATE WHERE THE PROJECT IS LOCATED, RESPONSIBLE FOR THE PREPARATION OF THESE DOCUMENTS.
- PLANS AND SPECIFICATIONS FOR DEFERRED SUBMITTAL ITEMS SHALL BE SUBMITTED TO THE PLANNING & DEVELOPMENT DEPARTMENT AFTER BEING REVIEWED FOR CONFORMANCE WITH THE BUILDING OR STRUCTURE'S DESIGN BY THE ARCHITECTURAL AND STRUCTURAL DESIGN FIRMS THAT ARE DEFERRING THE ELEMENT AND SHALL HAVE PDQ APPROVAL PRIOR TO FABRICATION, ERECTION, OR INSTALLATION IN THE FIELD. THE ARCHITECTURAL AND STRUCTURAL DESIGN FIRMS SHALL PROVIDE A NOTATION ON ALL DEFERRED DOCUMENTS THAT ACKNOWLEDGES REVIEW OF SUCH DOCUMENTS WITHIN THE REQUIREMENTS OF THE BUILDING INSPECTOR OR PLAN REVIEWER.

5. FACADE:
- a. DESIGN AND DETAILING OF THE FACADE SYSTEM AND CONNECTIONS TO THE STRUCTURE SHALL TAKE INTO CONSIDERATION THE FOLLOWING MOVEMENTS:
1. ±1/2" VERTICAL DIFFERENTIAL DEFLECTION OF SLAB EDGES ON ADJACENT FLOORS
2. ±1/2" HORIZONTAL DIFFERENTIAL DEFLECTION BETWEEN ADJACENT FLOORS
3. DIFFERENTIAL THERMAL EXPANSION / CONTRACTION BETWEEN FACADE SYSTEM AND SUPPORTING PRIMARY STRUCTURAL SYSTEM
4. THE FACADE SHALL NOT BE INSTALLED UNTIL AFTER THE SUPERSTRUCTURE HAS BEEN PLACED. ALL FRAMING HAVE BEEN INSTALLED, AND ALL SHORES AND RESHORES HAVE BEEN REMOVED.
5. WHERE STRUCTURAL STEEL HORIZONTAL GIRTS ARE PROVIDED, THOSE GIRTS MAY BE REQUIRED TO RESIST LATERAL LOADS. LATERAL LOADS MAY NOT BE IMPOSED BY THE FACADE UPON THE GIRT FRAMING UNLESS THE SECTIONS AND DETAILS ON THE STRUCTURAL DRAWINGS SPECIFICALLY INDICATE THAT FACADE GRAVITY LOAD CONNECTIONS MAY BE MADE TO THE GIRTS.

EARTHWORK

1. FOUNDATION DESIGN IS BASED ON GEOTECHNICAL REPORT NO. 2128/1119 DATED AUGUST 13 2019 BY WESTERN TECHNOLOGIES INC. FOLLOW UP RECOMMENDATIONS IN REPORT FOR ALL FOUNDATION WORK. REPORT IS ON FILE WITH THE ARCHITECT.
2. SOIL PROPERTIES:
- | | |
|-----------------------------|---------------------|
| COEFFICIENT OF FRICTION | 0.30 |
| EQUIVALENT FLUID PRESSURES: | |
| ACTIVE | 35 PSF/FT (DRAINED) |
| AT REST | 55 PSF/FT (DRAINED) |
| PASSIVE | 240 PSF/FT OF DEPTH |
3. SUBGRADE PREPARATION FOR FOOTINGS AND SLABS-ON-GRADE SHALL BE IN ACCORDANCE WITH THE RECOMMENDATIONS OF THE GEOTECHNICAL REPORT AND SHALL BE IN COMPLIANCE WITH THE APPLICABLE REQUIREMENTS OF THE GOVERNING AUTHORITIES HAVING JURISDICTION. THE CONTRACTOR SHALL DIRECT QUESTIONS REGARDING THE SUBGRADE PREPARATION REQUIREMENTS TO THE GEOTECHNICAL ENGINEER.
4. ANY TESTS, INSPECTIONS, FIELD OBSERVATIONS, OR APPROVAL FROM THE GEOTECHNICAL ENGINEER SHALL BE PERFORMED PRIOR TO PLACEMENT OF CONCRETE. ALTERATIONS TO SITE PREPARATION OR THE DESIGN OF COLD-FORMED STEEL FOUNDATION PLACING DRAWINGS THAT DO NOT PROVIDE THIS LIST OF TOLERANCES WILL BE REJECTED.

5. PROPERTY LINE LOCATIONS INDICATED ON FOUNDATION PLANS ARE APPROXIMATE. SEE ARCHITECTURAL AND/OR SITE DRAWINGS FOR LOCATION OF THE STRUCTURE ON THE SITE.
6. ALL EXCAVATIONS SHALL BE PROPERLY AND SAFELY BACKFILLED. DO NOT PLACE BACKFILL BEHIND RETAINING / BASEMENT WALLS BEFORE CONCRETE HAS ATTAINED SPECIFIED COMPRESSIVE STRENGTH. CONTRACTOR SHALL BRACE OR PROTECT ALL WALLS BELOW GRADE FROM LATERAL LOADS UNTIL SUPPORTING FLOORS ARE COMPLETELY IN PLACE AND HAVE ATTAINED 7-DAY STRENGTH MINIMUM. BACKFILLING IS NOT PERMITTED FOR FOUNDATION WALLS UNTIL SUPPORTED SLAB TOP AND BOTTOM ARE IN PLACE OR THE WALL IS ADEQUATELY BRACED TO RESIST LATERAL LOADS.
7. SOIL BEHIND RETAINING WALLS AND BASEMENT WALLS SHALL BE DRAINED TO ELIMINATE HYDROSTATIC PRESSURE BEHIND THE WALL. DESIGN OF SUCH WALL DRAINAGE SYSTEMS IS THE RESPONSIBILITY OF THE CONTRACTOR.
8. CONTRACTOR SHALL PROVIDE DE-WATERING OF EXCAVATIONS FROM SURFACE WATER, GROUNDWATER, OR SEEPAGE. DETAILS OF GROUNDWATER INFORMATION SHALL BE OBTAINED FROM THE GEOTECHNICAL REPORT. IF GROUNDWATER IS ENCOUNTERED DURING EXCAVATION, PROCEDURES SHALL BE IMPLEMENTED AS RECOMMENDED BY THE GEOTECHNICAL ENGINEER.
9. PROVIDE SHORING WHERE THERE IS INSUFFICIENT SPACE FOR STABLE-SLOPED EMBANKMENTS.
10. CONTRACTOR SHALL BE RESPONSIBLE FOR THE DESIGN AND INSTALLATION OF ALL CRIBBING, SHEETING, SHORING, ETC. REQUIRED FOR CONSTRUCTION OF THE PROJECT AND SHALL BE SOLELY RESPONSIBLE FOR ALL EXCAVATION PROCEDURES INCLUDING LAGGING, SHORING, AND PROTECTION OF ADJACENT PROPERTY, STRUCTURES, STREETS, AND UTILITIES. CONTRACTOR SHALL SUBMIT DESIGN CALCULATIONS AND SHOP DRAWINGS SIGNED AND SEALED BY AN ENGINEER LICENSED IN THE PROJECTS JURISDICTION.
11. CONTRACTOR SHALL INVESTIGATE SITE DURING CLEARING AND EARTHWORK OPERATIONS FOR FILL MATERIAL OR BURIED STRUCTURES SUCH AS CESSPOOLS, CISTERNS, AND FOUNDATIONS. IF ANY SUCH MATERIAL OR STRUCTURES ARE FOUND, ARCHITECT SHALL BE NOTIFIED IMMEDIATELY.
12. ANY REQUIRED IMPORT FILL SHALL HAVE A LOW POTENTIAL FOR EXPANSION AND SHALL BE APPROVED BY THE GEOTECHNICAL ENGINEER PRIOR TO IMPORTING.
13. UTILITY LINES SHALL NOT BE PLACED THROUGH OR BELOW FOUNDATIONS WITHOUT THE STRUCTURAL ENGINEER'S APPROVAL. BELOW GRADE UTILITY OR PIPE ELEVATIONS, WHERE SHOWN, ARE INDICATED FOR REFERENCE ONLY. REQUIRED ELEVATIONS SHALL BE DETERMINED BY THE CONTRACTOR AND COORDINATED WITH THE FOUNDATIONS.
14. WHERE GRADE ELEVATIONS ARE APPROXIMATELY EQUAL ON BOTH SIDES OF WALLS, BACKFILL SHALL BE PLACED SO THAT IT IS NOT UNBALANCED BY MORE THAN 2 FEET ON EITHER SIDE.
15. ALL REQUIRED BACKFILL AND UTILITY TRENCH BACKFILL WITHIN THE BUILDING AREA SHALL BE COMPACTED IN ACCORDANCE WITH THE GEOTECHNICAL REPORT.

SHALLOW FOUNDATIONS

1. SEE THE GEOTECHNICAL REPORT FOR SHALLOW FOUNDATION REQUIREMENTS.
2. SHALLOW FOUNDATIONS SHALL HAVE THE FOLLOWING MINIMUM NET ALLOWABLE SERVICE LOAD BEARING PRESSURES:
- | | |
|---------------------------------|------------------------------|
| SERVICE LOAD BEARING PRESSURES: | |
| NET ALLOWABLE BEARING PRESSURE | 2500 PSF, 1.5 FT BELOW GRADE |
| | 3000 PSF, 2.0 FT BELOW GRADE |
| | 3500 PSF, 2.5 FT BELOW GRADE |
3. FOUNDATION ELEVATIONS SHOWN INDICATE LOCATIONS WHERE ADEQUATE SOIL BEARING PRESSURE IS ANTICIPATED. IF INADEQUATE BEARING CAPACITY IS ENCOUNTERED, CONTACT STRUCTURAL ENGINEER FOR RESOLUTION. BEARING ELEVATIONS ARE ESTIMATED FROM SOIL BORING DATA INDICATED IN THE GEOTECHNICAL REPORT. DETERMINATION OF FINAL BEARING ELEVATIONS AND FIELD VERIFICATION OF SOIL BEARING PRESSURE SHALL BE MADE BY AN EXPERIENCED, QUALIFIED GEOTECHNICAL ENGINEER PRIOR TO PLACING FOUNDATIONS.
4. FOUNDATIONS SHALL BE PLACED ON UNDISTURBED SOIL OR COMPACTED STRUCTURAL FILL, AND CLEAN AND FREE OF LOOSE DEBRIS AND STANDING WATER AT TIME OF CONCRETE PLACEMENT.
5. THE SLOPE BETWEEN THE LOWER EDGES OF ADJACENT FOOTINGS SHALL NOT EXCEED 45 DEGREES WITH THE HORIZONTAL UNO IN THE GEOTECHNICAL REPORT. CONTACT STRUCTURAL ENGINEER WHERE ADEQUATE SLOPE IS NOT ACHIEVED.

REINFORCING STEEL

1. ALL REINFORCING STEEL SHALL BE DETAILED AND PLACED IN CONFORMANCE WITH ACI 318, "BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE," AND ACI 301, "SPECIFICATIONS FOR STRUCTURAL CONCRETE," UNO.
- | | | |
|--------------------------|------------------|----------------------------------|
| REINFORCING STEEL UNO | ASTM A615, GR 60 | Fy=60 KSI |
| WELDED REINFORCING STEEL | ASTM A706, GR 60 | Fy=60 KSI |
| WELDED WIRE REINFORCING | ASTM A1064 | Fy=65 KSI |
| DEFORMED BAR ANCHORS | ASTM A1064 | Fy=70 KSI |
| HEADED DEFORMED BARS | ASTM A970 | Fy=DEFORMED BAR TENSILE STRENGTH |
2. CONCRETE REINFORCING STEEL SHALL CONFORM TO THE FOLLOWING STANDARDS:

CONCRETE EXPOSURE	MEMBER	REINFORCE- MENT	COVER (IN)
CAST AGAINST AND PERMANENTLY IN CONTACT WITH GROUND	ALL	ALL	3
EXPOSED TO WEATHER OR IN CONTACT WITH GROUND	ALL	#6 TO #18	2
		#5 AND SMALLER	1 1/2
	SLABS, JOISTS, AND WALLS	#14 & #18	1 1/2
	#11 AND SMALLER	3/4	
NOT EXPOSED TO WEATHER OR IN CONTACT WITH GROUND	BEAMS, COLUMNS, PEDESTALS, STAIRS, TIES, SPIRALS, AND HOOPS	PRIMARY REINF. STIRRUPS, TIES, SPIRALS, AND HOOPS	1 1/2
	BOUNDARY ELEMENTS	ALL	1 1/2
OTHER	PARKING LEVEL SLABS	TOP BARS	1 1/2
		BOTTOM BARS	1

3. MINIMUM CONCRETE COVER SHALL BE PROVIDED AS FOLLOWS TO THE OUTERMOST REINFORCING BARS:
- REINFORCING STEEL SHALL BE INSTALLED TO WITHIN THE FOLLOWING TOLERANCES INDICATED TOLERANCES ARE PER ACI 117, "SPECIFICATIONS FOR TOLERANCES FOR CONCRETE CONSTRUCTION AND MATERIALS".
- ITEM FOR WHICH TOLERANCE IS BEING MEASURED
- PERMITTED TOLERANCE
- CONCRETE COVER FOR SLAB TOP AND BOTTOM BARS
- ±1/4"
- SPECIFIED SPACING BETWEEN PARALLEL BARS IN SLAB
- ± (SPECIFIED SPACING/4) BUT NOT TO EXCEED 1"
- HORIZONTAL DEVIATION FROM SPECIFIED LOCATION UNO
- ±3"
- SPACING AND LOCATION OF BEAM STIRRUPS
- ± (BEAM DEPTH IN INCHES/12) X 1"
- SPACING AND LOCATION OF COLUMN TIES
- ± (MINIMUM COLUMN DIMENSION IN INCHES/12) X 1"
- LOCATION OF ENDS OF BARS PERPENDICULAR TO SLAB EDGES
- ±1"
- HORIZONTAL LOCATION OF PRESTRESSED TENDONS
- ±1/2" FOR MEMBER DEPTH > 24" ±1" FOR MEMBER DEPTH > 24"
- VERTICAL LOCATION (DRAPE) OF PRESTRESSED TENDONS
- ±1/4" FOR MEMBER DEPTH = 8" ±3/8" FOR MEMBER DEPTH 8" <lt; D = 24" ±1/2" FOR MEMBER DEPTH > 24"

THE ABOVE LIST OF PERMITTED TOLERANCES SHALL BE PROVIDED ON ALL REINFORCING STEEL PLACING DRAWINGS AND ON ALL POST-TENSIONED TENDON PLACING DRAWINGS. PLACING DRAWINGS THAT DO NOT PROVIDE THIS LIST OF TOLERANCES WILL BE REJECTED.

5. FIELD BENDING OF REINFORCING STEEL IS NOT PERMITTED UNO.
6. WELDING OF REINFORCING STEEL OTHER THAN A706 IS PROHIBITED. WELDING OF REINFORCING BARS SHALL BE IN ACCORDANCE WITH AWS D1.4 OR D1.8.
7. ALL WELDED WIRE REINFORCING SHALL BE LAP SPICED 2 PANELS (1'-0" MIN).
8. SPLICING:
- a. SPLICES IN REINFORCING STEEL SHALL BE MADE ONLY AT THOSE LOCATIONS WHERE SPLICES ARE SHOWN ON THE STRUCTURAL DRAWINGS AND AT THOSE LOCATIONS WHERE SPLICES HAVE BEEN DETAILED ON THE REINFORCING STEEL PLACING DRAWINGS THAT HAVE BEEN REVIEWED AND APPROVED BY THE STRUCTURAL ENGINEER. ALL SPLICES SHALL BE CLASS B TENSION LAP SPLICES UNO.
- b. MECHANICAL SPICE COUPLERS MAY BE USED INSTEAD OF TENSION LAP SPLICES AT THE CONTRACTOR'S OPTION AT ANY LOCATION. MECHANICAL SPICE COUPLERS MUST BE USED WHERE SPLICES #14 AND LARGER BARS, INCLUDING WHERE SPLICES #14 AND LARGER BARS TO #11 AND SMALLER BARS. STAGGER MECHANICAL SPLICES IN ADJACENT BARS 30" MINIMUM.
- c. COMPRESSION LAP SPLICES MAY BE USED ONLY AT THOSE LOCATIONS WHERE SUCH SPLICES ARE SPECIFICALLY INDICATED. STAGGER SPLICES WHERE REQUIRED TO PROVIDE 1 1/2" MINIMUM CLEAR SPACING BETWEEN REINFORCING STEEL AT SPICE LOCATIONS.
9. ALL HOOKS SHALL BE STANDARD HOOKS OR STANDARD STIRRUP HOOKS UNO. STANDARD STIRRUP HOOKS SHALL HAVE CONTINUOUS BAR AT INSIDE CORNER OF HOOK.
10. VERTICAL REINFORCING STEEL IN CONCRETE AND MASONRY WALLS WITH ONE LAYER OF REINFORCING BARS SHALL BE INSTALLED IN THE CENTER OF THE WALL UNO.
11. STANDARD STIRRUP HOOKS FOR #3, #4, AND #5 BARS SHALL BE PROVIDED IN SLABS LESS THAN 9" THICK.
12. DOWELS SHALL MATCH GRADE, SIZE, SPACING, AND QUANTITY OF LAPPED REINFORCING STEEL UNO. EXTEND ALL DOWELS TO THE SUPPORTING ELEMENT AND PROVIDE HOOKS UNO. DOWELS SHALL NOT BE POST-INSTALLED INTO FRESH CONCRETE.
13. HEADED DEFORMED BARS MAY ONLY BE USED ON #11 AND SMALLER BARS. THREADED OR FORGED HEADS CAN BE USED AT THE FABRICATOR'S DISCRETION.
14. FIELD CUTTING OR REINFORCING STEEL IS PROHIBITED UNLESS INDICATED ON THE REINFORCING PLACING DRAWINGS.
15. HEATING OF BARS FOR BENDING IS PROHIBITED.
16. PROVIDE EPOXY-COATED REINFORCING STEEL IN ALL PARKING LEVEL SLABS, PLAZA SLABS, BALCONIES, TERRACES, AND ALL OTHER SLABS EXPOSED TO WEATHER AND ON COLUMN TIES AND ALL WALL REINFORCING STEEL WITHIN 4 FEET OF THE BOTTOM OF THOSE SLABS.
17. REINFORCING STEEL PLACING DRAWINGS SHALL BE PREPARED IN ACCORDANCE WITH THE REQUIREMENTS OF ACI 315. THE PLACING DRAWINGS SHALL SHOW ALL INFORMATION NECESSARY TO FABRICATE AND PLACE THE REINFORCING STEEL.
18. REINFORCING STEEL SPACINGS ARE CENTER-TO-CENTER DIMENSIONS UNO. REINFORCING STEEL SHOWN IN SECTION PERPENDICULAR TO THE CUT ARE CONTINUOUS UNO.
19. THE SPACING OF ALL REINFORCING STEEL MUST BE COMPUTED BY THE REINFORCING STEEL DETAILER AND MUST BE INDICATED ON THE PLACING DRAWINGS. EXTENT ARROWS MUST BE USED TO CLEARLY INDICATE THE LOCATIONS WHERE GROUPS OF REINFORCING BARS ARE TO BE INSTALLED.
20. A LIST OF ALL APPLICABLE REINFORCING STEEL PLACEMENT TOLERANCES SHALL BE INDICATED ON ALL REINFORCING STEEL PLACING DRAWINGS. PLACING DRAWINGS THAT DO NOT SHOW SUFFICIENT INFORMATION NEEDED TO PLACE THE REINFORCING STEEL WILL BE REJECTED.

CAST-IN-PLACE CONCRETE

1. CONCRETE MATERIALS SHALL CONFORM TO:
- | | |
|---------------------------|------------------------|
| PORTLAND LIMESTONE CEMENT | ASTM C595, TYPE II |
| FLY ASH | ASTM C618, TYPE C OR F |
| SLAB CEMENT | ASTM C595 |
| FINE AND COARSE AGGREGATE | ASTM C33 |
| LIGHTWEIGHT AGGREGATE | ASTM C330 |
| WATER | POTABLE |
| AIR-ENTRAINING ADMIXTURE | ASTM C260 |
| WATER REDUCING ADMIXTURE | ASTM C494 |
2. ALL CONCRETE WORK SHALL CONFORM TO THE REQUIREMENTS OF ACI 318, BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE, AND ACI 301, SPECIFICATIONS FOR STRUCTURAL CONCRETE UNO.
3. AIR ENTRAINMENT:
- | NOMINAL MAXIMUM AGGREGATE SIZE | REQUIRED AIR CONTENT PER EXPOSURE CATEGORY | | |
|--------------------------------|--|------|--|
| | F1 | F2 | |
| 3/8" | 6% | 7.5% | |
| 1/2" | 5.5% | 7% | |
| 3/4" | 5% | 6% | |
| 1" | 4.5% | 6% | |
- a. CONCRETE IN THESE LOCATIONS SHALL BE AIR ENTRAINMENT WITH THE APPROPRIATE PERCENTAGE AIR CONTENT LISTED IN THE TABLE ABOVE AS APPLICABLE FOR THE INDICATED EXPOSURE CLASS AND NOMINAL MAXIMUM AGGREGATE SIZE IN THE CONCRETE MIX. THE REQUIRED AIR CONTENT VALUE MAY BE REDUCED BY 1% FOR ALL CONCRETE WITH COMPRESSIVE STRENGTH GREATER THAN 5000 PSI. THE PERMITTED TOLERANCE ON THE REQUIRED AIR CONTENT IS ±1.0%.
- b. ALL LIGHTWEIGHT CONCRETE SHALL HAVE 4-7% AIR ENTRAINMENT.
- c. AIR ENTRAINMENT SHALL CONFORM TO UL RATING REQUIREMENTS FOR FIRE RESISTANCE.

LOCATION	f _c AT 28 DAYS (PSI)	MAX PERMITTED W/C	EXPOSURE CLASS
ALL FOUNDATION CONCRETE UNO	4500	0.45	
SLAB-ON-GRADE UNO	3000	0.55	
TOPPING SLABS	3000	0.55	
POURABLE FILL AND MUD SLABS	1000	N/A	
*LIGHTWEIGHT CONCRETE SHALL HAVE A 115 PCF DRY UNIT WEIGHT (±3PCF). LIGHTWEIGHT AGGREGATES SHALL BE PRESOAKED AS REQUIRED IN ACCORDANCE WITH ACI 301.			
** SEE SCHEDULE OR ELEVATIONS FOR AREAS OF DIFFERENT STRENGTH REQUIREMENTS. *** WALLS MONOLITHIC WITH COLUMNS OR PIERS SHALL BE CONSTRUCTED USING CONCRETE SPECIFIED FOR THE COLUMNS WITHIN 15 FEET MINIMUM TO EACH SIDE OF THE COLUMNS.			

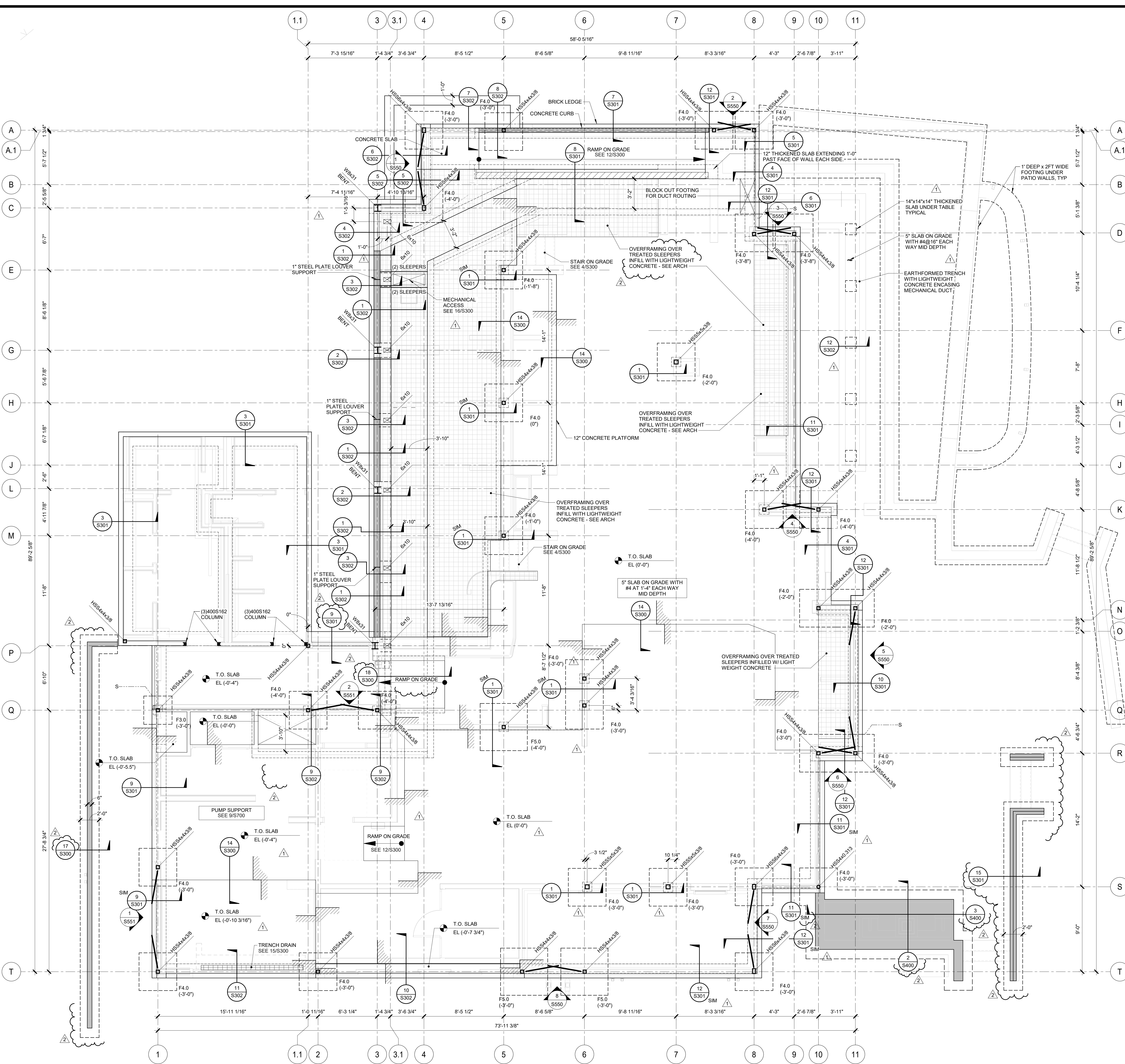
5. ALL FOUNDATION ELEMENTS SHALL BE CENTERED UNDER WALLS, PIERS, OR COLUMNS UNO.
6. CONTRACTOR SHALL SUBMIT PROPOSED LOCATIONS OF ALL CONSTRUCTION JOINTS WHERE JOINTS ARE NOT INDICATED ON THE DRAWINGS.
7. HORIZONTAL CONSTRUCTION JOINTS THROUGH CAST-IN-PLACE CONCRETE FRAMING ARE NOT PERMITTED EXCEPT WHERE SPECIFICALLY INDICATED ON THE STRUCTURAL DRAWINGS.
8. PROVIDE TEMPLETS TO SET EMBEDDED ITEMS.
9. PROJECTING CORNERS OF BEAMS, WALLS, COLUMNS, ETC. SHALL BE FORMED WITH A 3/4" CHAMFER UNO ON ARCHITECTURAL DRAWINGS.
10. SLOPE SLABS TO DRAINS. SEE ARCHITECTURAL AND MEP DRAWINGS FOR DRAIN LOCATIONS AND SLOPE REQUIREMENTS. SLAB THICKNESSES SHOWN ON DRAWINGS ARE MINIMUMS.
11. NOTIFY THE ARCHITECT 48 HOURS MINIMUM PRIOR TO ALL POURS.
12. CONTRACTOR SHALL SURVEY ALL CONCRETE WORK WITHIN 48 HOURS OF PLACING CONCRETE TO ENSURE PLACEMENT IS IN ACCORDANCE WITH PROJECT REQUIREMENTS.
13. ALL FORMWORK, SHORING, AND RESHORING SHALL BE DESIGNED BY THE CONTRACTORS ENGINEER OR LICENSED ENGINEER IN THE PROJECT'S JURISDICTION. ALL SUBMISSIONS SHALL BE SIGNED AND SEALED.
14. CURING OF CONCRETE IS NOT PERMITTED UNLESS APPROVED BY THE STRUCTURAL ENGINEER. THE CONTRACTOR SHALL SUBMIT PROPOSED CONCRETE CURING METHODS.
15. REINFORCING STEEL SHALL NOT BE DAMAGED WHEN DRILLING CONCRETE.
16. ADHERE TO ACI 305R AND ACI 308R FOR HOT AND COLD WEATHER CONCRETE CONSTRUCTION.
17. DRYPACK AND GROUT SHALL HAVE A MINIMUM 28-DAY STRENGTH OF 7000 PSI.
18. THE PROPOSED MATERIALS AND MIX DESIGN SHALL BE FULLY DOCUMENTED AND REVIEWED BY THE TESTING AND INSPECTION AGENCY. RESPONSIBILITY FOR OBTAINING THE REQUIRED DESIGN STRENGTH IS THE CONTRACTOR'S. SUBMIT TEST DATA ON EACH PROPOSED MIX FOR REVIEW IN ACCORDANCE WITH THE APPLICABLE CODE. MIX DESIGNS SUBMITTED WITHOUT THE REQUIRED TEST DATA WILL BE RETURNED WITHOUT REVIEW.
19. SEE ARCHITECTURAL DRAWINGS FOR DIMENSIONS, LOCATIONS, AND DETAILS OF ALL ARCHITECTURAL FEATURES IN THE CONCRETE. SEE ARCHITECTURAL DRAWINGS AND PROJECT SPECIFICATIONS FOR REQUIREMENTS FOR ALL CONCRETE FINISHES.

STEEL

1. STRUCTURAL STEEL SHALL BE DETAILED IN ACCORDANCE WITH THE AMERICAN INSTITUTE OF STEEL CONSTRUCTION, INC. (AISC) DETAILING FOR STEEL CONSTRUCTION AND FABRICATED AND ERECTED IN ACCORDANCE WITH THE "SPECIFICATION FOR STRUCTURAL STEEL BUILDINGS".
2. STRUCTURAL STEEL SHALL CONFORM TO ASTM STANDARDS AS NOTED BELOW.[ENGR NOTE: EDIT STEEL MATERIALS TO SUIT PROJECT.]
- | | | |
|----------------------------|--------------------|------------|
| WIDE FLANGE SHAPES | ASTM A992 | Fy=50 KSI |
| OTHER ROLLED SHAPES | ASTM A36 | Fy=36 KSI |
| PIPE SECTIONS | ASTM A53, GR B | Fy=35 KSI |
| HSS SECTIONS, ROUND | ASTM A500, GR C | Fy=46 KSI |
| HSS SECTIONS, SQUARE | ASTM A500, GR C | Fy=50 KSI |
| BASE AND CONNECTION PLATES | ASTM A36 | Fy=36 KSI |
| ANCHOR RODS | ASTM F1554, GR 36 | Fy=36 KSI |
| HIGH STRENGTH BOLTS | ASTM F3125, GR A35 | Fy=120 KSI |
| HEAVY HEX NUTS | ASTM A563 | |
| WASHERS | ASTM F436 | |
| ELECTRODES FOR ARC WELDING | AWS S. 1, E70XX | |
3. HIGH STRENGTH BOLTS SHALL BE INSTALLED IN ACCORDANCE WITH AISC "SPECIFICATIONS FOR STRUCTURAL JOINTS USING HIGH-STRENGTH BOLTS". SEE DETAILS FOR BOLT SIZE AND MATERIAL ASTM DESIGNATION.
4. ALL BOLTED CONNECTIONS SHALL BE GRADE A325N BEARING TYPE BOLTS UNO. ALL BOLTS SHALL BE INSTALLED PERPENDICULAR TO THE PLACING DRAWINGS.
5. FIELD CONNECTIONS SHALL BE WELDED OR BOLTED. SHOP CONNECTIONS SHALL BE WELDED UNO. WELDS INDICATED WITH A SHOP WELD SYMBOL MAY BE MADE IN THE FIELD UNO. WELDS SHALL BE MADE BY THE CONTRACTOR. LOCATIONS OF ALL FIELD WELDS SHALL BE CLEARLY SHOWN ON THE SHOP DRAWINGS. WELDS SHALL BE DESIGNED TO BE FULLY EQUIVALENT IN STRENGTH TO BOLTED CONNECTIONS DETAILED TO MINIMIZE CONNECTION SIZE AND CONNECTION TO BE MADE.
6. WELD LENGTHS INDICATED ON THE DRAWINGS ARE THE NET EFFECTIVE LENGTH REQUIRED, WHERE WELD LENGTH IS NOT SPECIFIED, PROVIDE WELD ALONG ENTIRE INTERSECTION OF THE JOINED PARTS. WHERE FILLET WELD SYMBOL IS GIVEN WITHOUT INDICATION OF SIZE, USE MINIMUM WELD SIZE AS SPECIFIED IN AISC 360, TABLE J2.4.
7. ALL WELDING OF STRUCTURAL STEEL SHALL BE PERFORMED BY CERTIFIED WELDERS WITH EXPERIENCE AND CERTIFICATION IN THE TYPES OF WELDING INDICATED. WELDERS SHALL HAVE BEEN RECENTLY IN COMPLIANCE AS PRESCRIBED IN "QUALIFICATION PROCEDURES" OF THE AMERICAN WELDING SOCIETY (AWS).
8. BEAMS SHALL BE CAMBERED UPWARD WHERE SHOWN ON THE DRAWINGS. WHERE NO UPWARD CAMBER IS INDICATED, ANY MILL CAMBER SHALL BE DETAILED UPWARD IN THE BEAMS.
9. SPLICING OF STEEL MEMBERS WHERE NOT DETAILED ON THE DRAWINGS IS PROHIBITED WITHOUT THE PRIOR APPROVAL OF THE STRUCTURAL ENGINEER AS TO LOCATION, TYPE OF SPICE, AND CONNECTION TO BE MADE.
- <

- S002

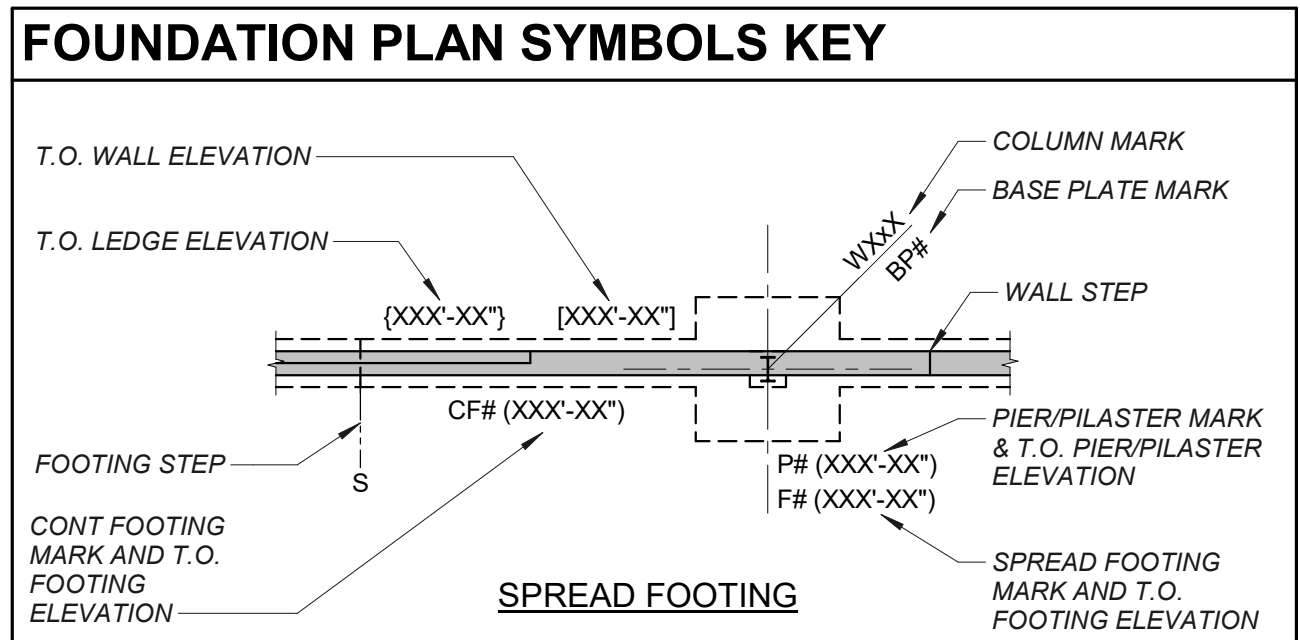
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- FOUNDATION NOTES:**
1. DATUM ELEVATION (0'-0") EQUALS LEVEL 1 TOP OF SLAB ELEVATION SEE CIVIL FOR USGS DATUM.
 2. ALL SLAB-ON-GRADES ARE 5" THICK WITH #4 AT 1'-4" EACH WAY, UON.
 3. CONTRACTOR TO COORDINATE ALL DIMENSIONS AND DETAILS WITH ARCHITECTURAL DRAWINGS.
 4. GENERAL CONTRACTOR SHALL FOLLOW ALL REQUIREMENTS LISTED IN SOILS REPORT.
 5. ALL SITE WALLS NOT SHOWN ON THIS PLAN SHALL BE BUILT PER ARCHITECTURAL LANDSCAPE PLANS, DETAILS, AND ELEVATIONS. SITE WALLS SHALL NOT BE CONNECTED TO THE BUILDING UNLESS NOTED OTHERWISE.
 6. REFER ARCHITECTURAL DRAWINGS FOR HANDRAIL/GUARDRAIL LOCATIONS. AT VERTICAL POSTS PROVIDE 1/2"x4"x1'-0" EMBED PLATE WITH (3) 5/8"x4" X4" AT 4" GAGE. SPACE POSTS AT 4'-0" MAX.

GENERAL SYMBOLS:	
SYMBOL	DESCRIPTION
	STRUCTURAL STEP

SPREAD FOOTING SCHEDULE						
MARK	LENGTH	WIDTH	DEPTH	REINFORCING		REMARKS
				LONG DIRECTION	SHORT DIRECTION	
F3.0	3'-0"	3'-0"	1'-0"	(3)#5 BOT	(3)#5 BOT	
F4.0	4'-0"	4'-0"	1'-0"	(4)#5 BOT	(4)#5 BOT	T&B AT FRAMES
F5.0	5'-0"	5'-0"	1'-2"	(5)#5 BOT	(5)#5 BOT	
F14X4	14'-0"	4'-0"	1'-2"	(12)#5 T&B	(4)#5 T&B	



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1" = 1'-0"

1 2 3
INCHES PROJECT WIDENESS 11'

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architect's stamp

consultant

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#	DATE	ISSUE RECORD
1	04/30/2025	CHECKSET
2	10/09/2025	PLAN CHECK SUBMITTAL 02
3	01/01/2026	PC COMMENTS

PROJECT TITLE:

3101 EAST CAMELBACK ROAD
PHOENIX, ARIZONA 85016

PC/PM: LG
DRAWN BY: STAFF
JOB NO.: 24004991.01

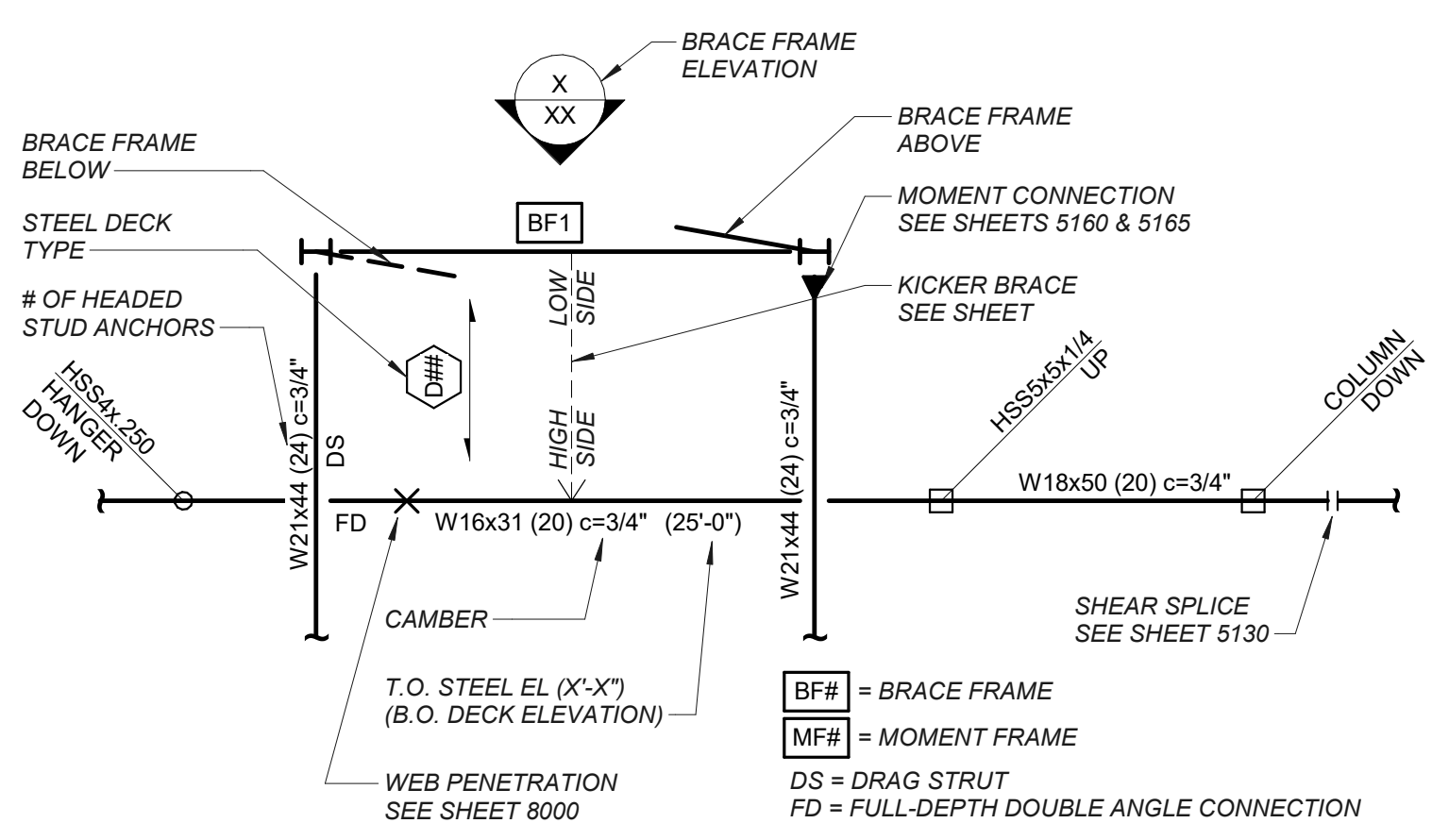
DRAWING TITLE:
FOUNDATION PLAN

SHEET NO.:
S110

STEEL ROOF FRAMING NOTES:

1. SLOPE METAL ROOF DECK, JOISTS, AND STEEL BEAMS UNIFORMLY BETWEEN ELEVATIONS SHOWN ON PLAN.
2. DECK DESIGN ASSUMES A MINIMUM OF 2-SPAN DECK CONDITION.
3. DECK SHALL BEAR A MINIMUM OF 2" ON STRUCTURAL FRAMING AT ALL LOCATIONS, UNLESS DETAIL OTHERWISE.
4. STEEL BEAMS OR JOISTS SHALL BE EQUALLY SPACED BETWEEN GRID LINES UNLESS DIMENSIONED OTHERWISE.

STEEL PLAN SYMBOLS KEY



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ROOF FRAMING PLAN

$$1/4'' = 1'-0''$$




1

CARPORT FOUNDATION PLAN

1/4" = 1'-0"



2

CARPORT ROOF FRAMING PLAN

1/4" = 1'-0"

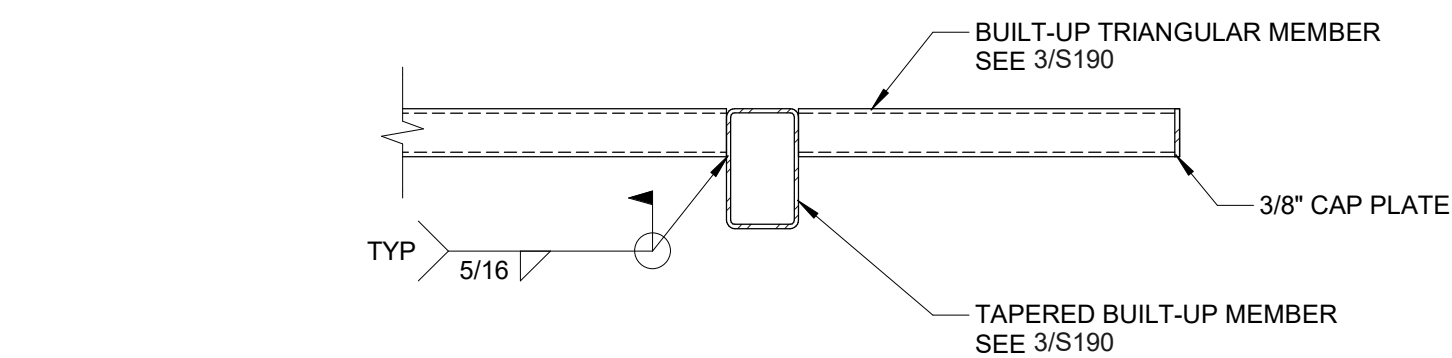
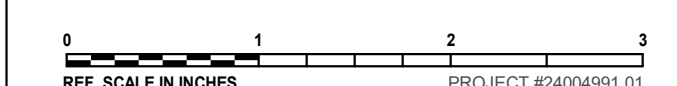
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CARPORT EDGE DETAIL

3/4" = 1'-0"



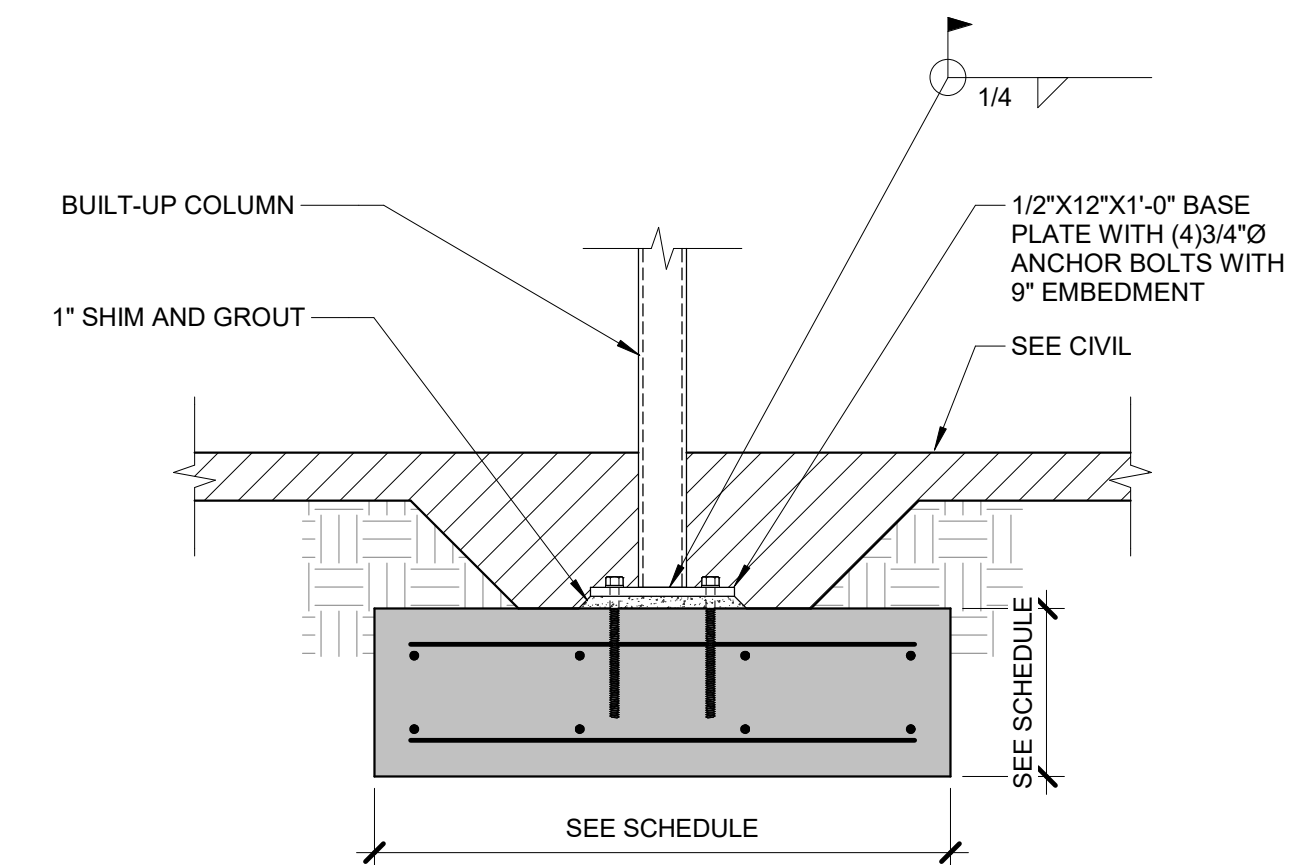
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4

CARPORT BASE DETAIL

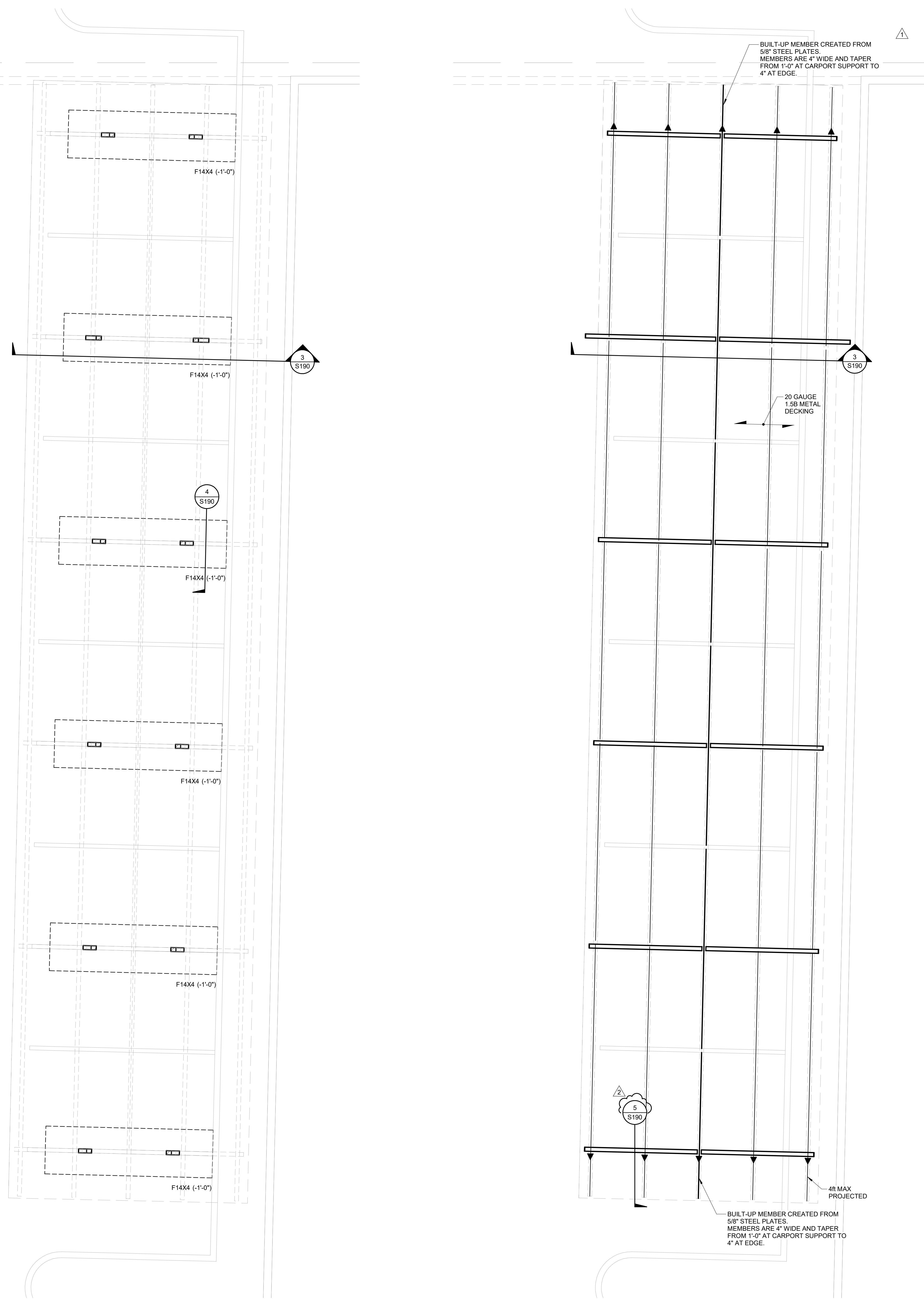
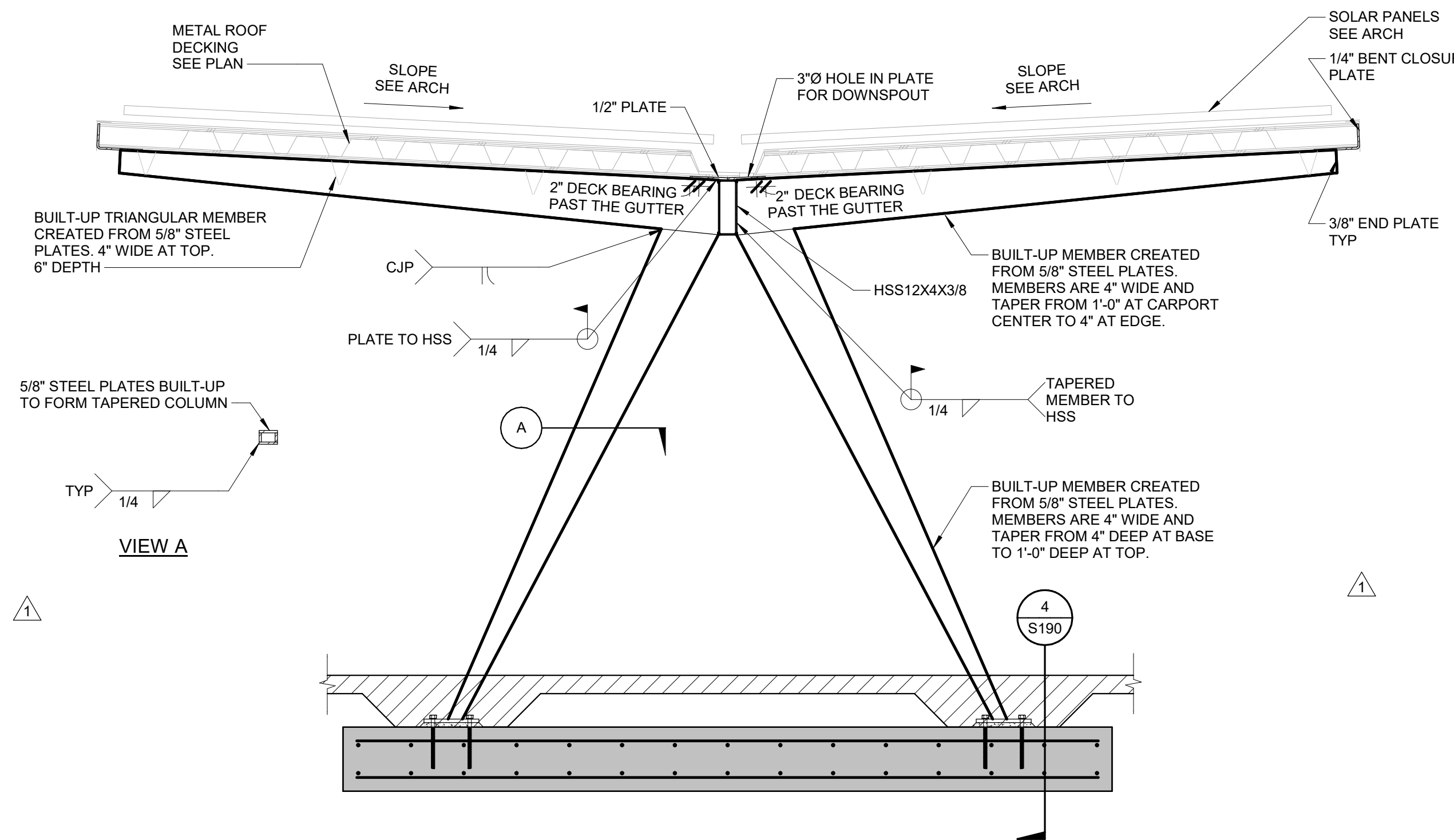
3/4" = 1'-0"



3

CARPORT SECTION

1/2" = 1'-0"



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#	DATE	ISSUE RECORD
1	04/20/2025	CHECKSET
2	01/21/2025	PLAN CHECK SUBMITTAL 02
3	01/21/2025	PC COMMENTS
4	01/21/2025	PC COMMENTS
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60	01/21/2025	PC COMMENTS
61	01/21/2025	PC COMMENTS
62	01/21/2025	PC COMMENTS
63	01/21/2025	PC COMMENTS
64	01/21/2025	PC COMMENTS
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68	01/21/2025	PC COMMENTS
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70	01/21/2025	PC COMMENTS
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72	01/21/2025	PC COMMENTS
73	01/21/2025	PC COMMENTS
74	01/21/2025	PC COMMENTS
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84	01/21/2025	PC COMMENTS
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90	01/21/2025	PC COMMENTS
91	01/21/2025	PC COMMENTS
92	01/21/2025	PC COMMENTS
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94	01/21/2025	PC COMMENTS
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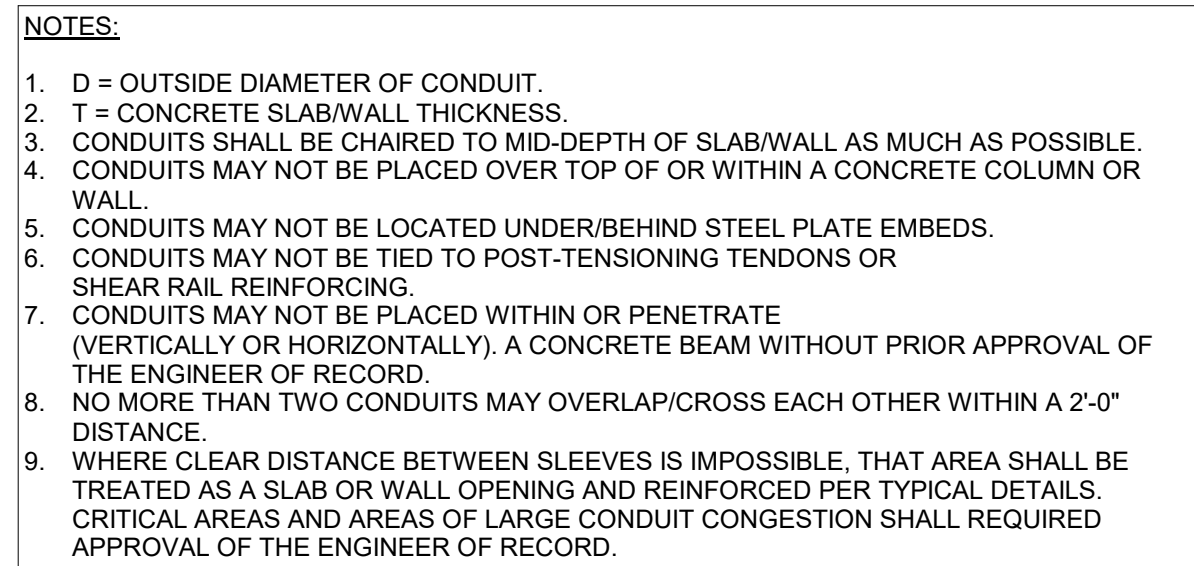
PROJECT TITLE:

3101 EAST CAMELBACK ROAD
PHOENIX, ARIZONA 85016
PC/PM: LG
DRAWN BY: STAFF
JOB NO.: 24004991.01

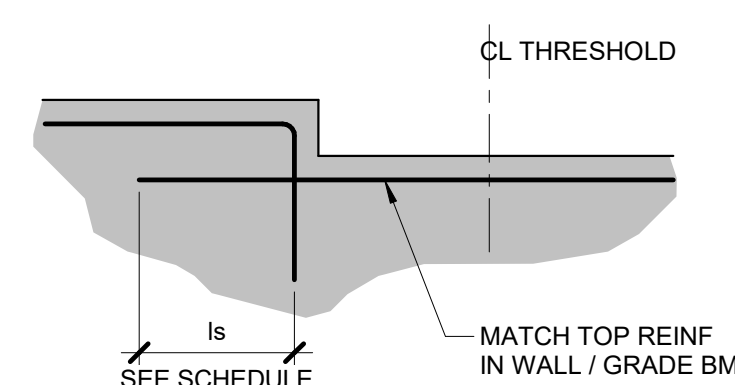
DRAWING TITLE:
CARPORT PLANS

SHEET NO.:

S190



2 FOUNDATION AT THRESHOLD



1'-0"

6"

2'-0"

1/2" ISOLATION JOINT

#5x2'-0" @ 1'-0" OC

#5x2'-0" @ 1'-0" OC EACH WAY

(2) #5 AT BOTTOM - TYP

1/2"

SEE ARCH

SEE ARCH

NOTE: WHERE STEPS OCCUR AT CORNER, PROVIDE CORNER BARS FOR CONTINUOUS REINFORCING

STANDARD HOOK

TIES AS OCCUR

LAP LENGTH

BARS TO MATCH TYPICAL REINFORCING

TYPICAL TOP REINFORCING

TYPICAL BOTTOM REINFORCING

SEE PLAN

SEE PLAN

SEE PLAN

3'-0"

3"

3"

CLASS B TENSION LAP SPLYCE LENGTH											
REIN STEEL Type	BAR SIZE	BAR LOCATION	CONCRETE STRENGTH, f_c								
			3 KSI	4 KSI	5 KSI	6 KSI	7 KSI	8 KSI	9 KSI	10 KSI	
60 KSI	#3	TOP	2'-4"	2'-1"	1'-10"	1'-8"	1'-7"	1'-6"	1'-5"	1'-4"	
		OTHER	1'-11"	1'-7"	1'-5"	1'-4"	1'-3"	1'-2"	1'-1"	1'-0"	
	#4	TOP	3'-2"	2'-9"	2'-5"	2'-3"	2'-1"	1'-11"	1'-10"	1'-9"	
		OTHER	2'-5"	2'-1"	1'-11"	1'-9"	1'-7"	1'-6"	1'-5"	1'-4"	
	#5	TOP	3'-11"	3'-5"	3'-0"	2'-9"	2'-7"	2'-5"	2'-3"	2'-2"	
		OTHER	3'-0"	2'-7"	2'-4"	2'-2"	2'-0"	1'-10"	1'-9"	1'-8"	
	#6	TOP	4'-8"	4'-1"	3'-8"	3'-4"	3'-1"	2'-11"	2'-9"	2'-7"	
		OTHER	3'-7"	3'-1"	2'-10"	2'-7"	2'-4"	2'-3"	2'-1"	2'-0"	
	#7	TOP	6'-9"	5'-4"	5'-1"	4'-5"	4'-4"	4'-2"	3'-11"	3'-9"	
		OTHER	5'-3"	4'-6"	4'-1"	3'-9"	3'-5"	3'-3"	3'-0"	2'-11"	
	#8	TOP	7'-9"	6'-6"	6'-0"	5'-6"	5'-1"	4'-9"	4'-6"	4'-3"	
		OTHER	6'-0"	5'-2"	4'-8"	4'-3"	3'-11"	3'-8"	3'-6"	3'-3"	
#9	TOP	8'-9"	7'-6"	6'-9"	6'-2"	5'-9"	5'-4"	5'-1"	4'-10"		
	OTHER	6'-9"	5'-0"	5'-3"	4'-9"	4'-5"	4'-2"	3'-11"	3'-8"		
#10	TOP	9'-10"	8'-6"	7'-8"	7'-0"	6'-5"	6'-0"	5'-8"	5'-5"		
	OTHER	7'-9"	6'-5"	5'-10"	5'-4"	5'-0"	4'-5"	4'-2"	4'-0"		
#11	TOP	10'-11"	9'-6"	8'-6"	7'-9"	7'-2"	6'-8"	6'-4"	6'-0"		
	OTHER	8'-5"	7'-3"	6'-6"	5'-11"	5'-6"	5'-2"	4'-10"	4'-7"		

- NOTES:**
1. SPlice LENGTHS SHOWN ARE APPLICABLE FOR SPLICES OCCURRING UNDER THE FOLLOWING CONDITIONS:
 - a. NORMAL-WEIGHT CONCRETE
 - b. MINIMUM BAR SPACING REQUIREMENTS:
 - i. CLEAR SPACING BETWEEN BARS AT SPlice LOCATION > BAR DIAMETER, CLEAR COVER TO BARS > BAR DIAMETER, AND TIES OR STIRRUPS OCCUR PER CODE SPACING WITHIN LENGTH OF SPlice; OR
 - ii. CLEAR SPACING BETWEEN BARS AT SPlice $\geq 2 \times$ BAR DIAMETER AND CLEAR COVER > BAR DIAMETER
 2. INDICATED SPlice LENGTHS SHALL BE INCREASED BY THE LISTED FACTORS WHERE THE FOLLOWING CONDITIONS EXIST:

CONDITION	SPlice LENGTH/MULTIPLIER*
A. BAR SPACING OR CLEAR COVER IS LESS THAN REQUIRED PER NOTE 1	1.5
B. LIGHTWEIGHT CONCRETE	1.3
C. EPOXY COATED REINF WITH COVER < 3.0 db OR CLEAR SPACING < 1.5 DB	1.5
D. ALL OTHER EPOXY COATED BARS	1.2

*WHERE MULTIPLE CONDITIONS EXIST, APPLY EACH OF THE APPLICABLE FACTORS TO THE TABULATED TENSION LAP SPlice LENGTH TO OBTAIN REQUIRED SPlice LENGTH
 3. TOP BARS ARE HORIZONTAL BARS LOCATED WHERE MORE THAN 12" OF REQUIRED CONCRETE IS CAST IN THE MEMBER BELOW THE BARS.
 4. SMALLER BAR SIZE TO DETERMINE LENGTH WHERE SPLICING BARS OF DIFFERENT SIZES.

THICKNESS (t)

SAWCUT JOINT WITHIN 24 HOURS OF CONCRETE PLACEMENT

SLAB ON GRADE SEE PLAN

1/4"

1/4" MINIMUM

- NOTES:
1. REFERENCE SPECIFICATIONS FOR MATERIAL AND COMPACTION REQUIREMENTS.

Diagram illustrating the cross-section of a concrete slab with reinforcement details:

- THICKNESS (SEE PLAN)**: Indicated on the left side of the slab.
- UNBONDED ON THIS SIDE**: Indicated on the left side of the slab.
- SMOOTH DWELL BAR OR DIAMOND SHAPE LOAD PLATES - SEE SCHEDULE**: Indicated on the right side of the slab.
- FIRST POUR**: The initial concrete pour.
- SUBSEQUENT POUR**: The concrete pour following the first pour.

SLAB THICKNESS	SMOOTH DWELL BAR	DIAMOND SHAPE LOAD PLATES
4"	1/2"Ø x 1'-4" @ 18" OC	1/4" x 4 1/2" x 0'-4 1/2" @ 18" OC
5"	3/4"Ø x 1'-4" @ 18" OC	

2'-6"

2'-0"

CORNER BARS TO MATCH LONGITUDINAL FOOTING REINFORCING

LONGITUDINAL FOOTING REINFORCING - SEE PLAN

The diagram illustrates the reinforcing details for single curtain walling, divided into two main sections: CORNER and INTERSECTION.

CORNER: This detail shows a 90-degree corner. Key components include:

- (1) #5 VERT UNO:** Vertical unadorned reinforcement bars.
- (2) #5 VERT UNO:** Vertical unadorned reinforcement bars.
- ALTERNATE HOOKS:** Hooks used for the reinforcement bars.
- LAP SPICE PER GENERAL NOTES, TYP:** Indicated for the lap joints.
- SEE PLAN AND SECTIONS FOR DIMENSION, TYP:** Reference to other drawings for dimensions.

INTERSECTION: This detail shows the intersection of two wall sections. Key components include:

- (1) #5 VERT UNO:** Vertical unadorned reinforcement bars.
- (2) #5 VERT UNO:** Vertical unadorned reinforcement bars.
- 0" TYP:** A dimension indicating a typical offset or gap.
- LAP SPICE PER GENERAL NOTES, TYP:** Indicated for the lap joints.

DOUBLE CURTAIN REINFORCING: This section shows a detail for double curtain walling, featuring:

- (4) #5 VERT UNO:** Vertical unadorned reinforcement bars.
- HORIZONTAL REINFORCING, TYP:** Typical horizontal reinforcement bars.
- (2) #5 VERT UNO:** Vertical unadorned reinforcement bars.
- LAP SPICE PER GENERAL NOTES, TYP:** Indicated for the lap joints.

T.O. SLAB
EL. (SEE PLAN)

4"

8" TYP

OPTION A-
THREADED
WITH NUT
(PREFERRE
D)

OPTION B-
HEADED

AS6 PLATE
1/4x3x3
OPTION C-
PLATED

#4x CONTINUOUS @ 2'-0" OC
EXTENDING THRU RAMP AT CENTER

#4 DOWEL @ 1'-6" OC

T.O. SLAB

SEE PLAN

Diagram illustrating the reinforcement detail for a slab opening, showing the layout of reinforcement bars and the location of the opening.

Labels and Callouts:

- T.O. SLAB EL (SEE PLAN)
- SEE SCHEDULE
- TRIM BARS AT ALL SLAB PENETRATIONS/OPENINGS THAT EXCEED 1'-0"x1'-0"
- AT EACH CORNER, (1)-TOP, (1)-BOTTOM, (2)-#5x4'-0" DIAGONALLY
- ADD 1/2 NUMBER OF BARS INTERRUPTED BY OPENING *1 (@ 3")
- BARS INTERRUPTED
- (2) - #5xCONT ON EACH SIDE
- (1) - TOP
- (1) - BOTTOM

SEE ARCH AND MEP DRAWINGS FOR EXACT LOCATIONS, EXTENTS AND DIMENSIONS OF DEPRESSIONS

2 x 'A'

'A'

1'-0"

#3 x 3'-6" @ 18", TYP

#4 @ 6"

#3 @ 18"

(2) #4

3'-0" MAX PERMITTED

6"

DEPRESSION < SLAB THICKNESS

DEPRESSION > SLAB THICKNESS

Diagram illustrating the cross-section of a trench drain assembly. Key components and dimensions shown include:

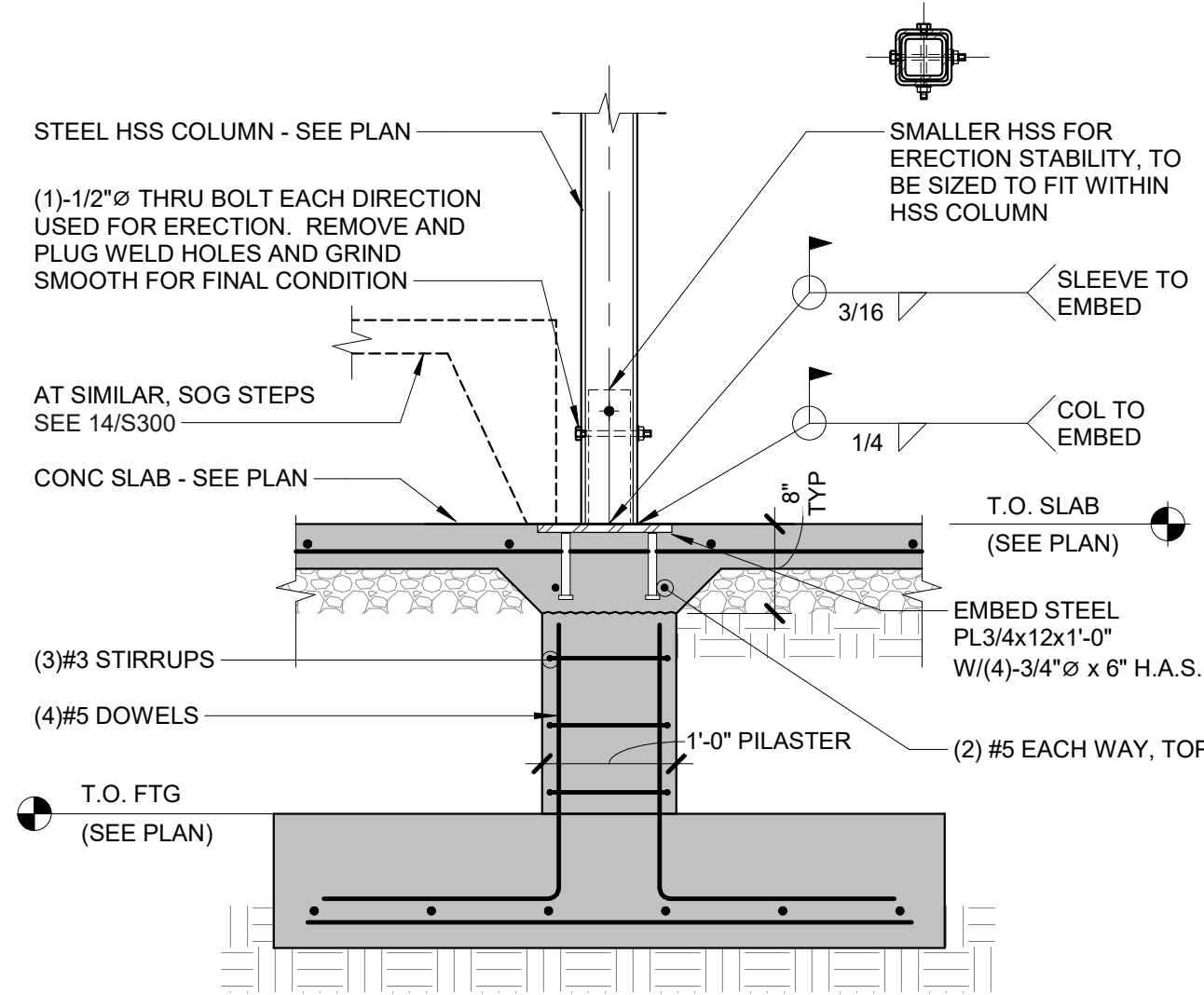
- PREFABRICATED TRENCH DRAIN (SEE MEP & ARCH DWGS FOR ADDL INFORMATION)
- 3" CLR
- #4x4-0" @ 12"
- #4 @ 12"
- (5) #4
- 8"
- 1/2"
- 1/2"
- 1
- to

RAMP SLAB
SEE 12/S300

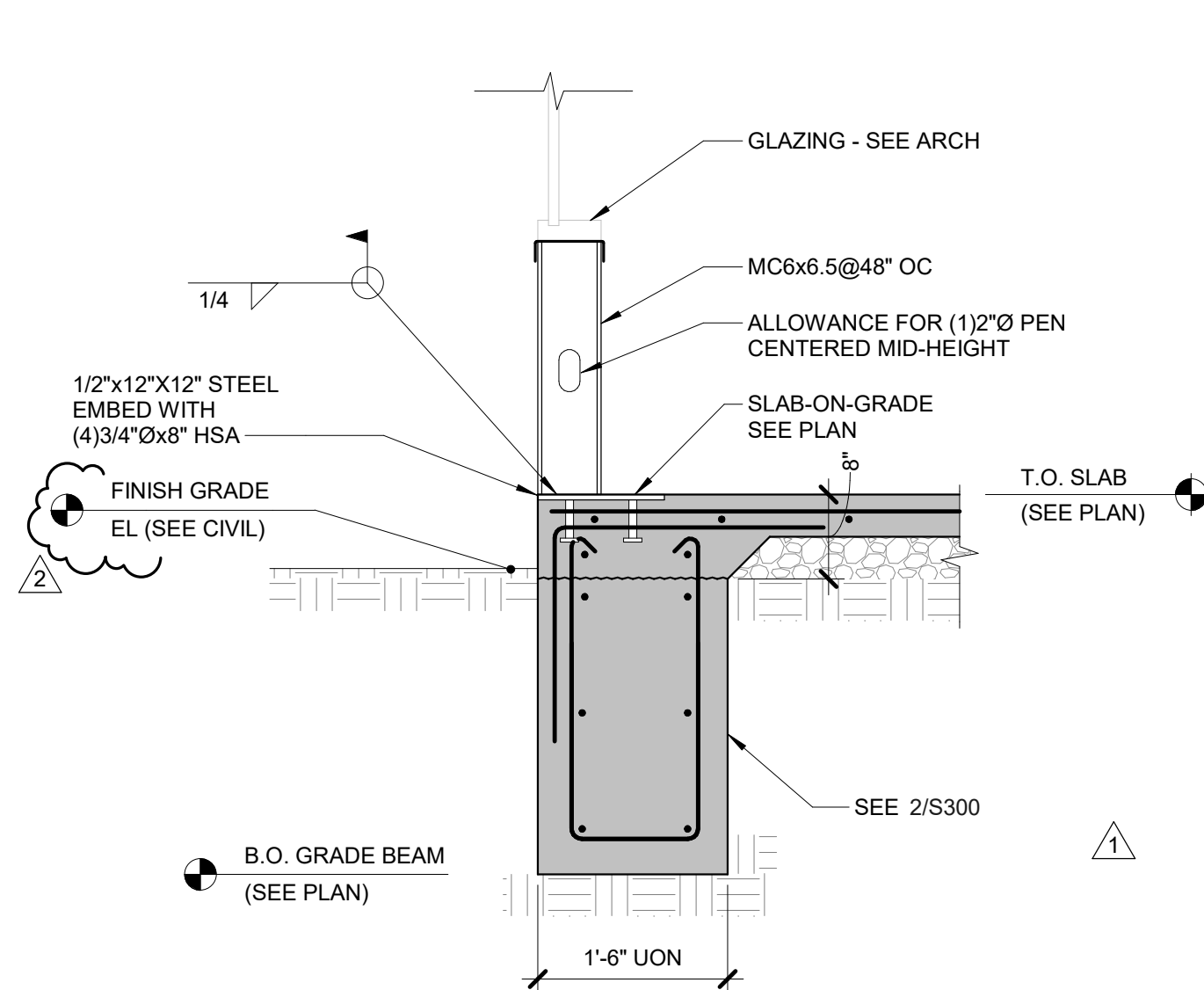
MECH TRENCH
SEE 16/S300

SEE PLAN

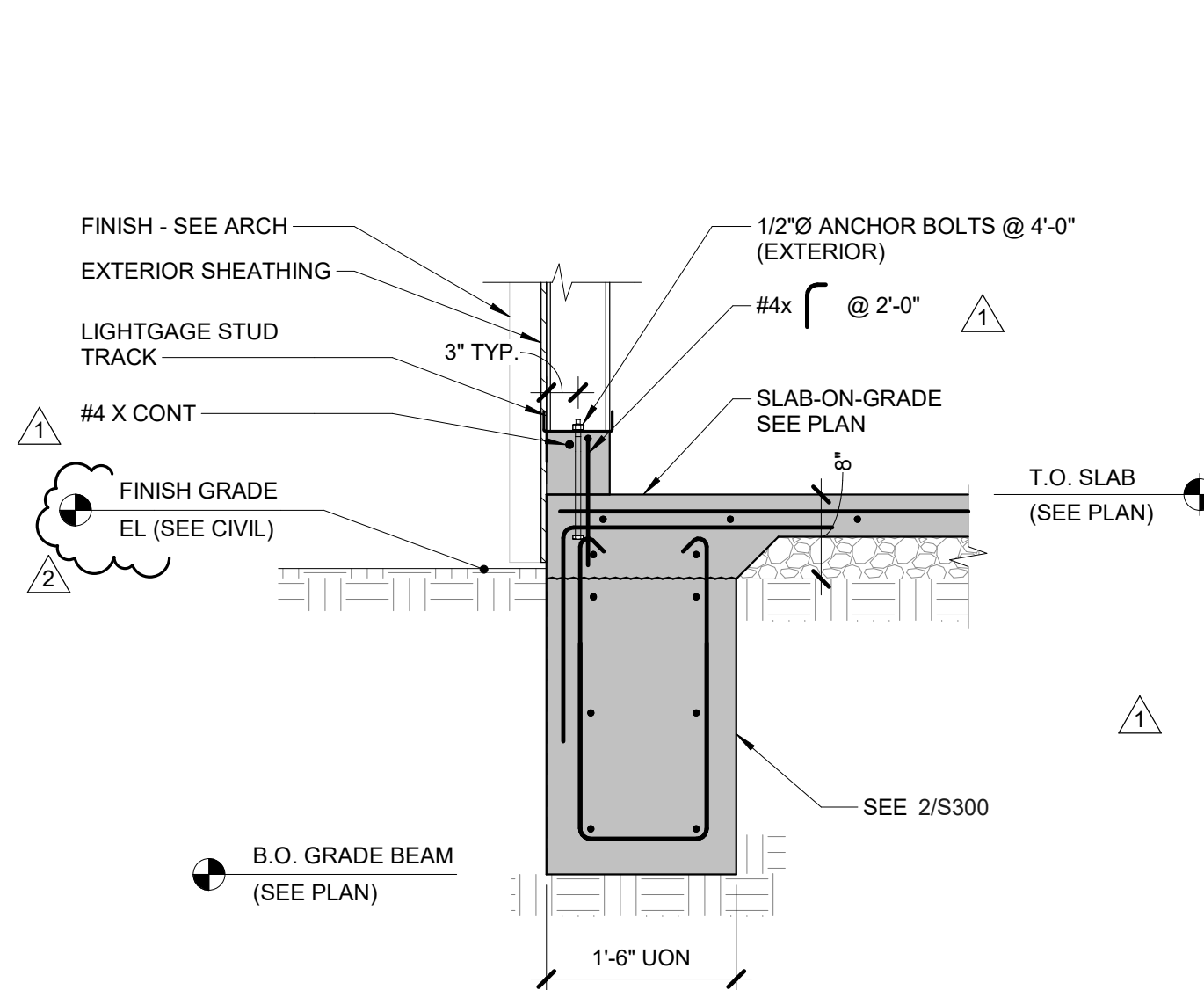
8" 8"



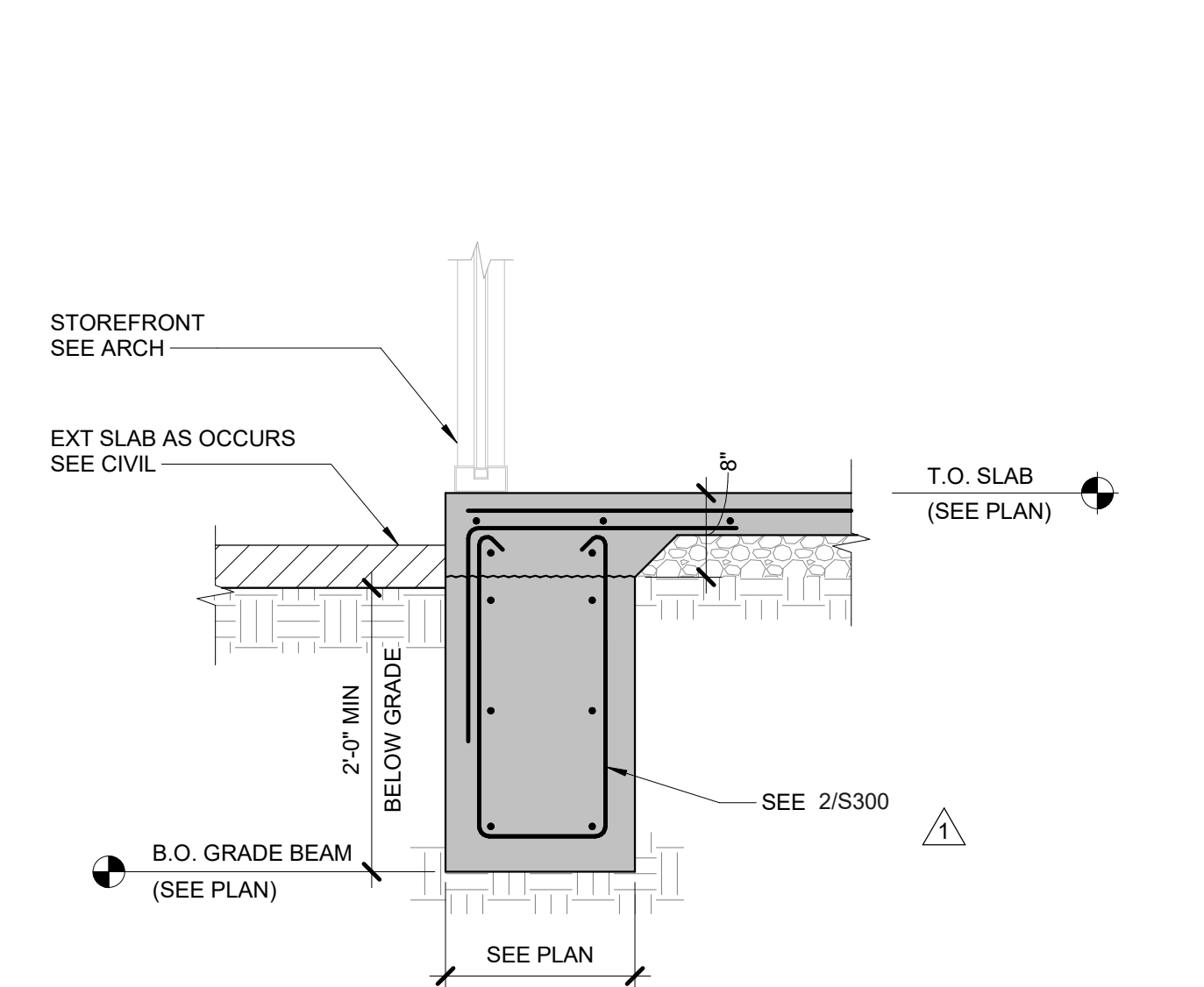
1 TYPICAL STEEL COLUMN TO
CONCRETE SLAB DETAIL
3/4" = 1'-0"



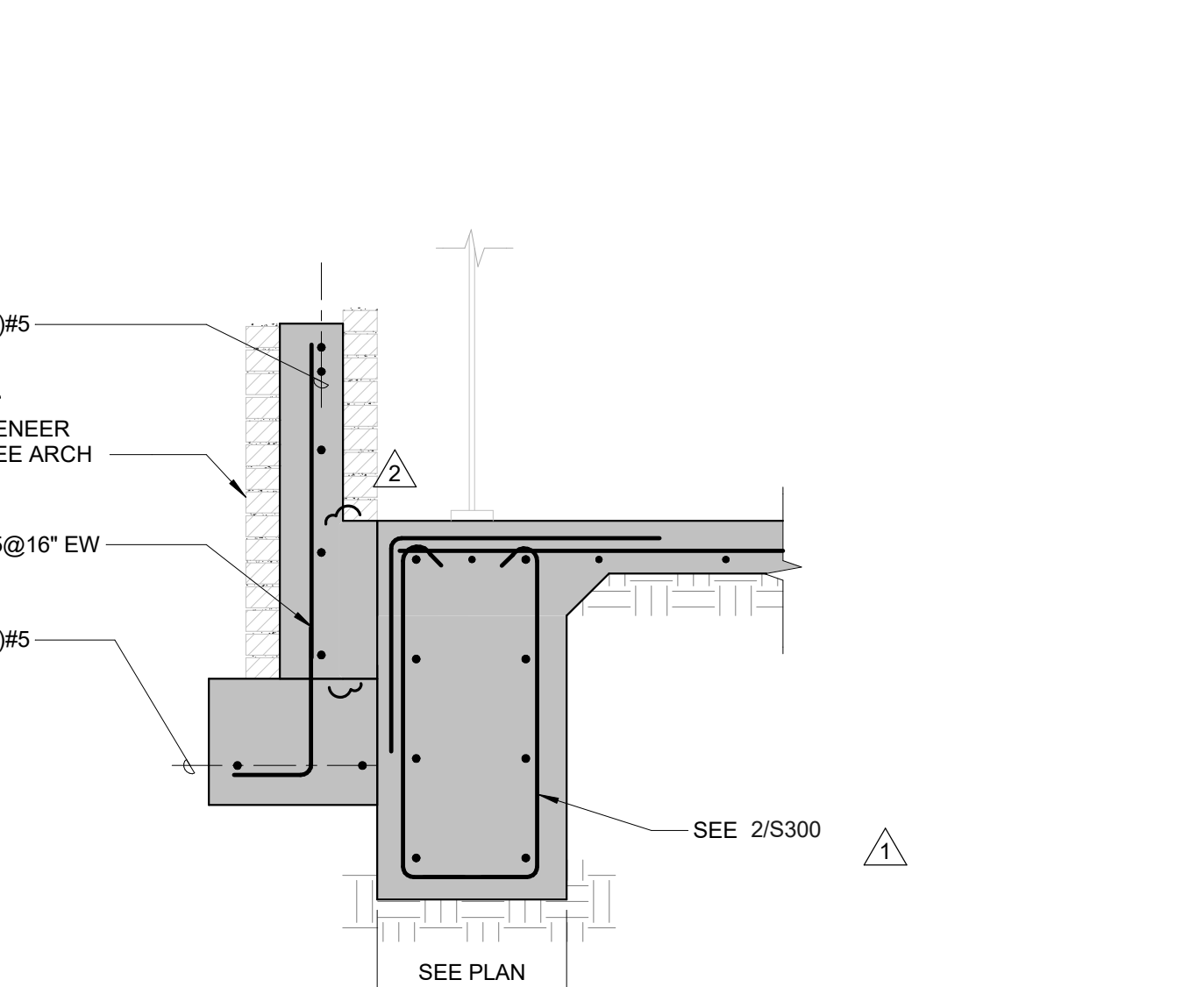
2 FOUNDATION AT EXTERIOR WALL
BELOW OPENINGS
3/4" = 1'-0"



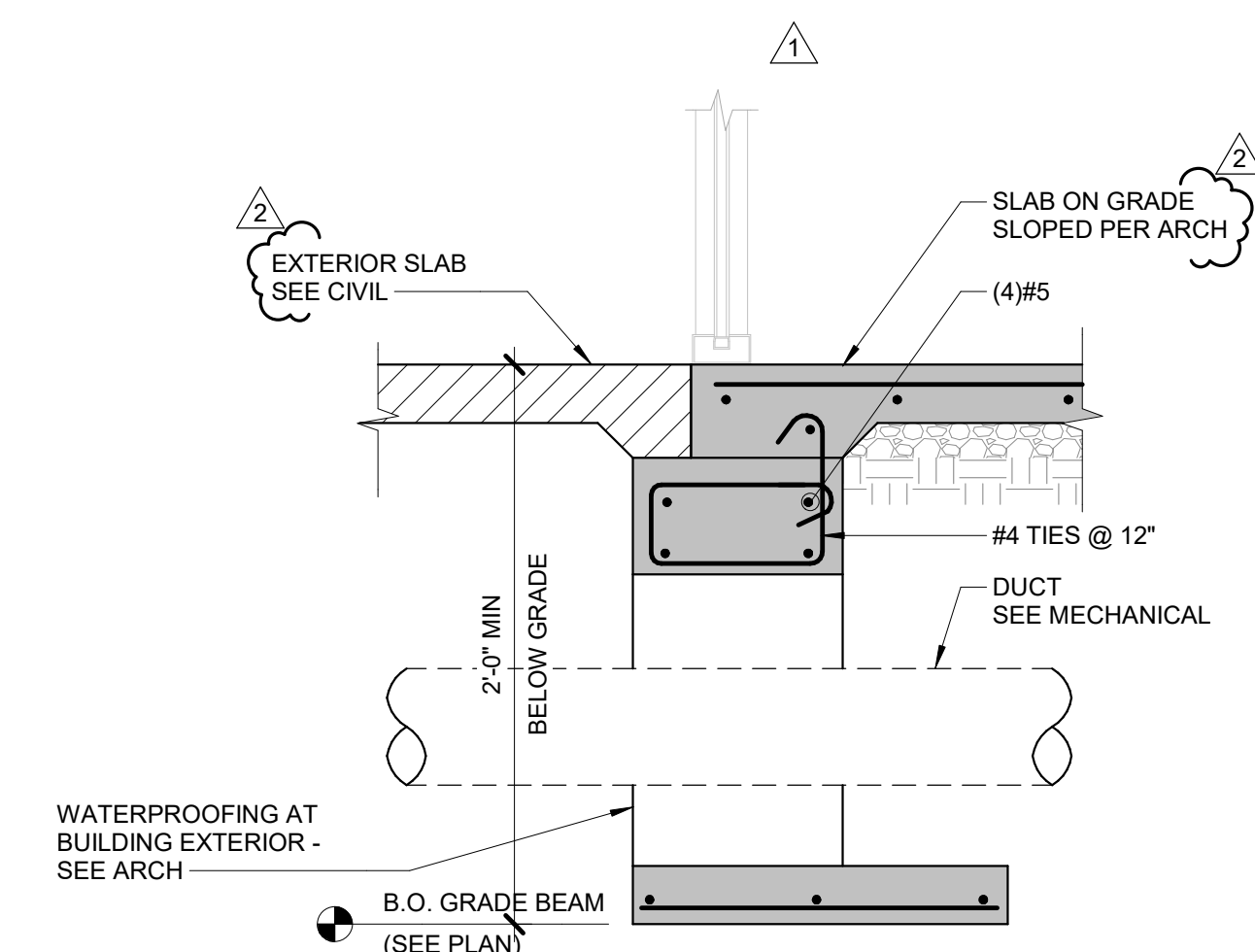
3 FOUNDATION AT EXTERIOR WALL
WITH STUCCO
3/4" = 1'-0"



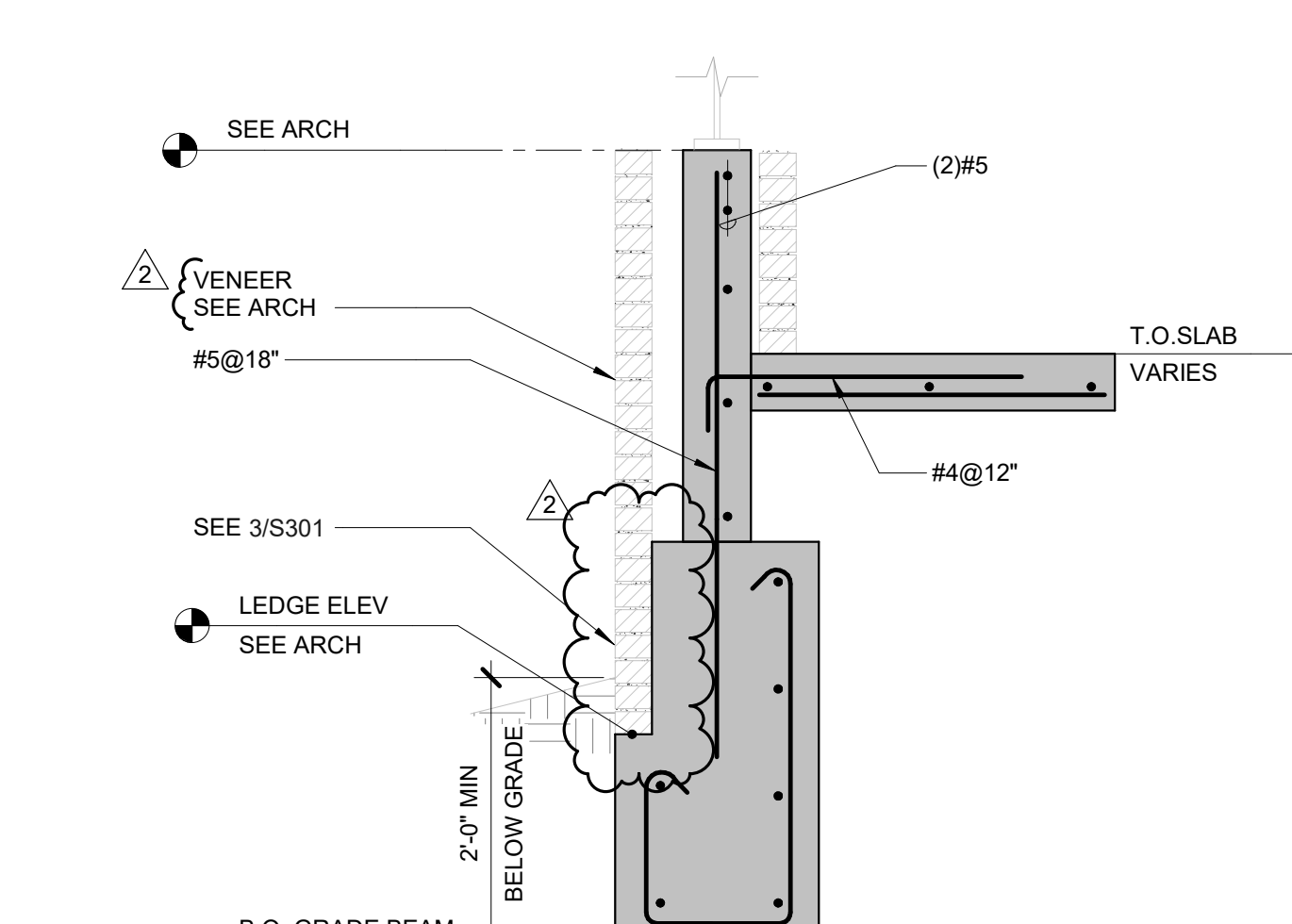
4 FOUNDATION WALL, SOG,
STOREFRONT
3/4" = 1'-0"



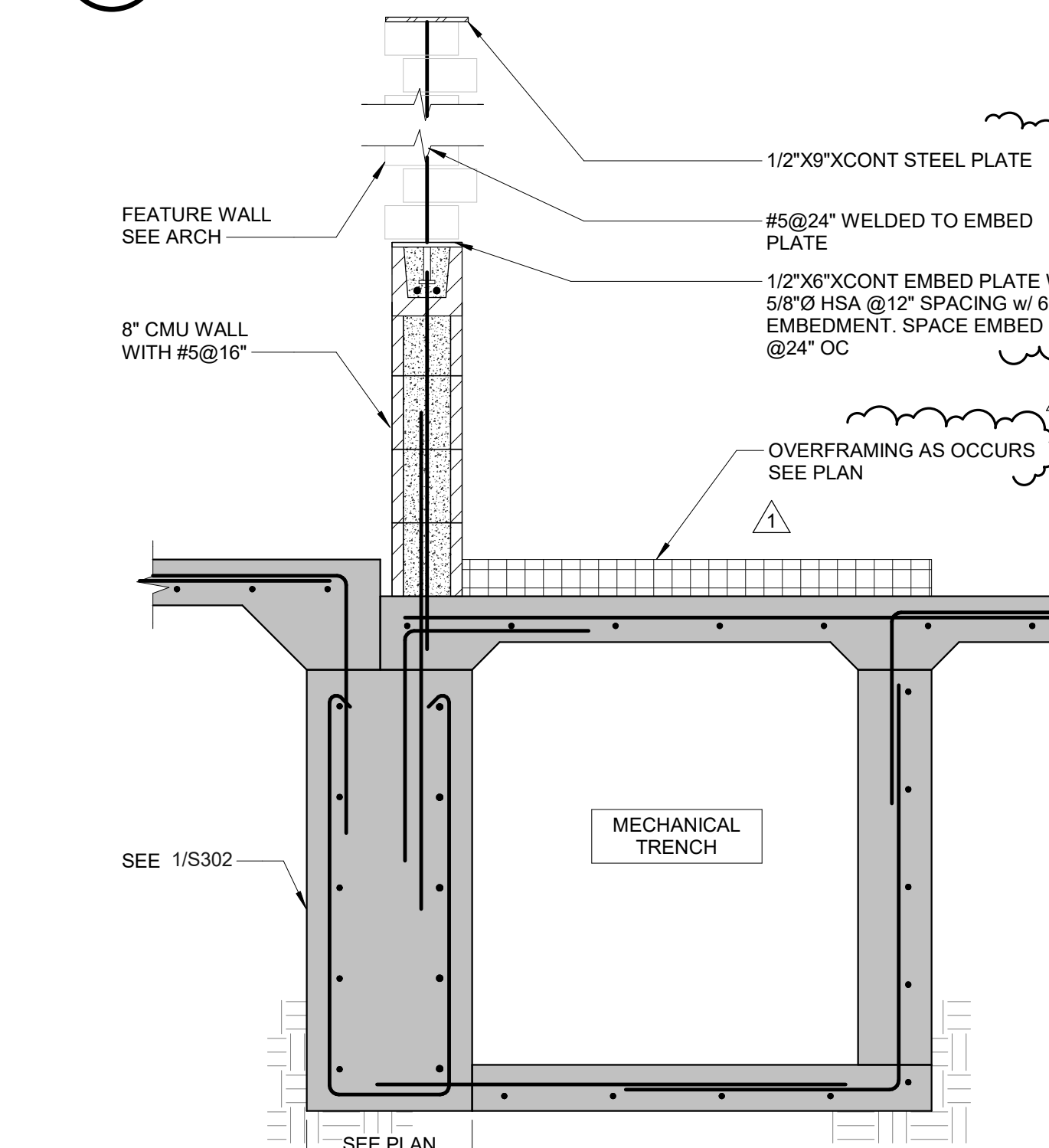
5 FOUNDATION AT SITE WALL AT
STOREFRONT
3/4" = 1'-0"



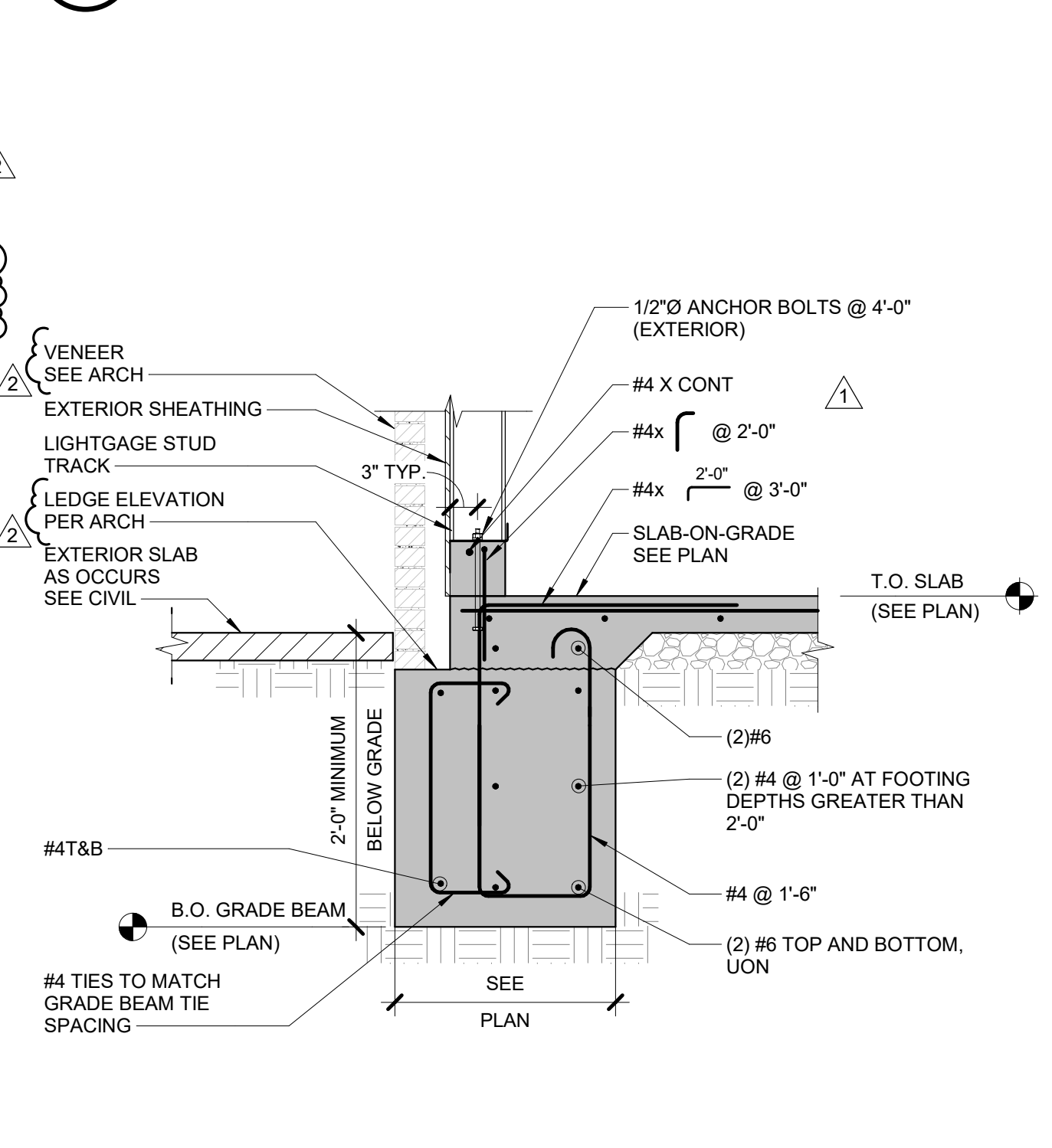
6 FOUNDATION AT MECH TRENCH
EXIT
3/4" = 1'-0"



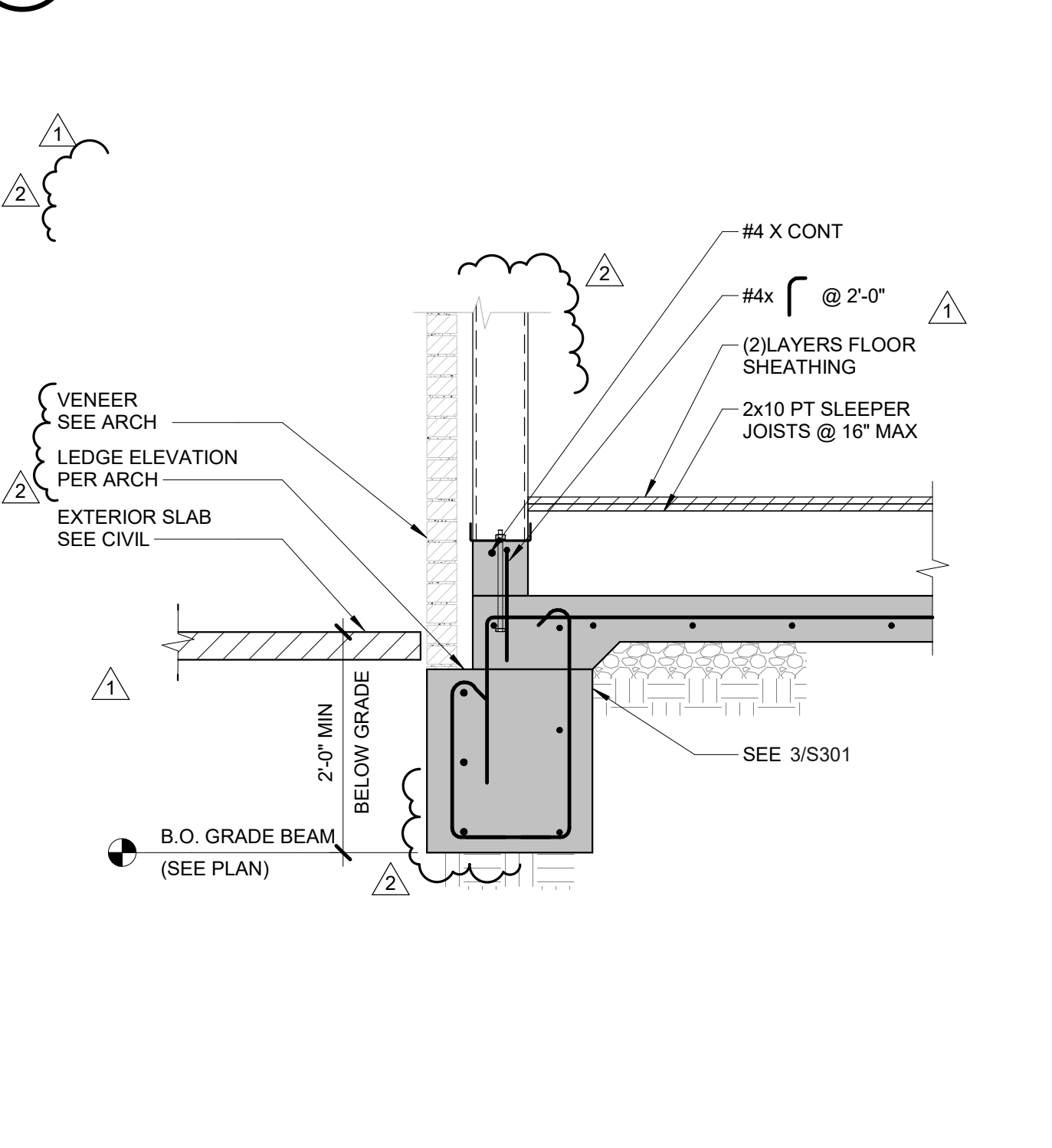
7 FOUNDATION AT RAISED DINING
3/4" = 1'-0"



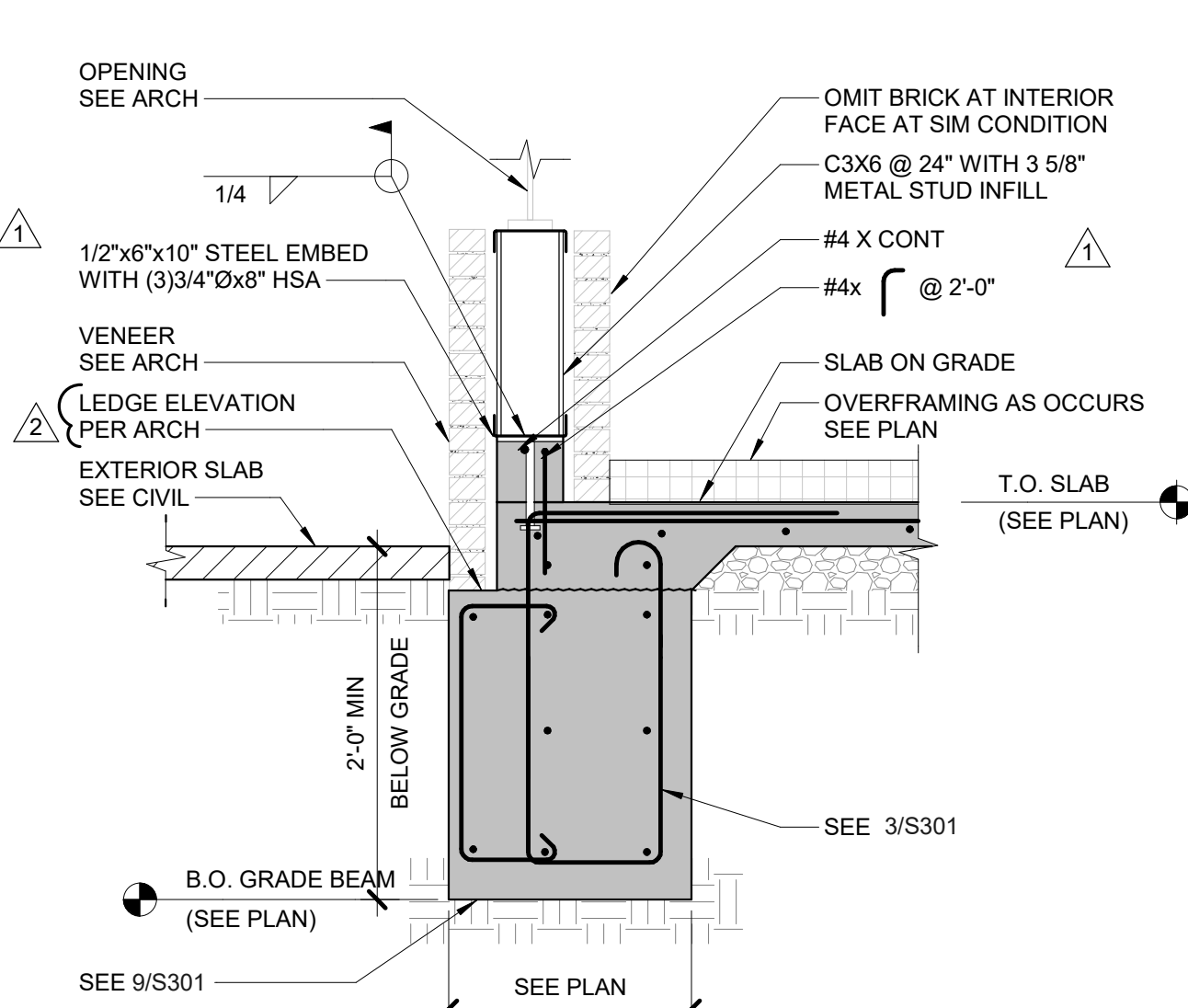
8 FOUNDATION AT FEATURE WALL
AT MECH TRENCH
3/4" = 1'-0"



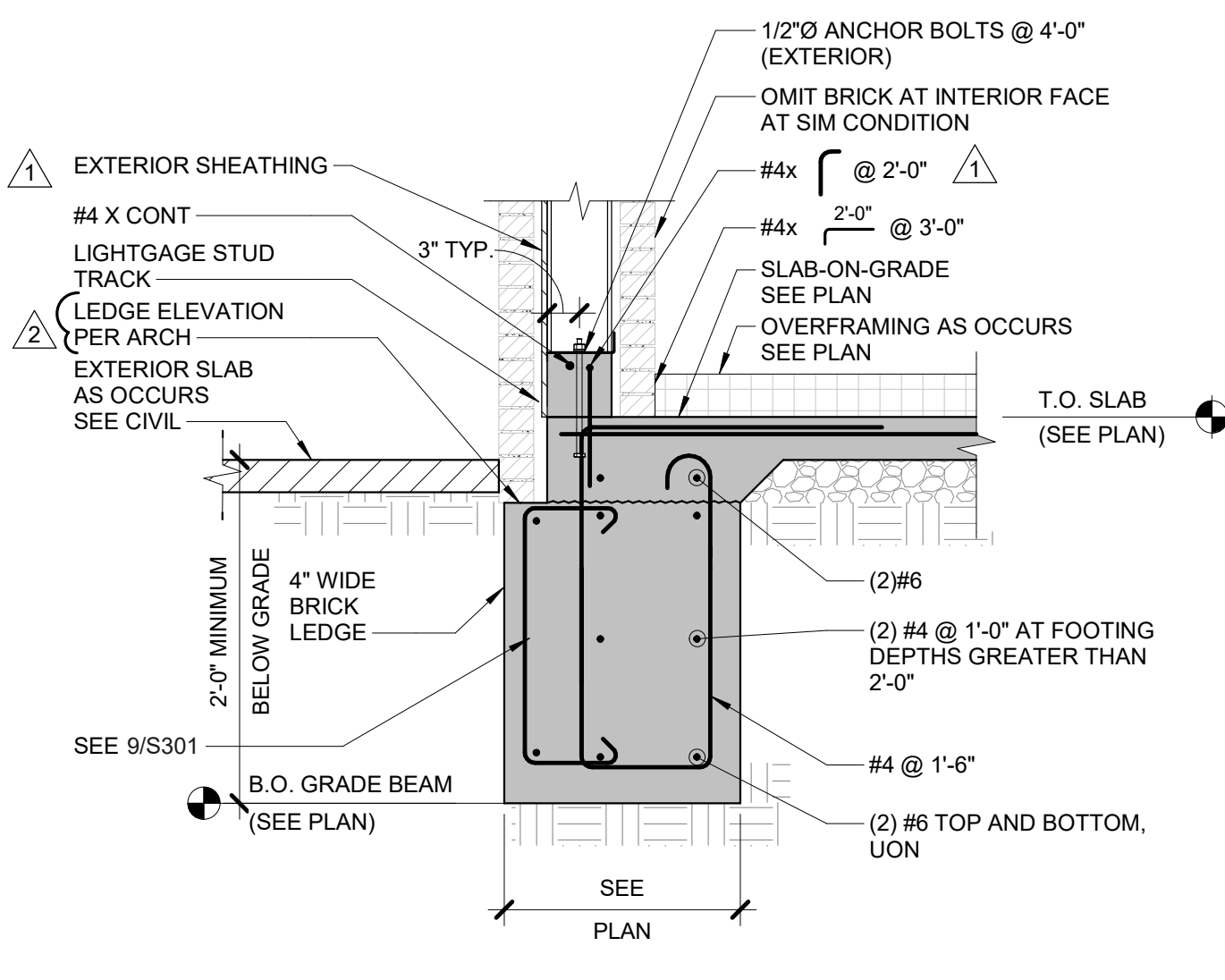
9 FOUNDATION AT EXTERIOR WALL
WITH BRICK LEDGE
3/4" = 1'-0"



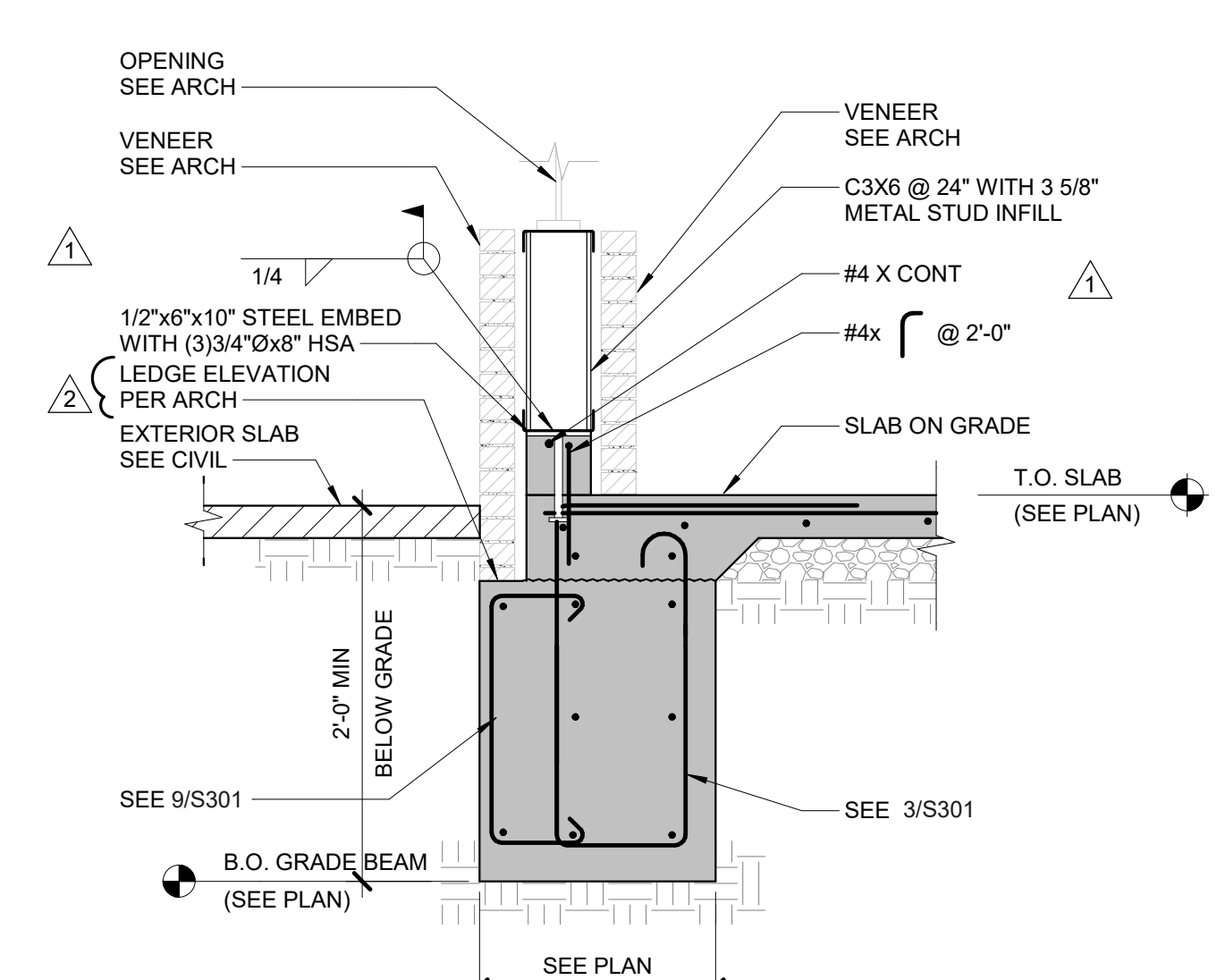
10 FOUNDATION AT EXT w/
PLATFORM
3/4" = 1'-0"



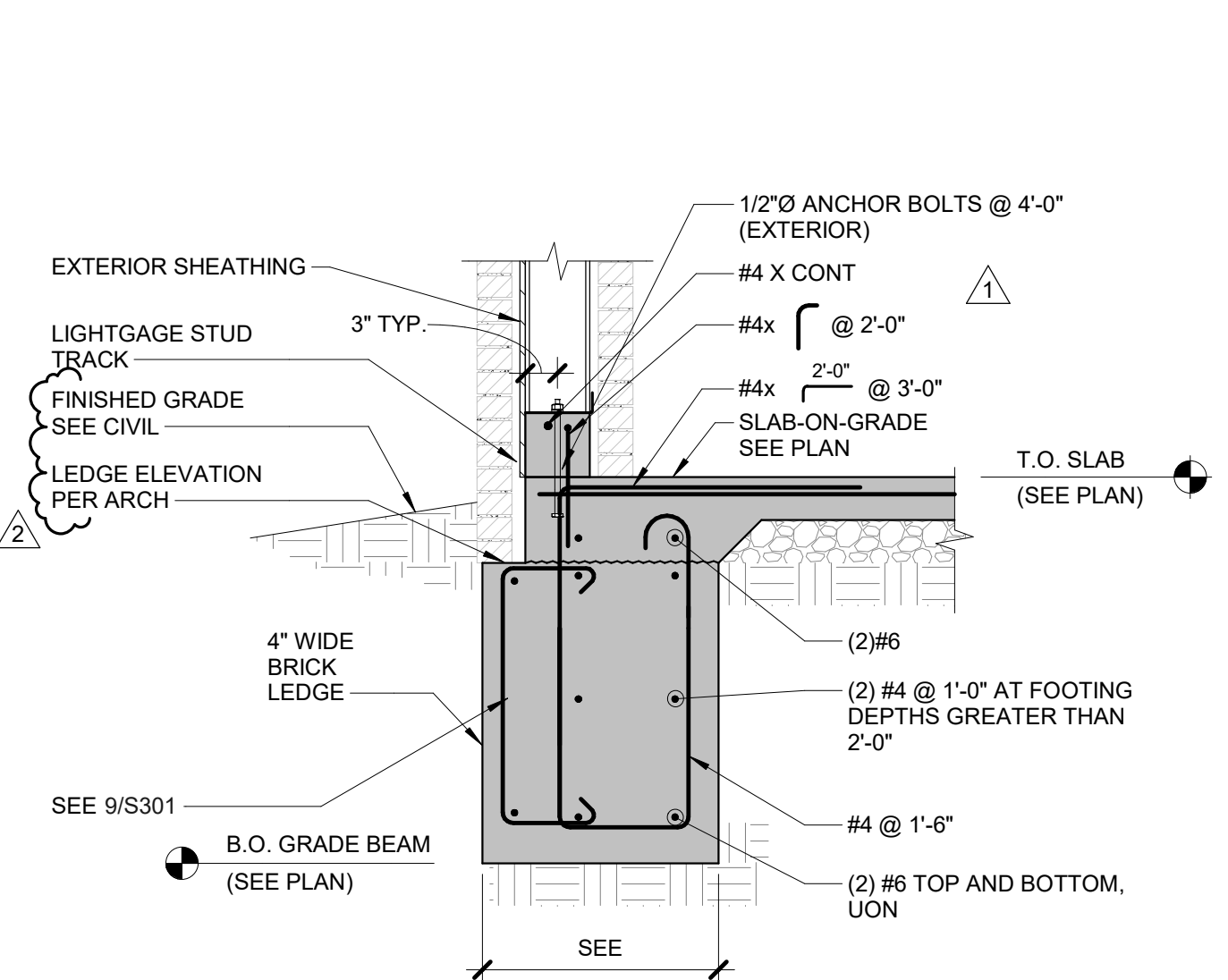
11 FOUNDATION AT EXTERIOR
3/4" = 1'-0"



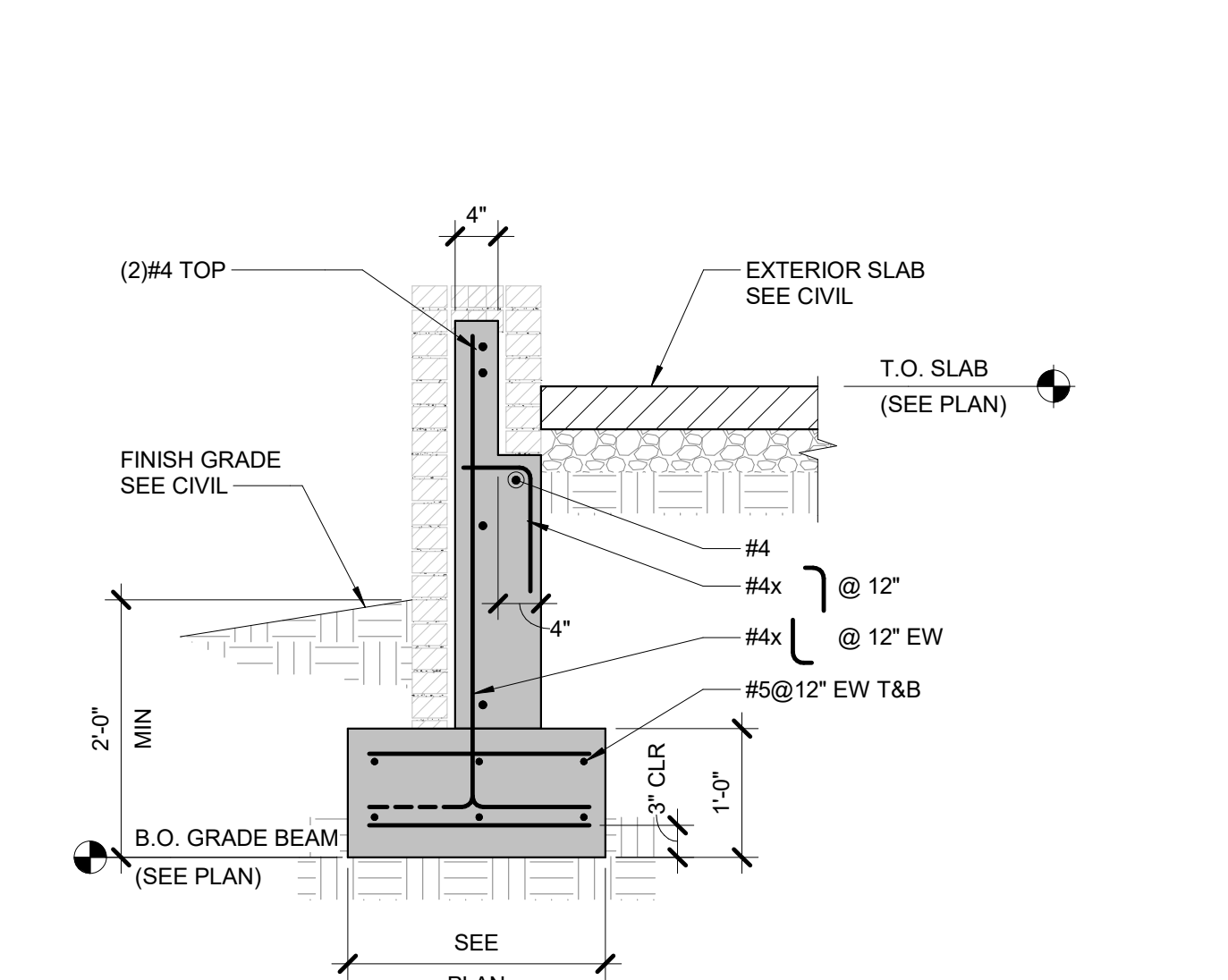
12 FOUNDATION AT EXTERIOR WALL
WITH BRICK
3/4" = 1'-0"



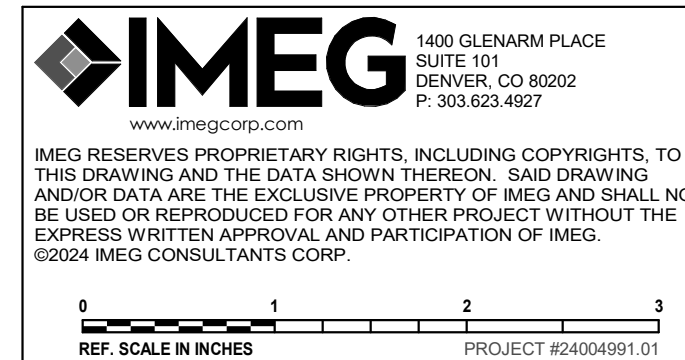
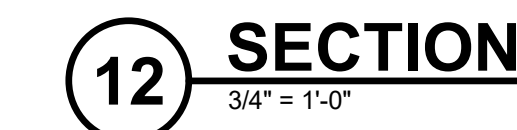
13 FOUNDATION AT EXTERIOR
3/4" = 1'-0"



14 FOUNDATION AT EXTERIOR WALL
WITH BRICK
3/4" = 1'-0"

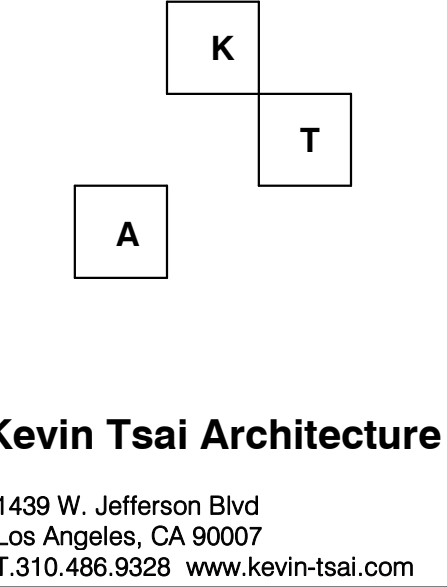
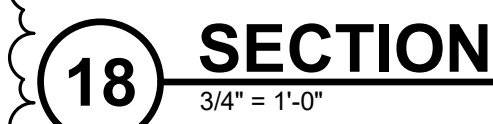
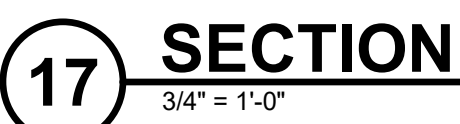
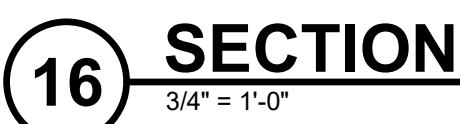
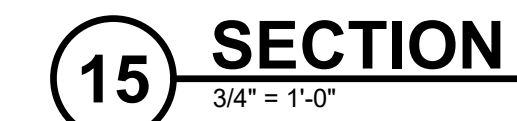
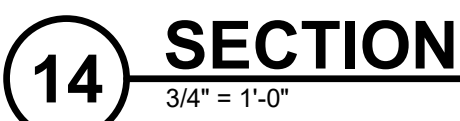
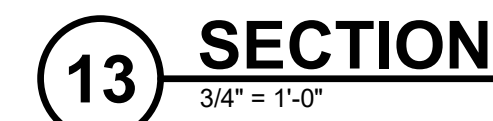
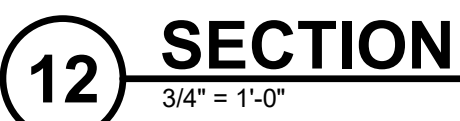
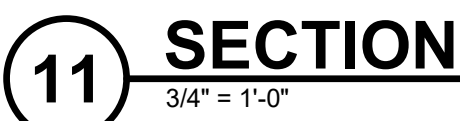
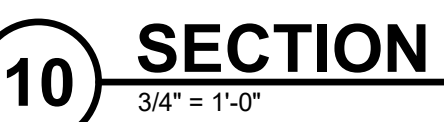
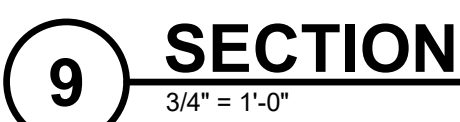
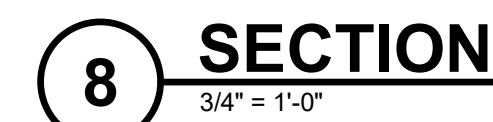
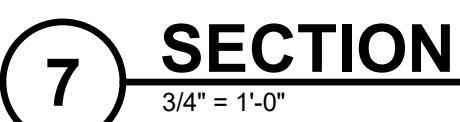


15 FOUNDATION AT EXTERIOR WALL
WITH BRICK
3/4" = 1'-0"

PROJECT TITLE



6 SECTION
3/4" = 1'-0"



architect's stamp

consultant

KEY PLAN:

designs, ideas, arrangements and plans indicated by these drawings and specifications are the property and copyright of the ARCHITECT and shall neither be used on any other work nor be used by any other person for any use whatsoever without written permission. Written dimensions shall take precedence over scaled dimensions and shall be verified at the job site. Any dimensional discrepancy shall be brought to the attention of the ARCHITECT prior to the commencement of work.

[illegible]

PROJECT TITLE:

01 EAST CAMELBACK ROAD
PHOENIX, ARIZONA 85016

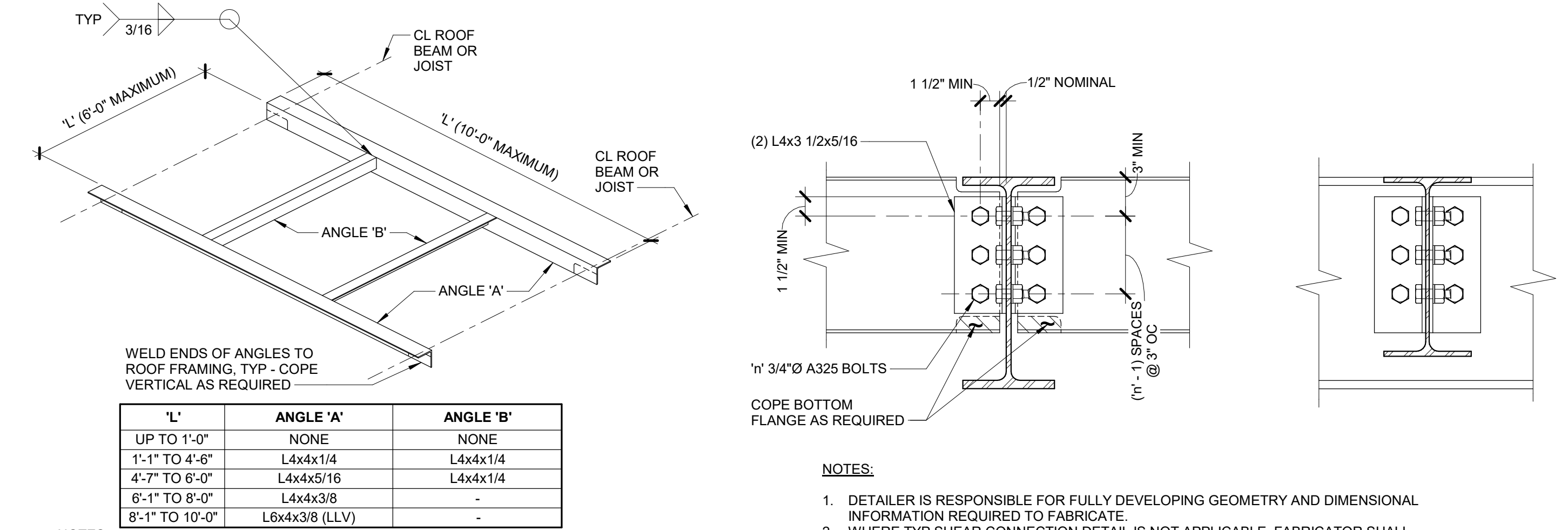
Q/PM:	LG
DRAWN BY:	STAFF
JOB NO.:	24004991.01

DRAWING TITLE:
STEEL SECTIONS

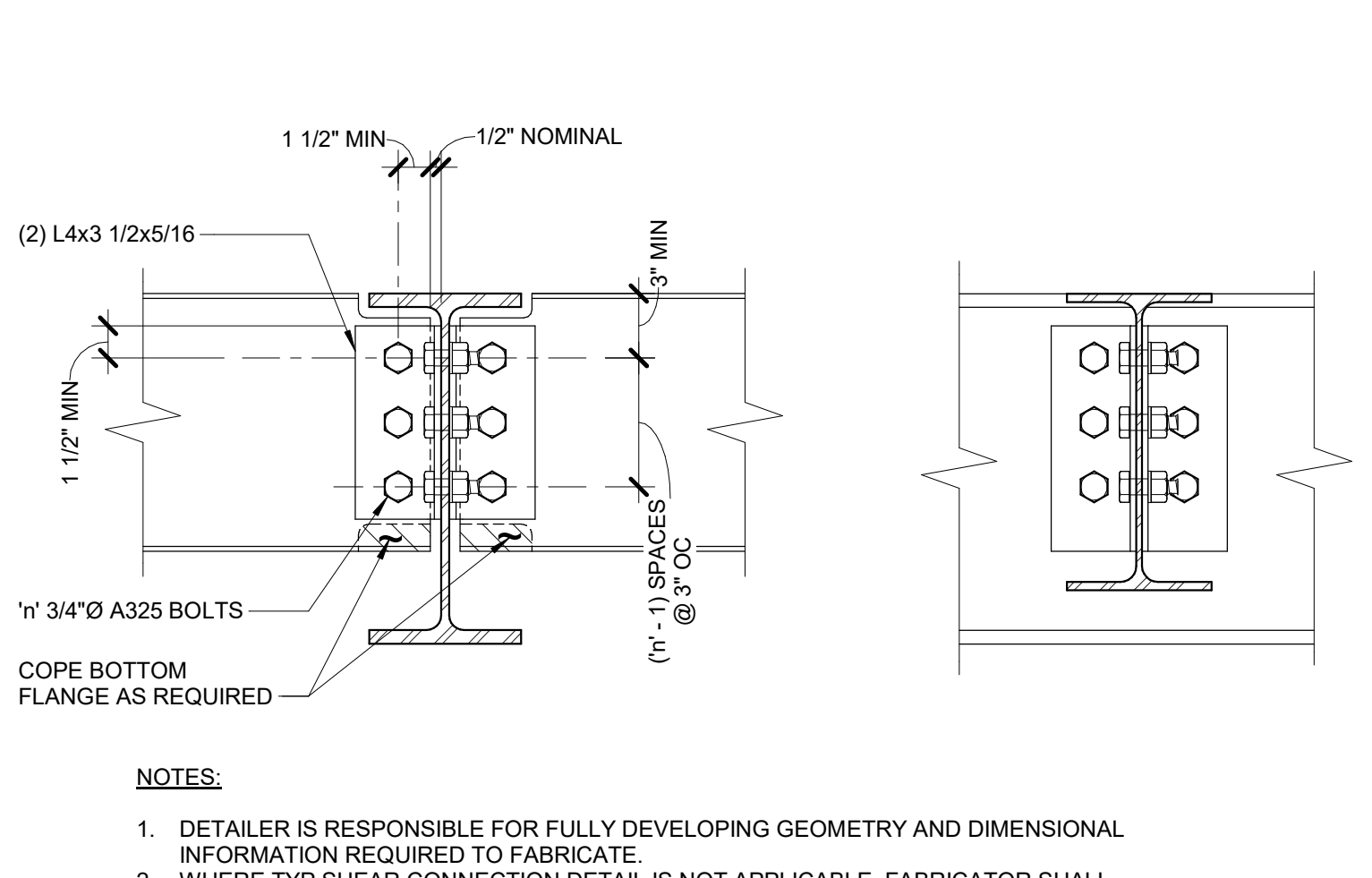
SHEET NO.:

S503

11/21/2026 4:54:53 PM

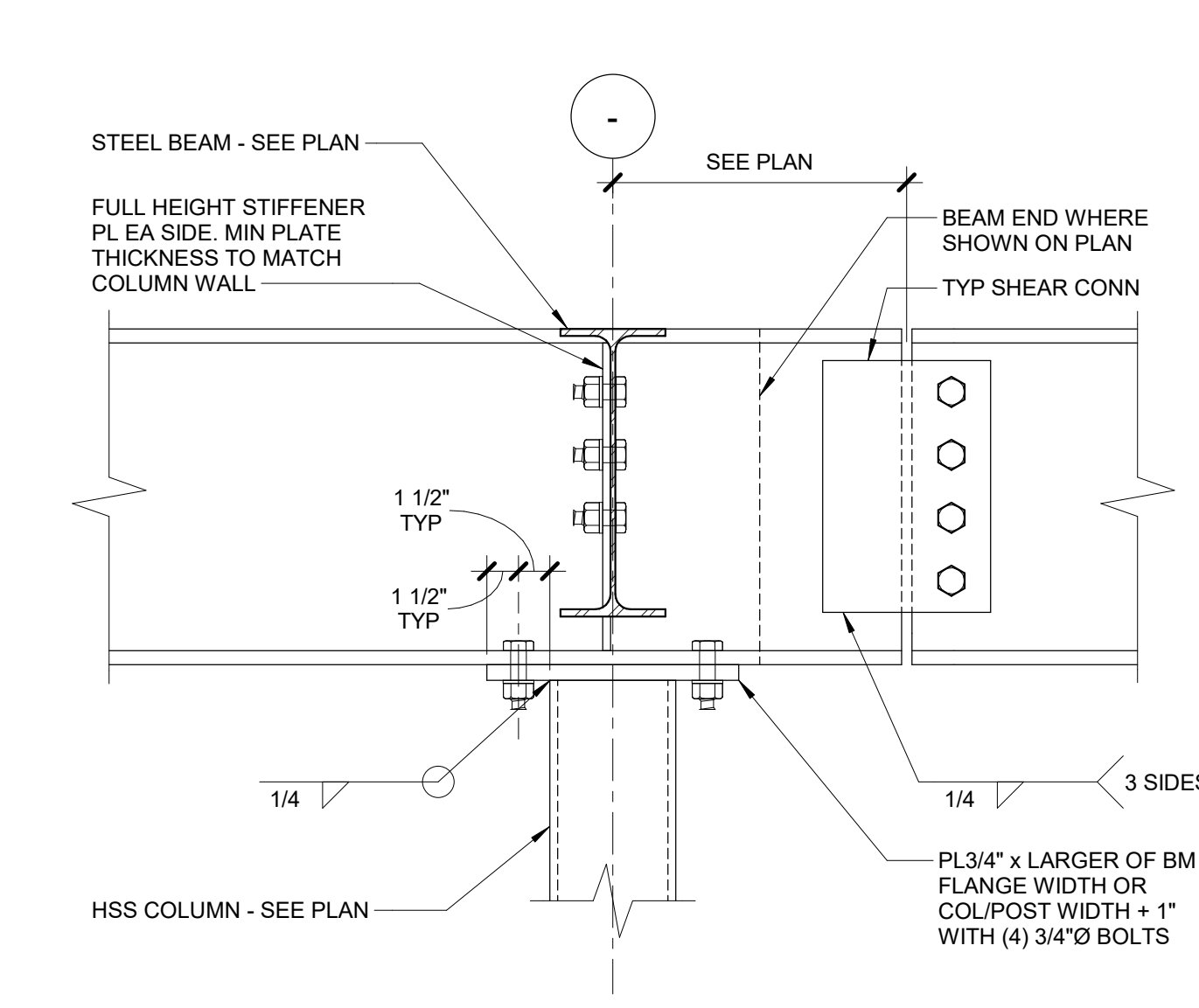


1 ROOF OPENING FRAMING
3/4" = 1'-0"

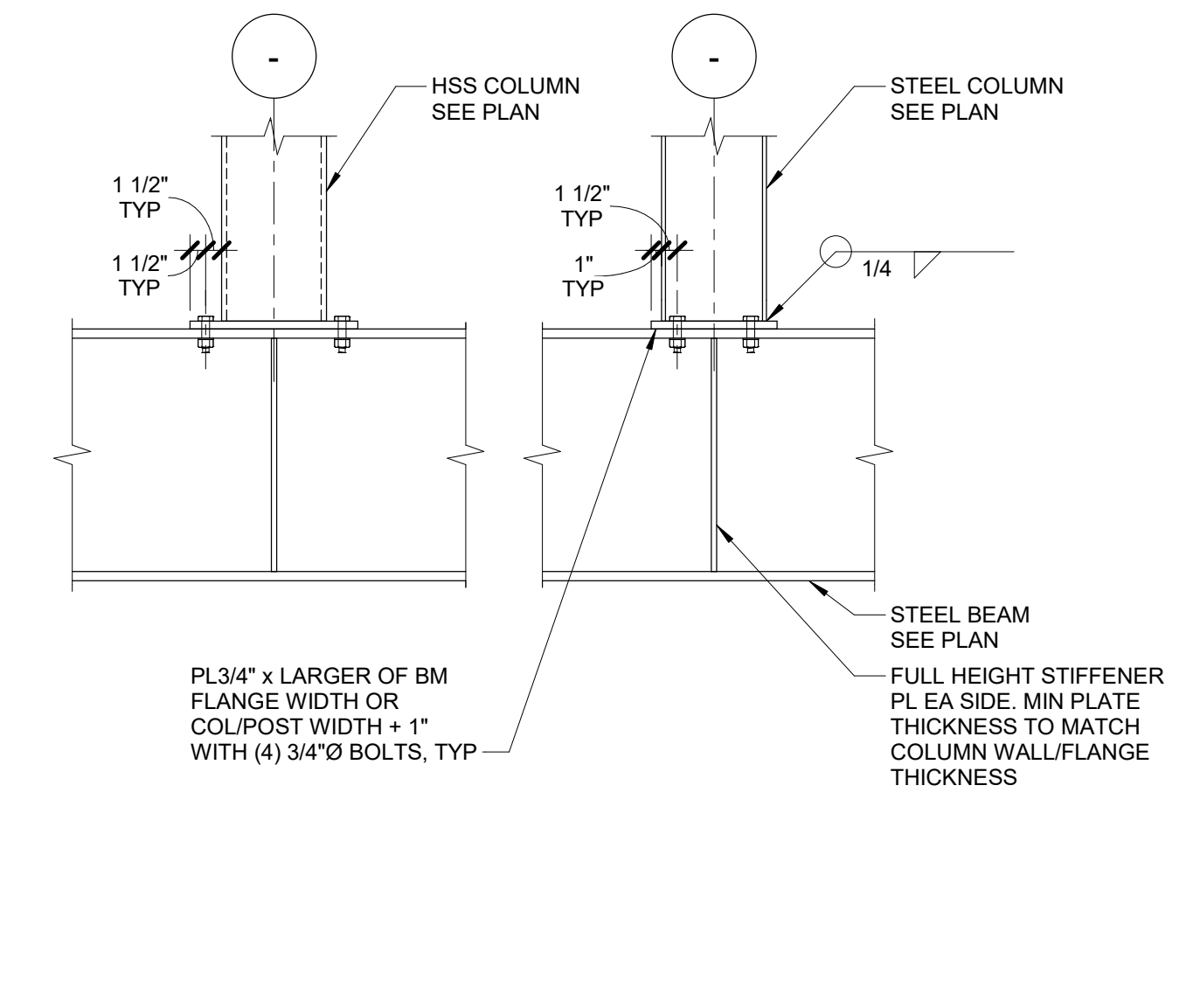


2 DOUBLE ANGLE SHEAR CONNECTION
1 1/2" = 1'-0"

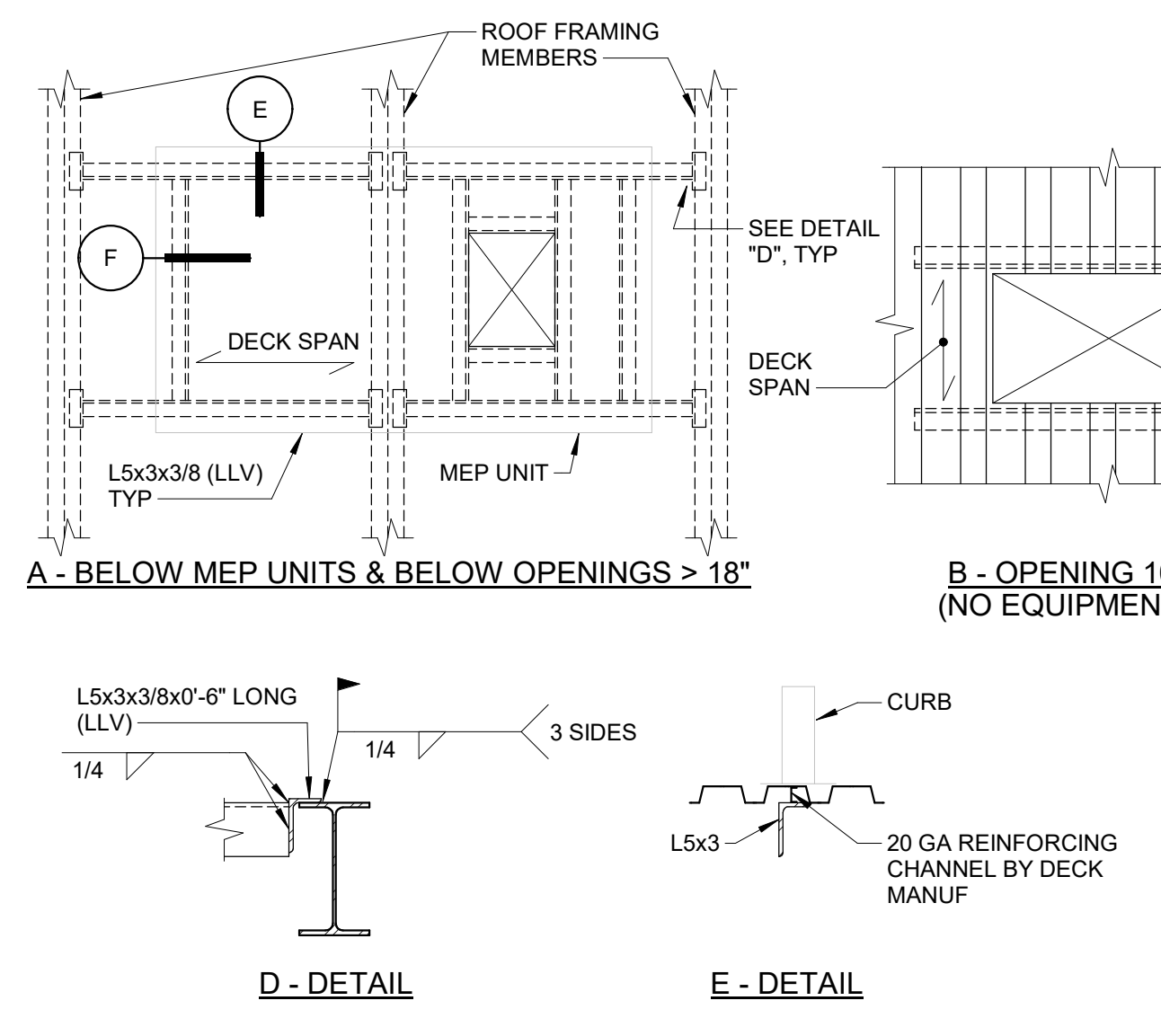
SUPPORTED BEAM	"I"
W8, W10	2
W12, W14	3
W16	4
W18	5
W21	6
W24, W27	7
W30	8
W33	9
W36	10
W40	11
W44	12



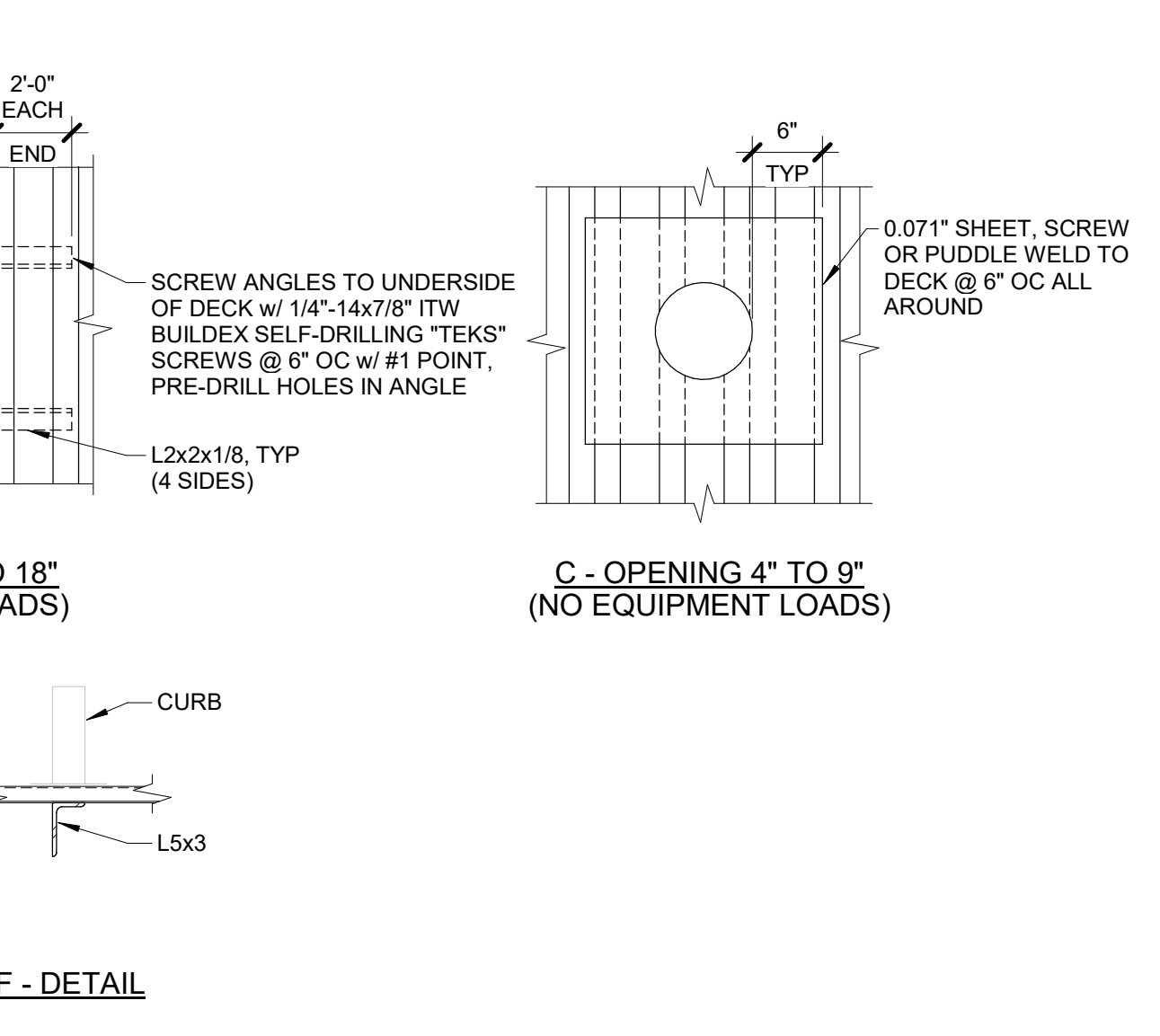
3 BEAM BEARING ON HSS COLUMN
1 1/2" = 1'-0"



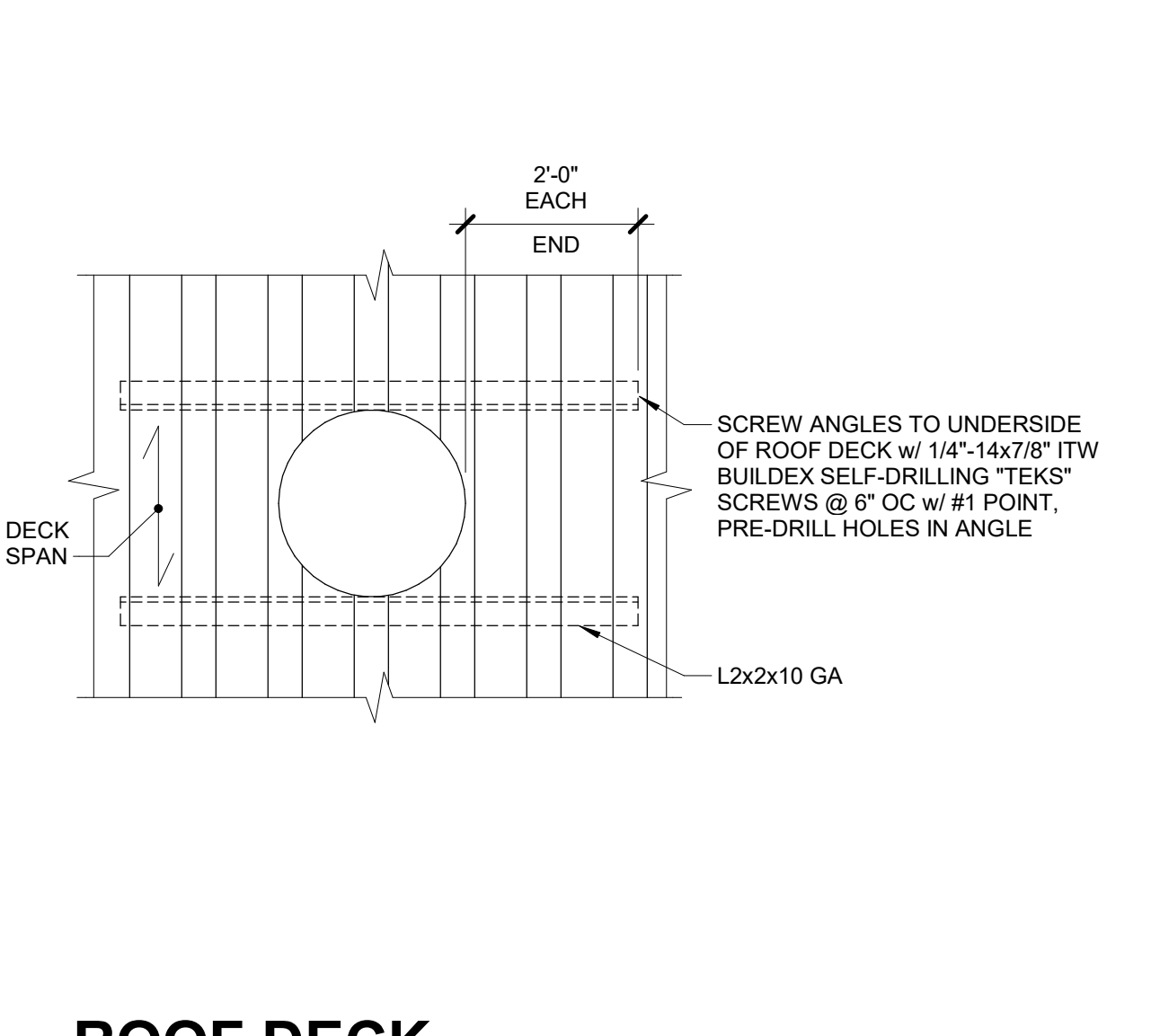
4 COLUMN BEARING ON BEAM
3/4" = 1'-0"



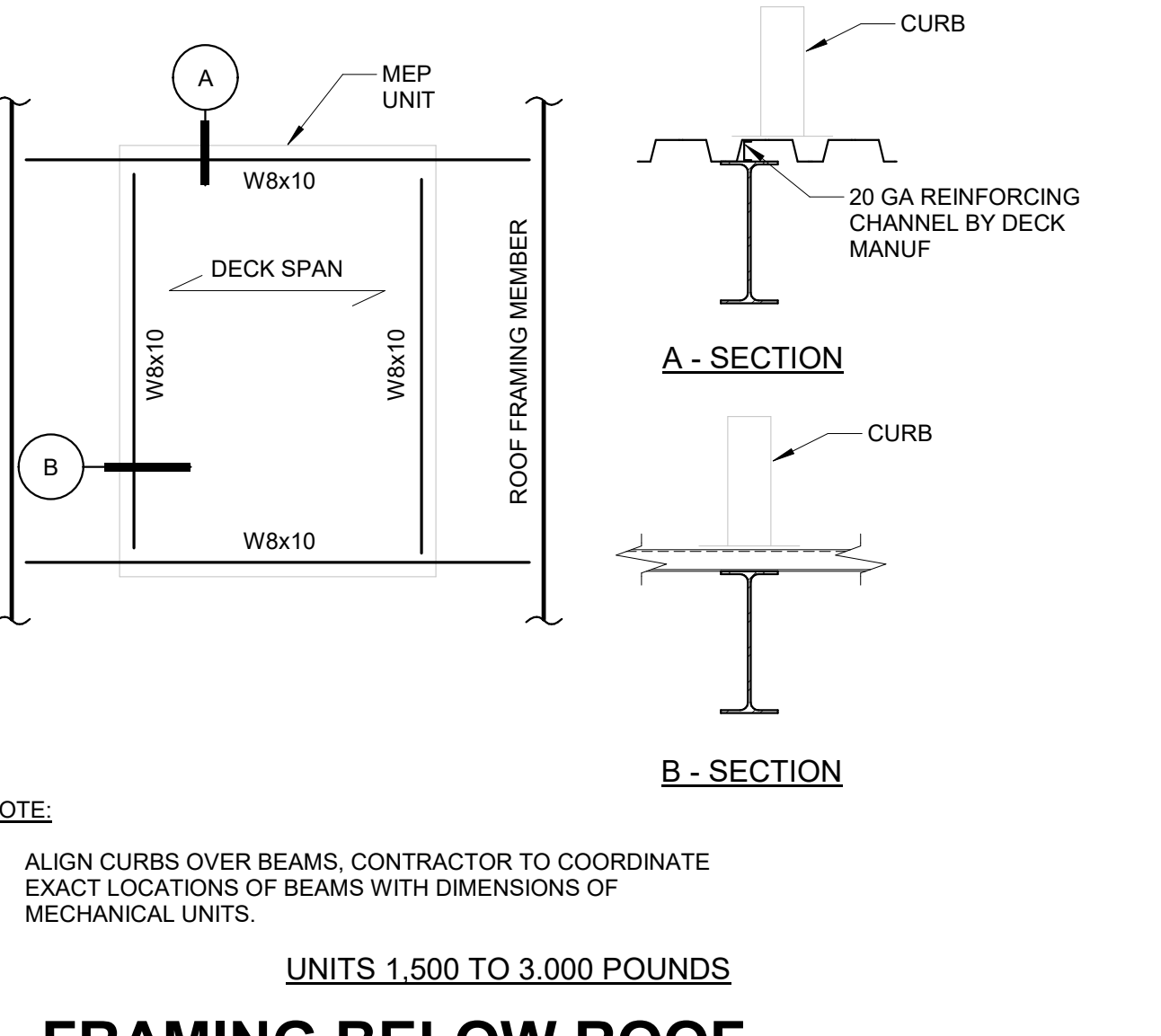
5 ROOF OPENING FRAMING FOR MEP EQUIP WT LESS THAN 1500 LBS
NO SCALE



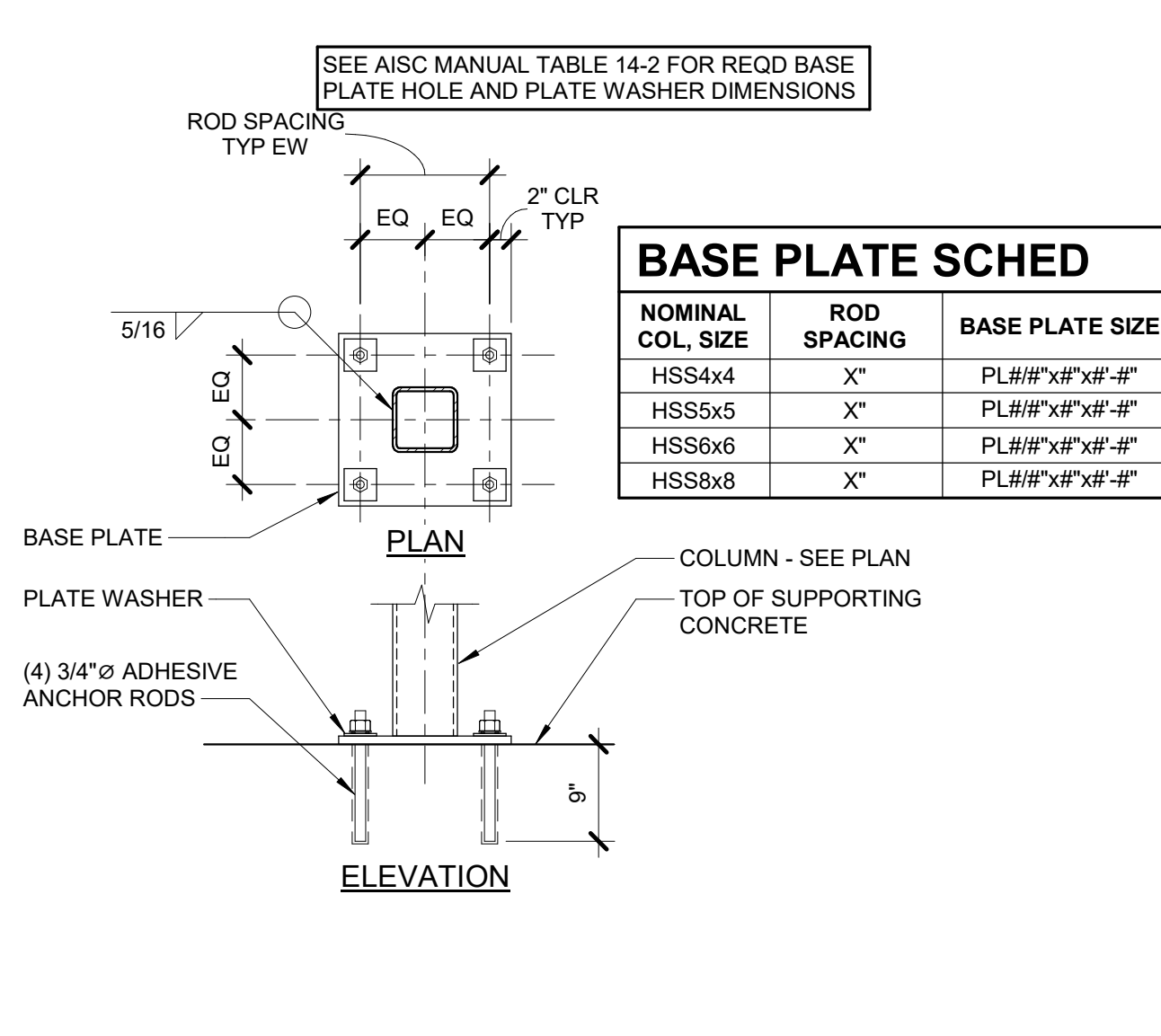
6 ROOF DECK REINFORCING AT ROOF DRAIN
1" = 1'-0"



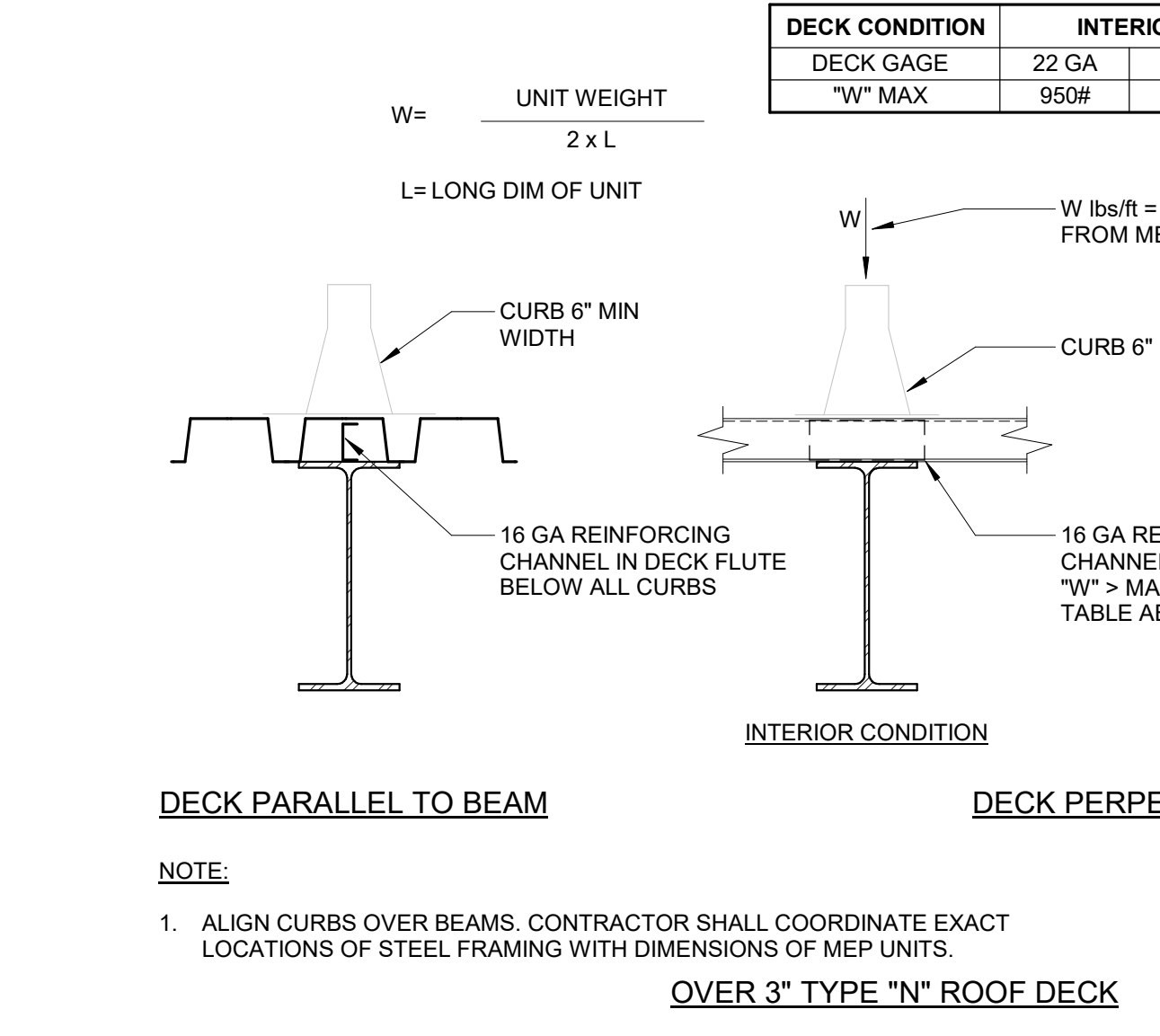
7 FRAMING BELOW ROOF TOP MECHANICAL UNITS
1" = 1'-0"



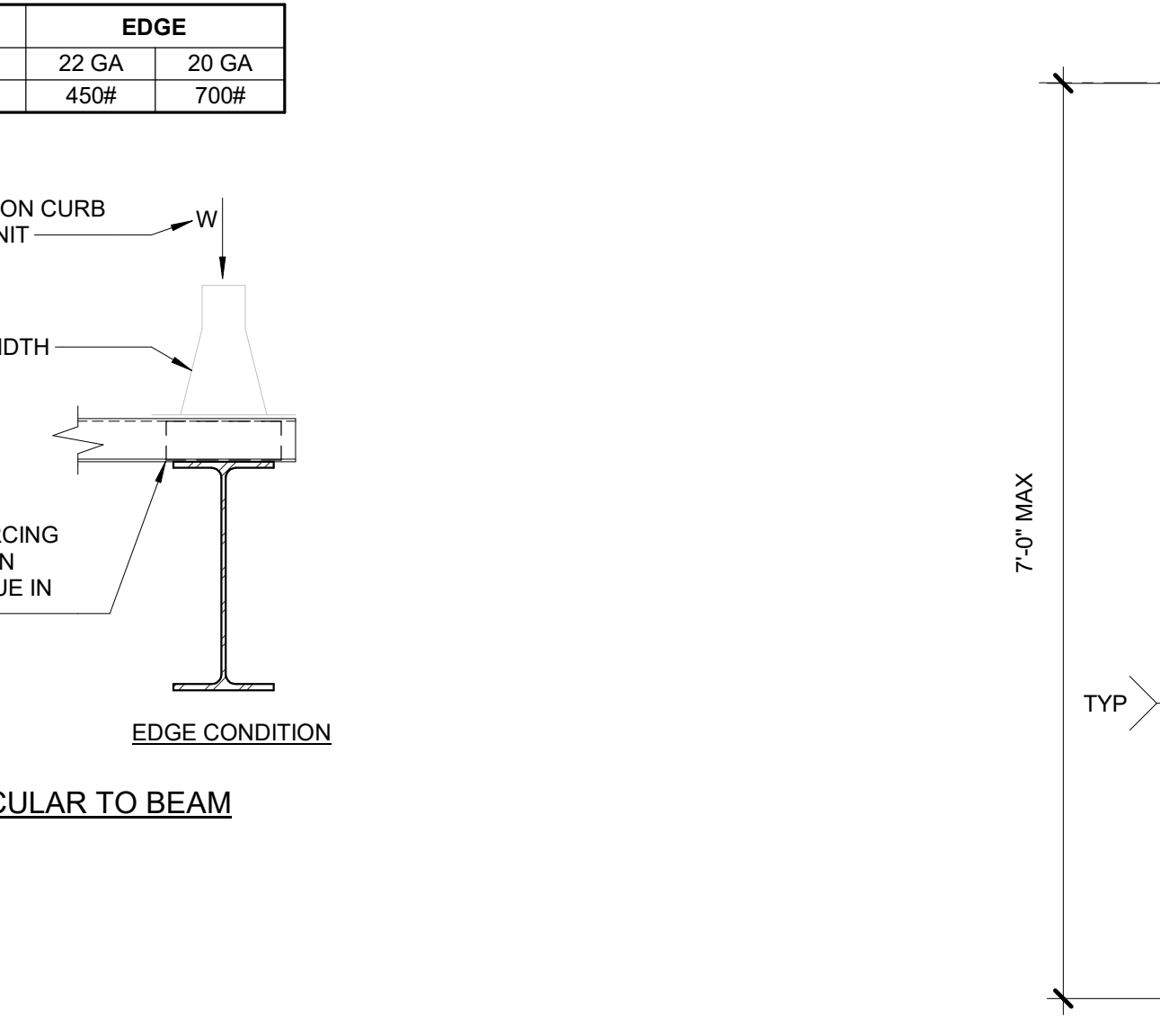
8 EDGE OF ROOF DECK
1" = 1'-0"



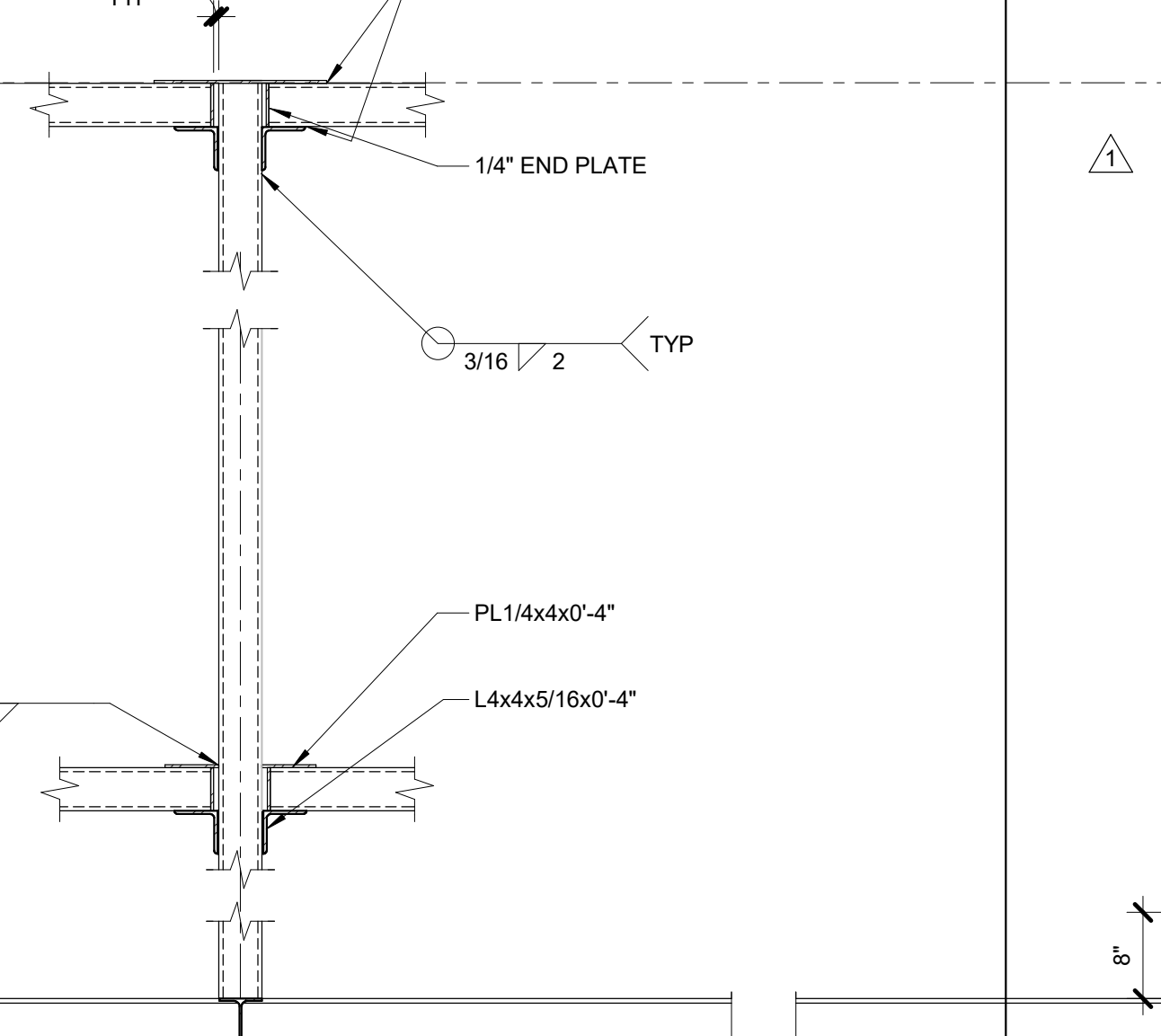
9 HSS COLUMN BASE PLATE AT EXISTING
3/4" = 1'-0"



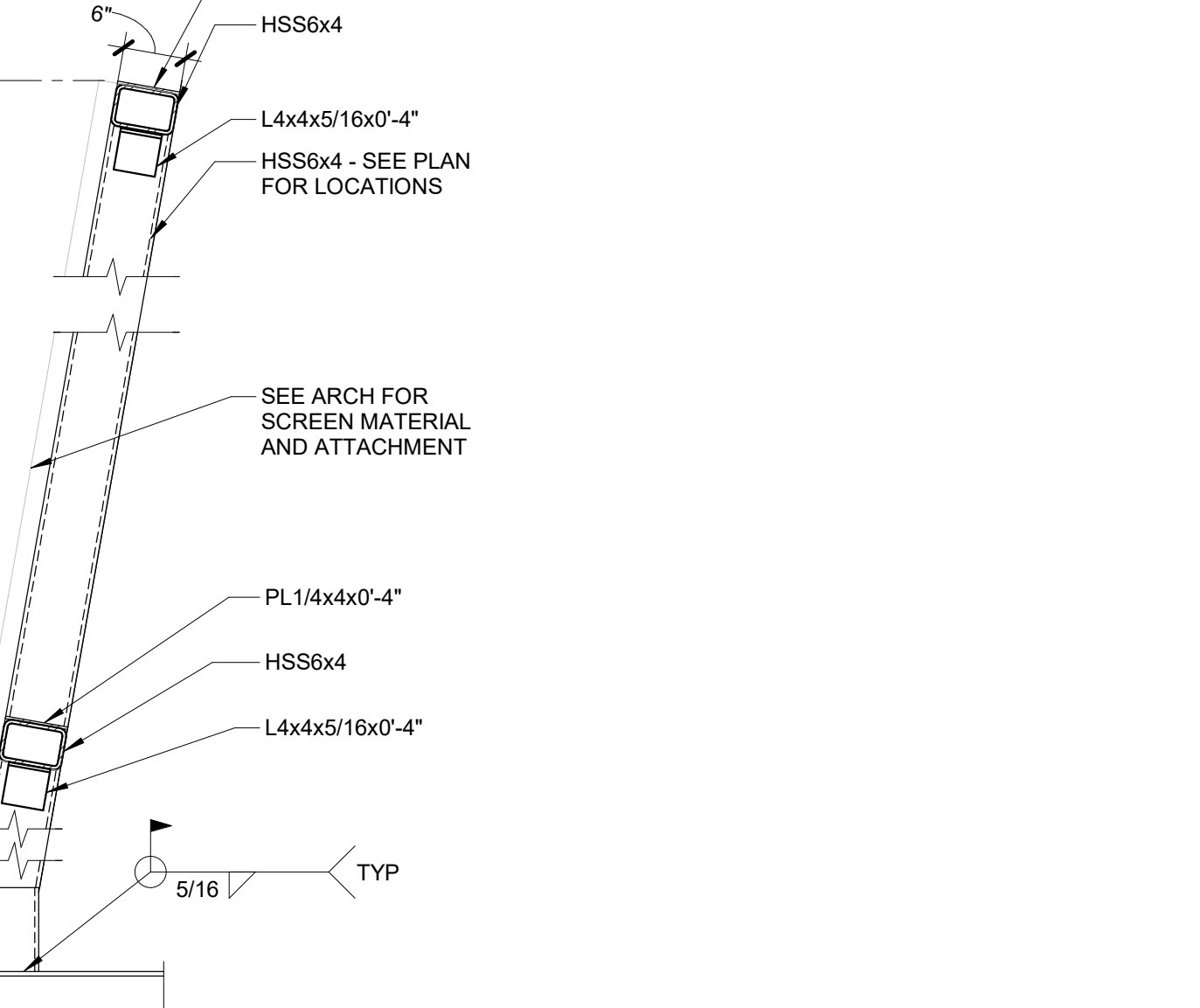
10 ROOFTOP MEP UNIT CURB CONNECTION
1" = 1'-0"



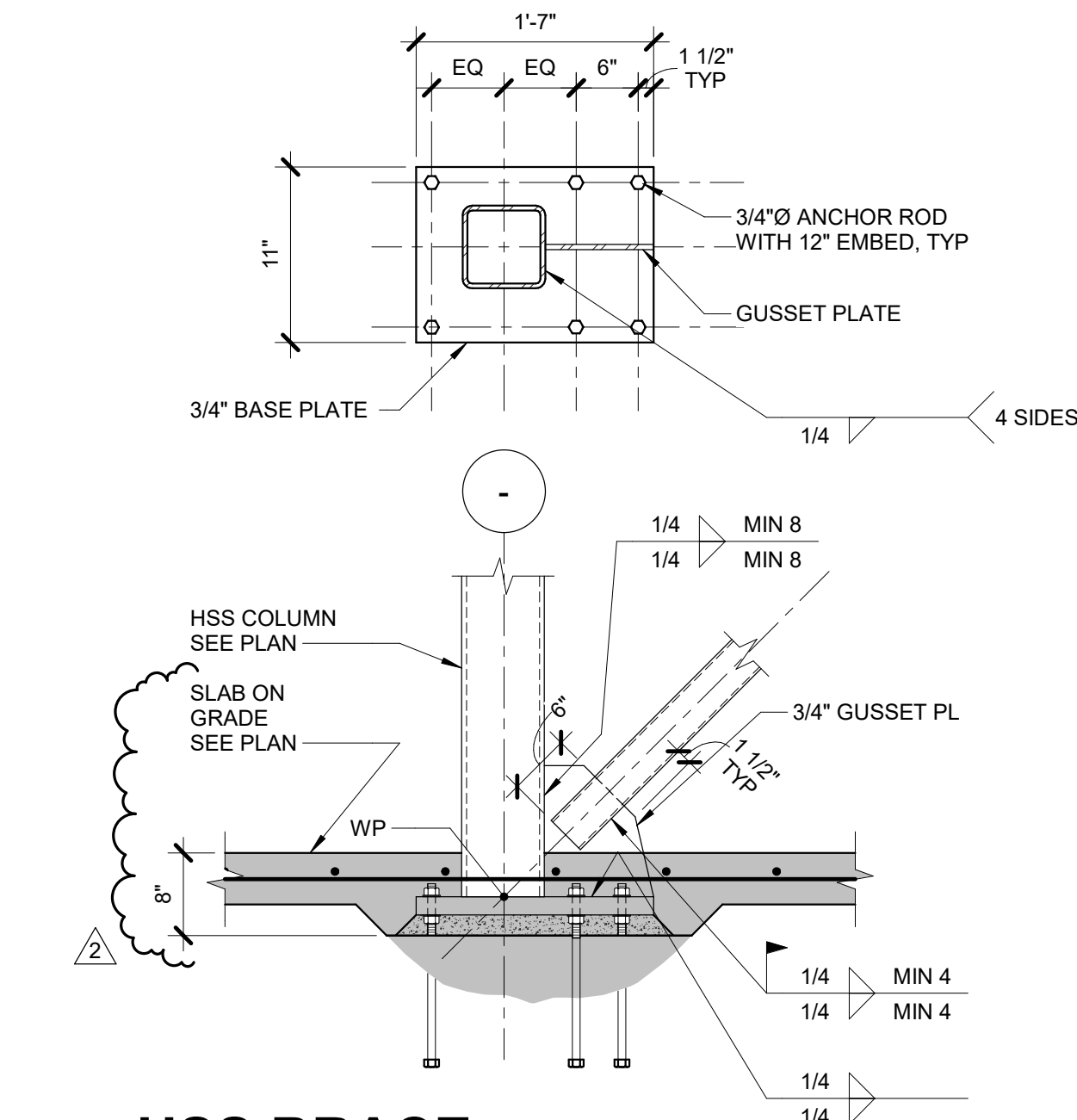
11 CANTILEVER SCREEN WALL SUPPORT AT STEEL BEAM
3/4" = 1'-0"



12 CANTILEVER SCREEN WALL SUPPORT AT STEEL BEAM
3/4" = 1'-0"



13 CANTILEVER SCREEN WALL SUPPORT AT STEEL BEAM
3/4" = 1'-0"



**HSS BRACE
CONNECTION AT BASE PLATE
WITH SLAB POUROVER**

3/4" = 1'-0"

HOLES IN BASEPLATES FOR ANCHORS

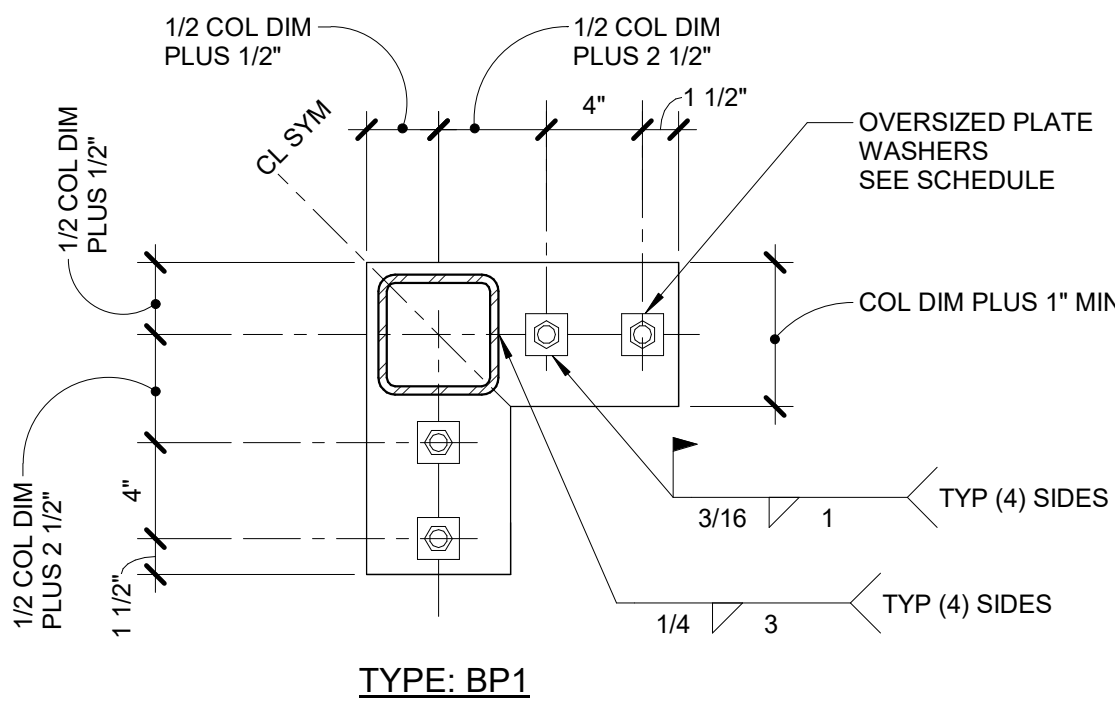
ANCHOR ROD DIAMETER	MAXIMUM HOLE DIAMETER	MINIMUM WASHER SIZE	MINIMUM WASHER THICKNESS	REMARKS
3/4"	1 5/16"	2"	1/4"	
7/8"	1 9/16"	2 1/2"	5/16"	
1"	1 13/16"	3"	3/8"	
1 1/4"	2 1/16"	3"	1/2"	
1 1/2"	2 5/16"	3 1/2"	1/2"	
1 3/4"	2 3/4"	4"	5/8"	
2"	3 1/4"	5"	3/4"	
2 1/2"	3 3/4"	5 1/2"	7/8"	

NOTES:

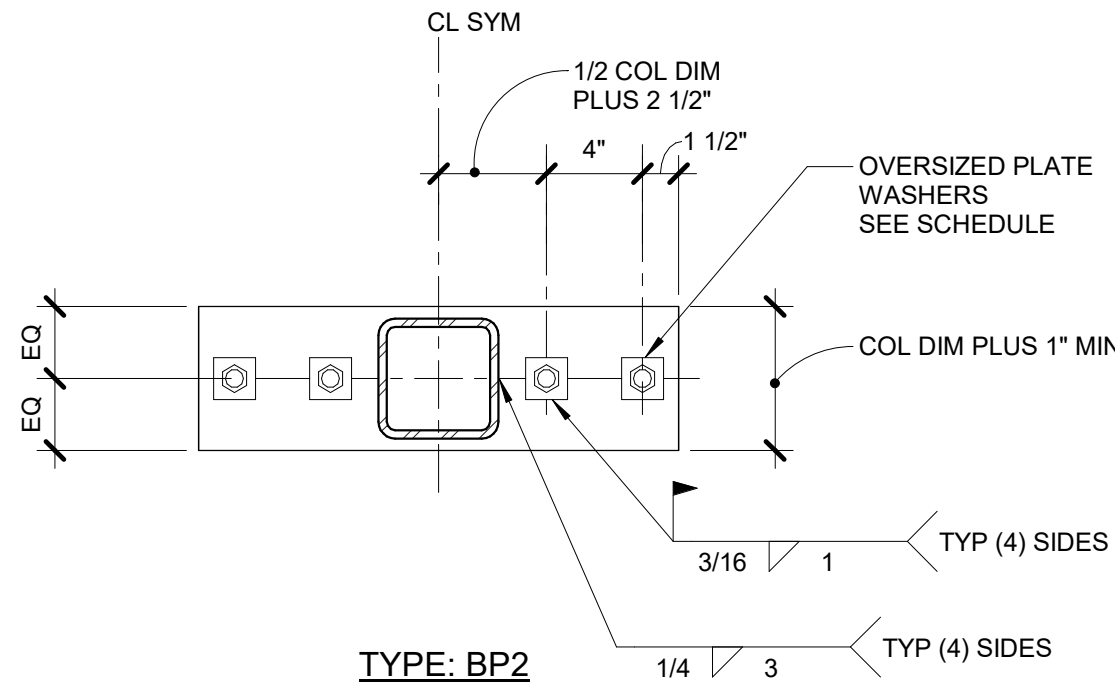
- CIRCULAR OR SQUARE WASHERS ARE ACCEPTABLE.
- COLOR DESIGNATIONS FOR ASTM F1554 ANCHOR RODS ARE AS FOLLOWS:
A. BLUE - 36 ksi
B. YELLOW - 55 ksi (WELDABLE)
C. RED - 105 ksi
- ERECTION PROVISIONS OF OSHA 1926.75 REQUIRE THAT THE GENERAL CONTRACTOR COMPLY WITH THE FOLLOWING:
A. ANCHOR RODS SHALL NOT BE REPAIRED OR REPLACED OR FIELD MODIFIED WITHOUT THE APPROVAL OF THE STRUCTURAL ENGINEER OF RECORD.
B. APPROVAL MUST STATE IF REPAIR/MODIFICATION SHALL REQUIRE GUYING OR BRACING THE COLUMN.
C. CONTRACTOR SHALL PROVIDE WRITTEN NOTIFICATION TO ERECTOR OF ANY REPAIR OR MODIFICATION.
- FIELD SURVEY OF ANCHOR RODS IS RECOMMENDED PRIOR TO COL/BASE PLATE FABRICATION.

**BASE PLATE SCHEDULE - WITH
NOTES**

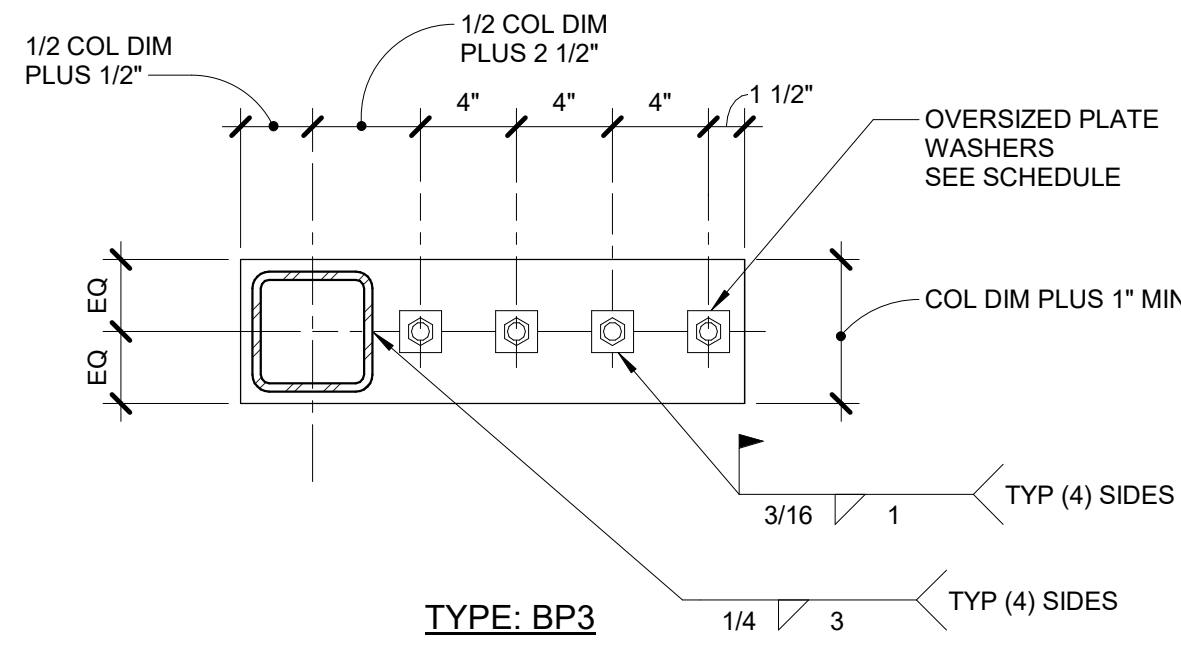
3/4" = 1'-0"



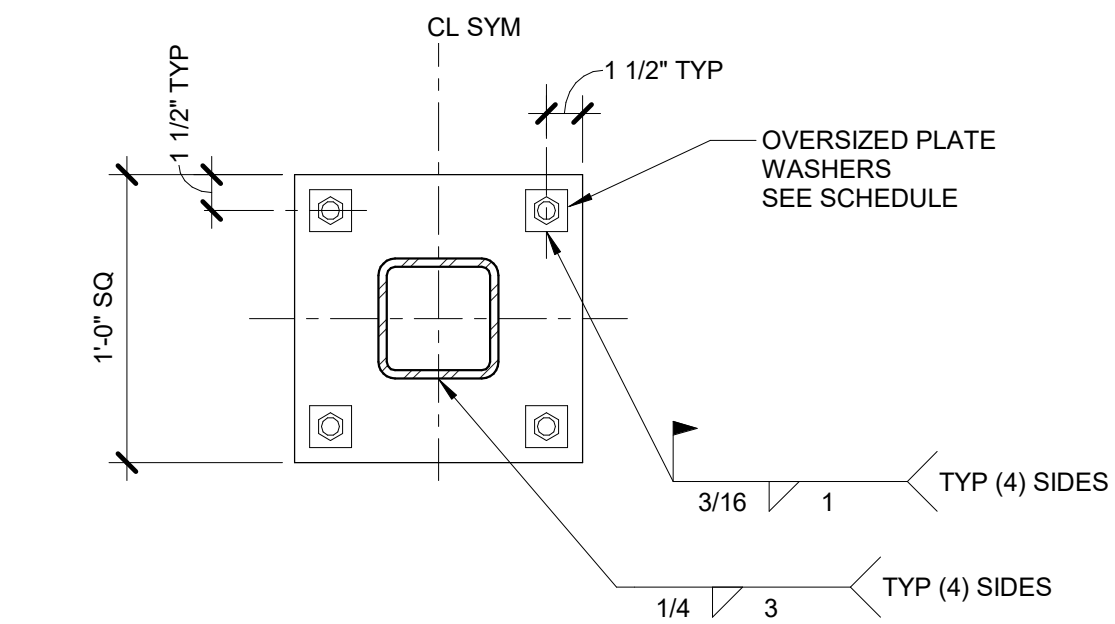
TYPE: BP1



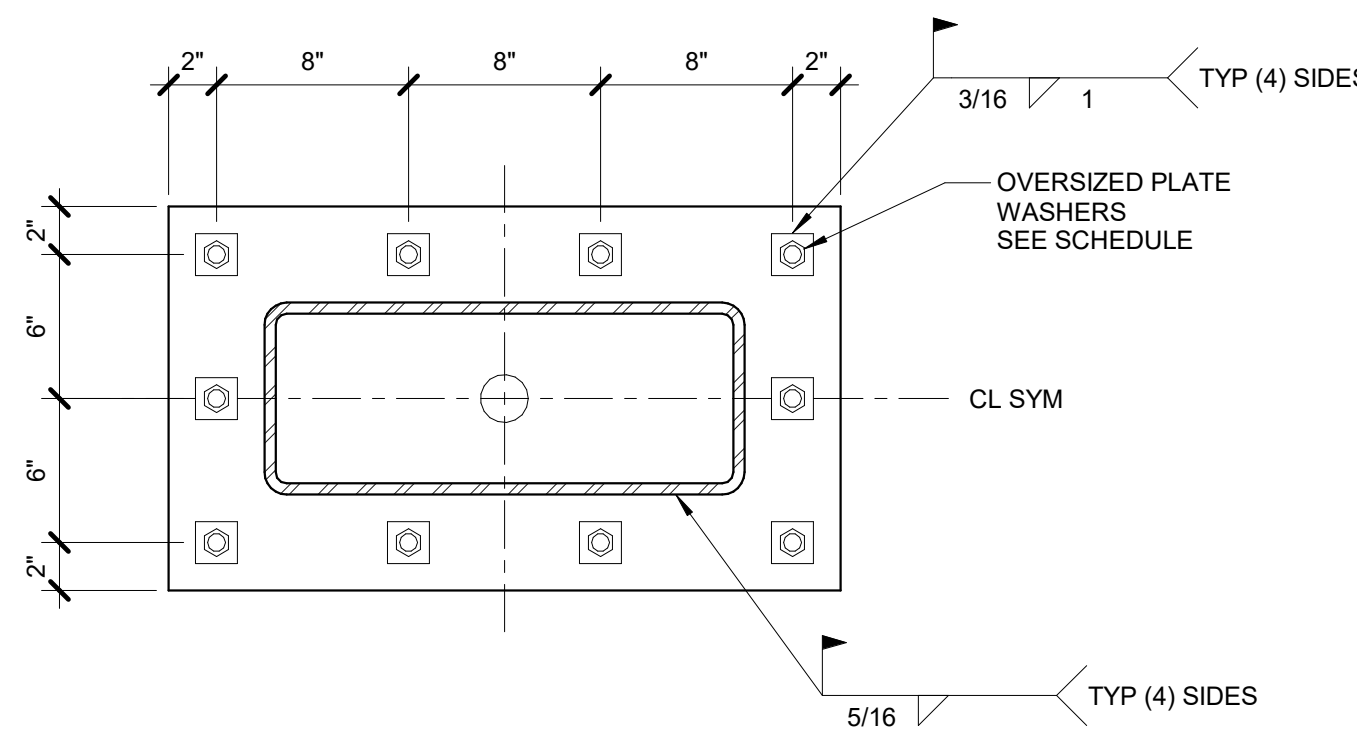
TYPE: BP2



TYPE: BP3



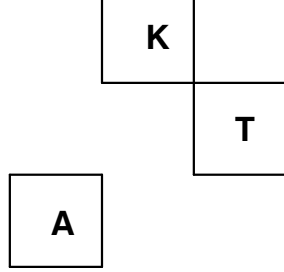
TYPE: BP4



TYPE: BP5

BASE PLATES, RESIDENTIAL

1 1/2" = 1'-0"



Kevin Tsai Architecture

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architect's stamp

consultant

KEY PLAN:

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#	DATE	ISSUE RECORD
1	04/20/2025	CHECKSET
2	01/21/2026	ISS COMMENTS

PROJECT TITLE:

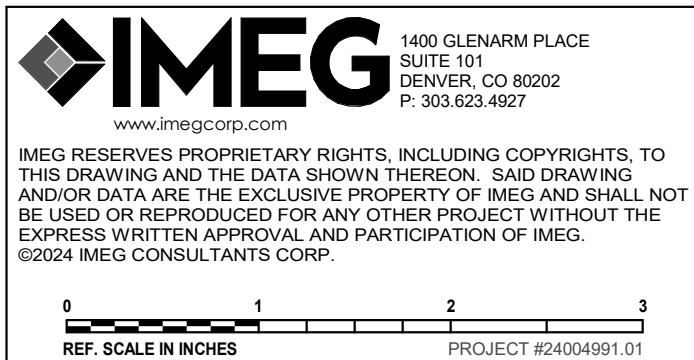
3101 EAST CAMELBACK ROAD
PHOENIX, ARIZONA 85016

PC/PM: LG
DRAWN BY: STAFF
JOB NO.: 24004991.01

DRAWING TITLE:
**STEEL BASE PLATE
DETAILS**

SHEET NO.:

S530



2

$$1\frac{1}{2}'' = 1'-0''$$


3/4" = 1'-0"



2


$$\frac{3}{4}'' = 1'-0''$$


$\frac{3}{4}'' = 1'-0''$

Kevin Tsai Architecture

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KEY PLAN:

DATE ISSUE RECORD

PROJECT TITLE:

SHEET NO.:

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#	DATE	ISSUE RECORD
	04/30/2025	CHECKSET
2	01/21/2025	PC COMMENTS

PROJECT TITLE:

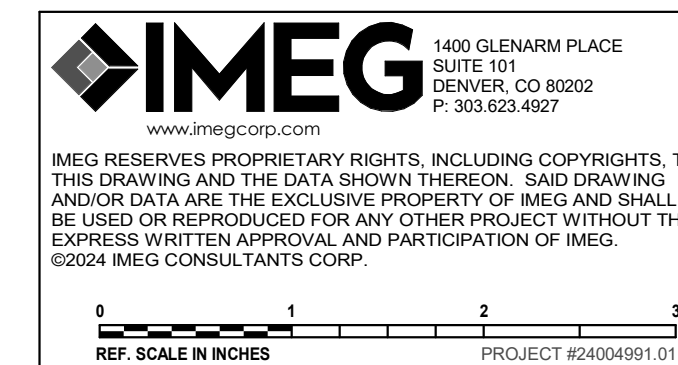
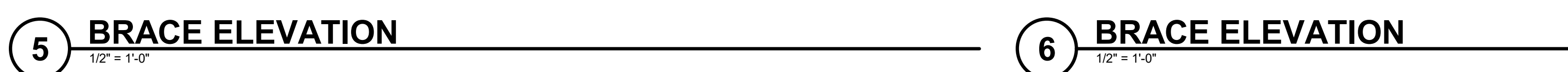
3101 EAST CAMELBACK ROAD
PHOENIX, ARIZONA 85016

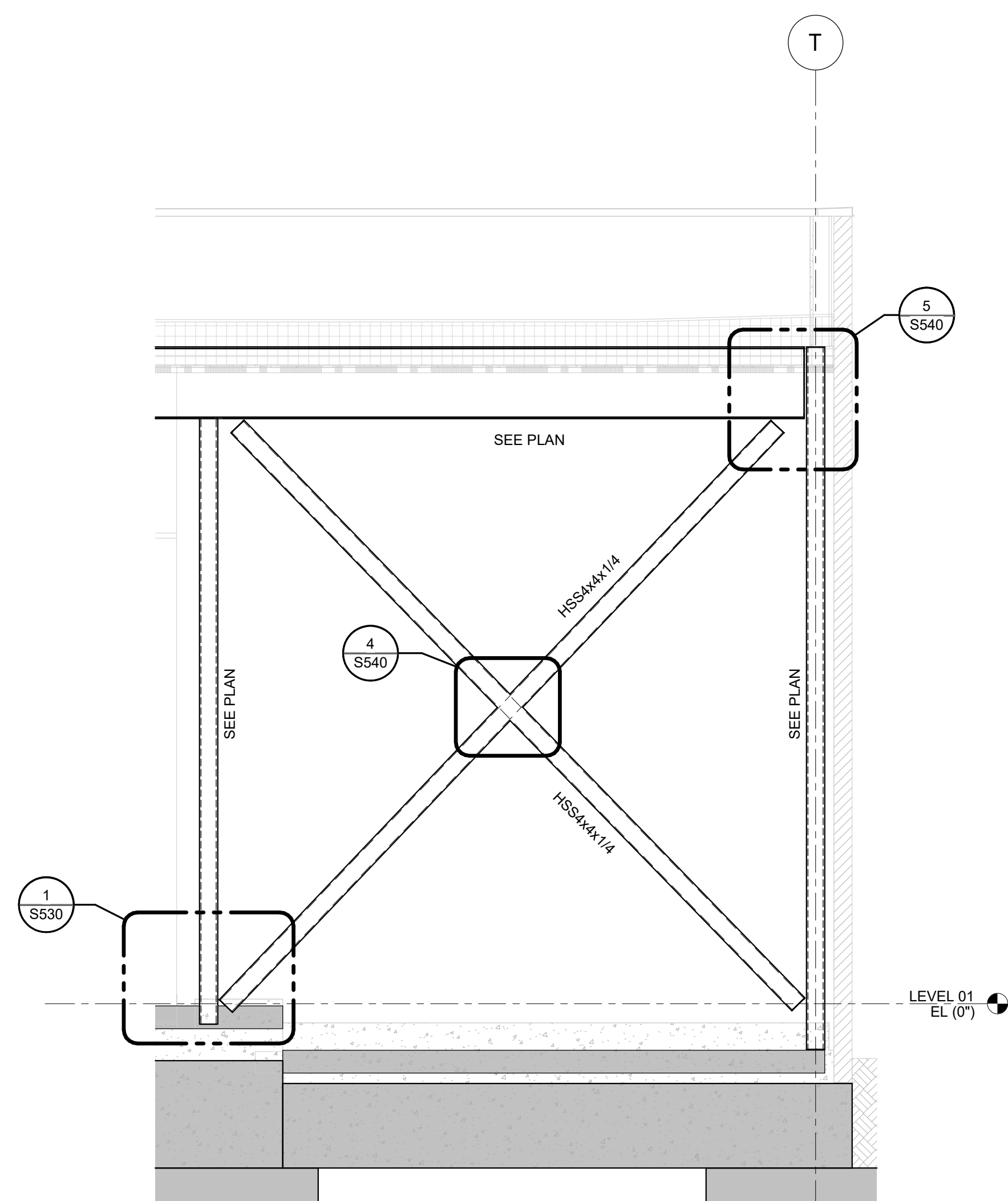
PC/PM: LG
DRAWN BY: STAFF
JOB NO.: 24004991.01

DRAWING TITLE:
**STEEL BUILDING
FRAME
ELEVATIONS**

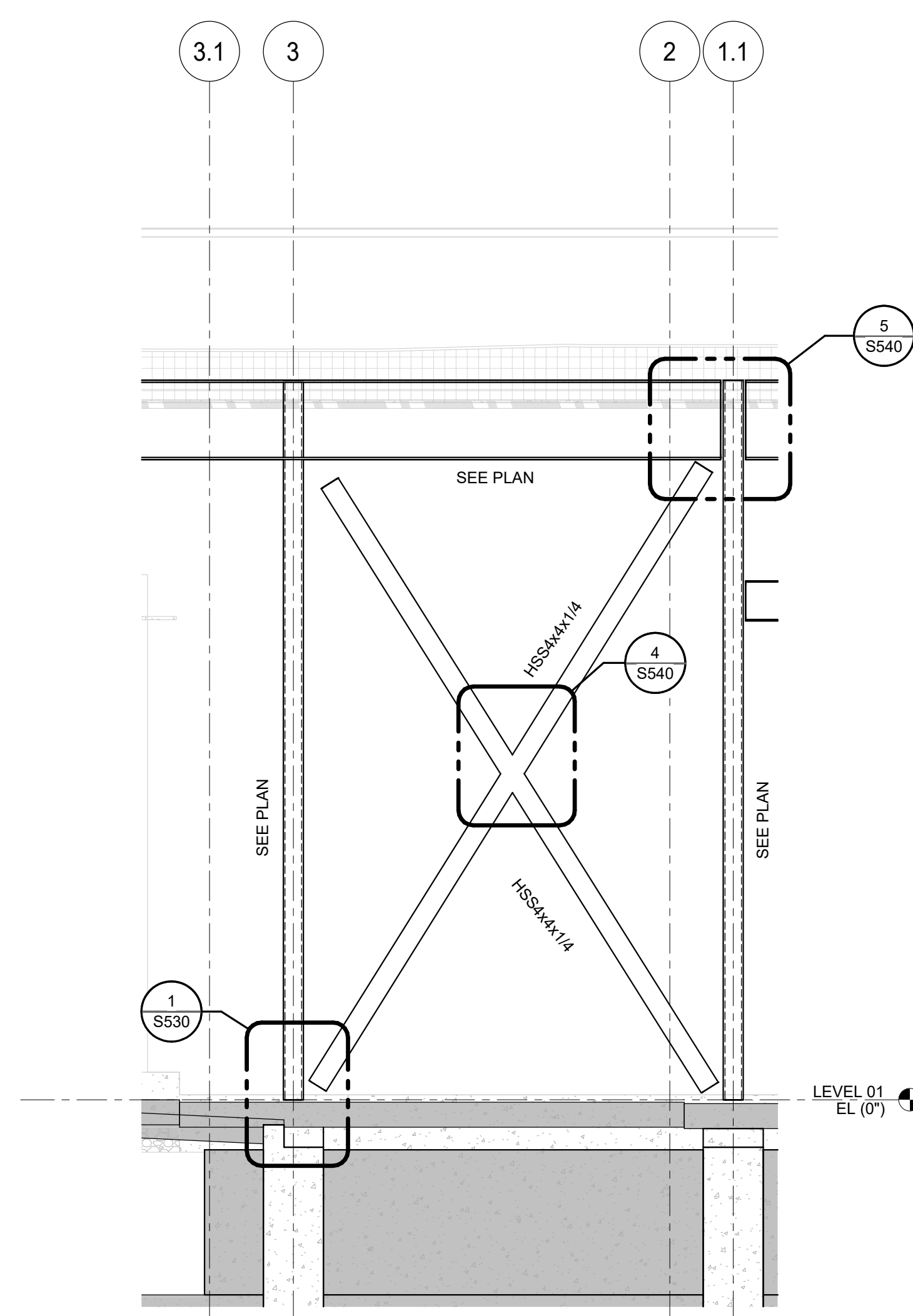
SHEET NO.:

S550



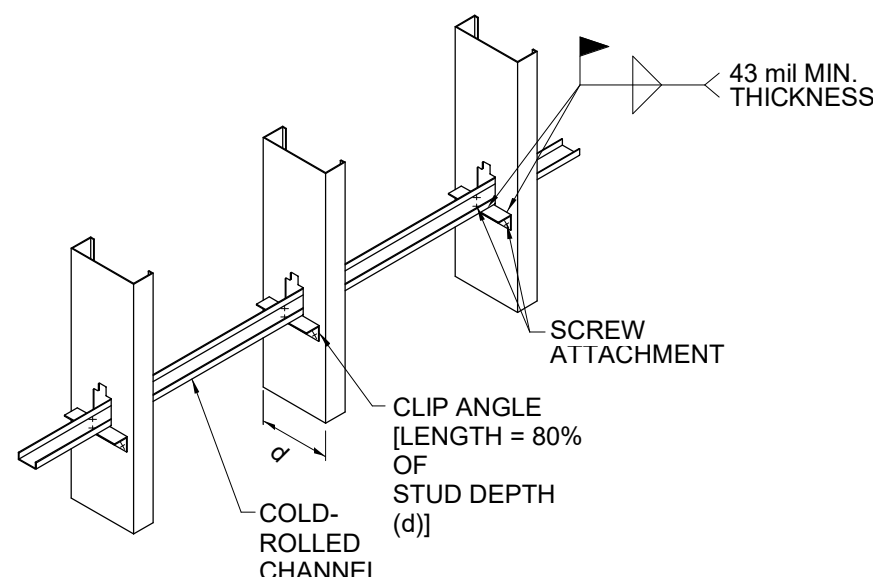


1 BRACE ELEVATION

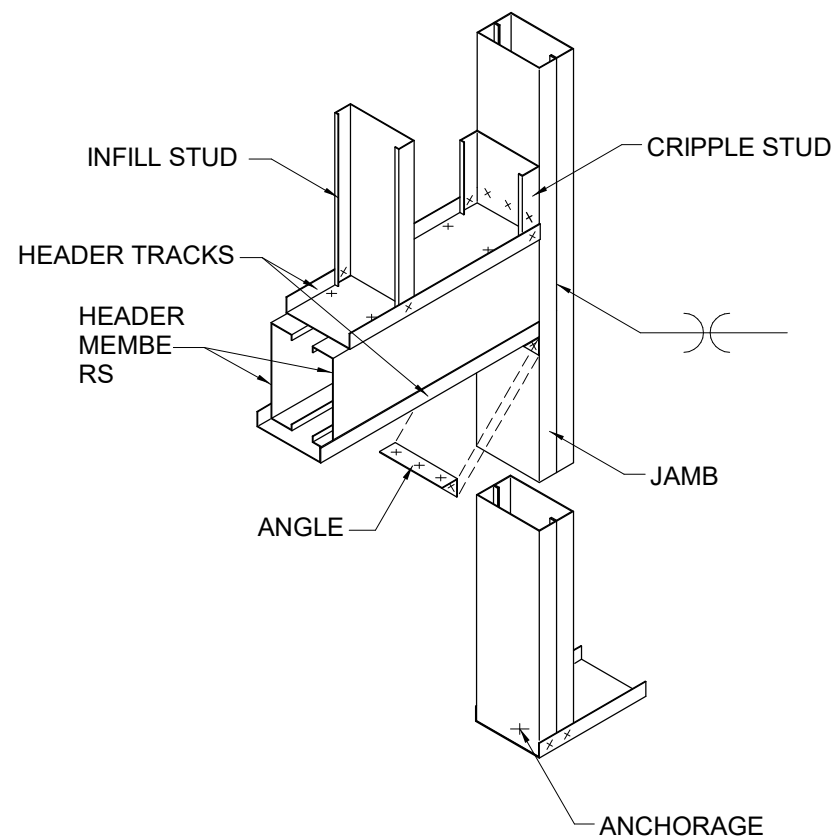


2 BRACE ELEVATION

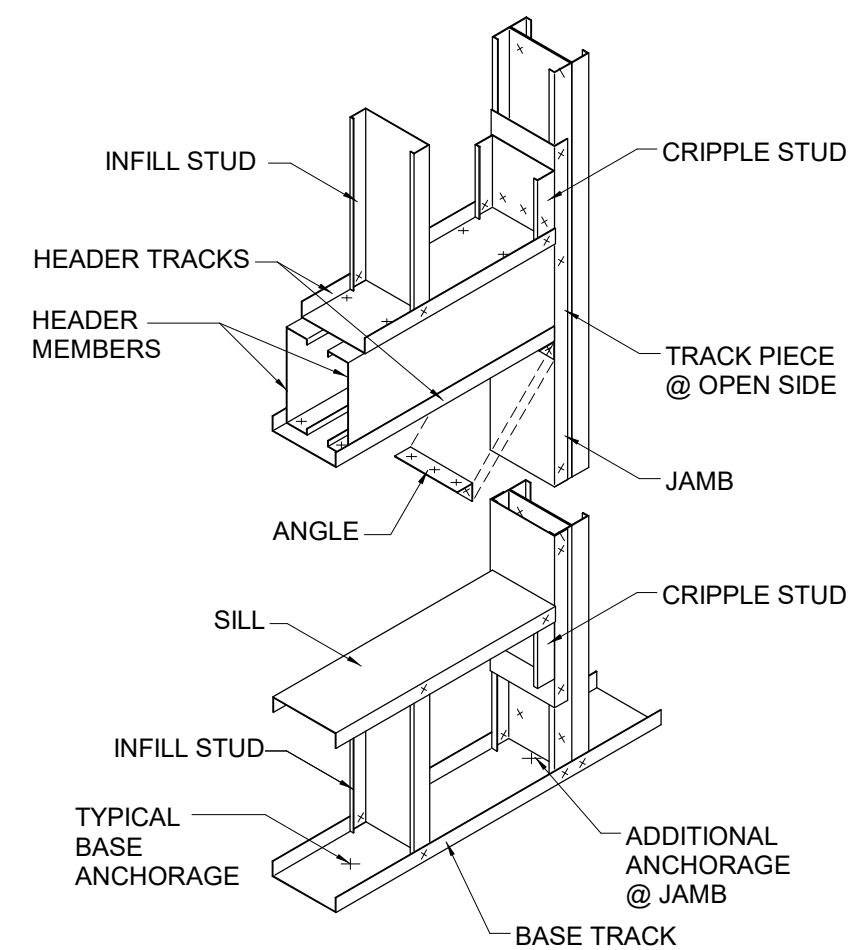
[illegible]



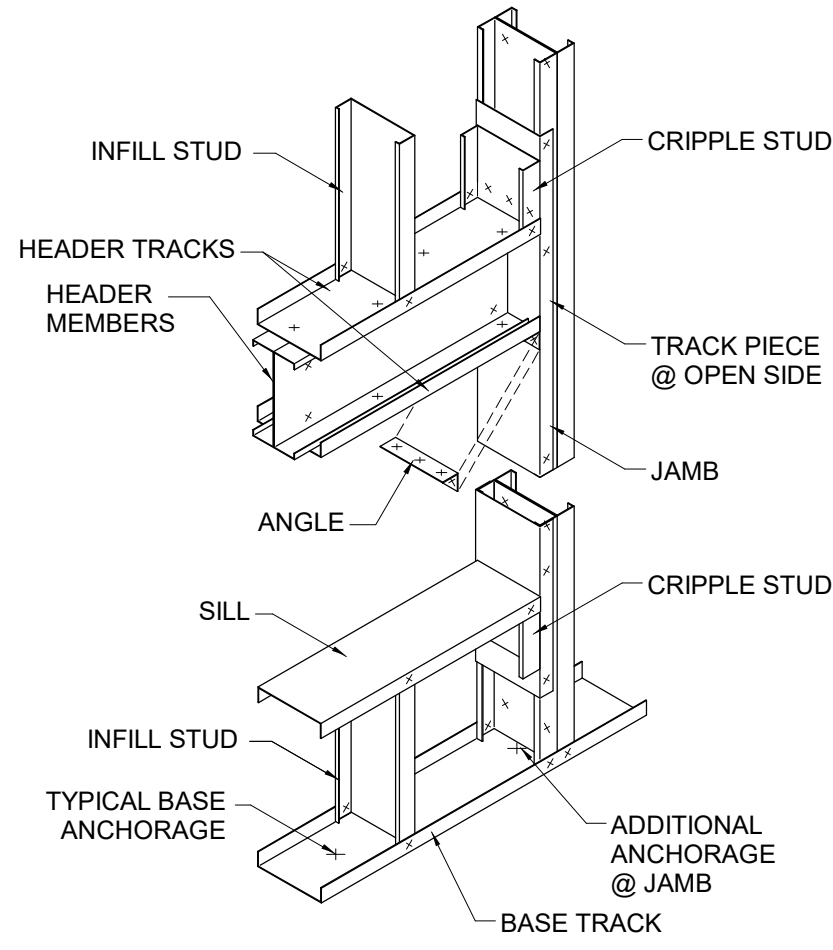
COLD FORMED METAL STUD BRIDGING ISO



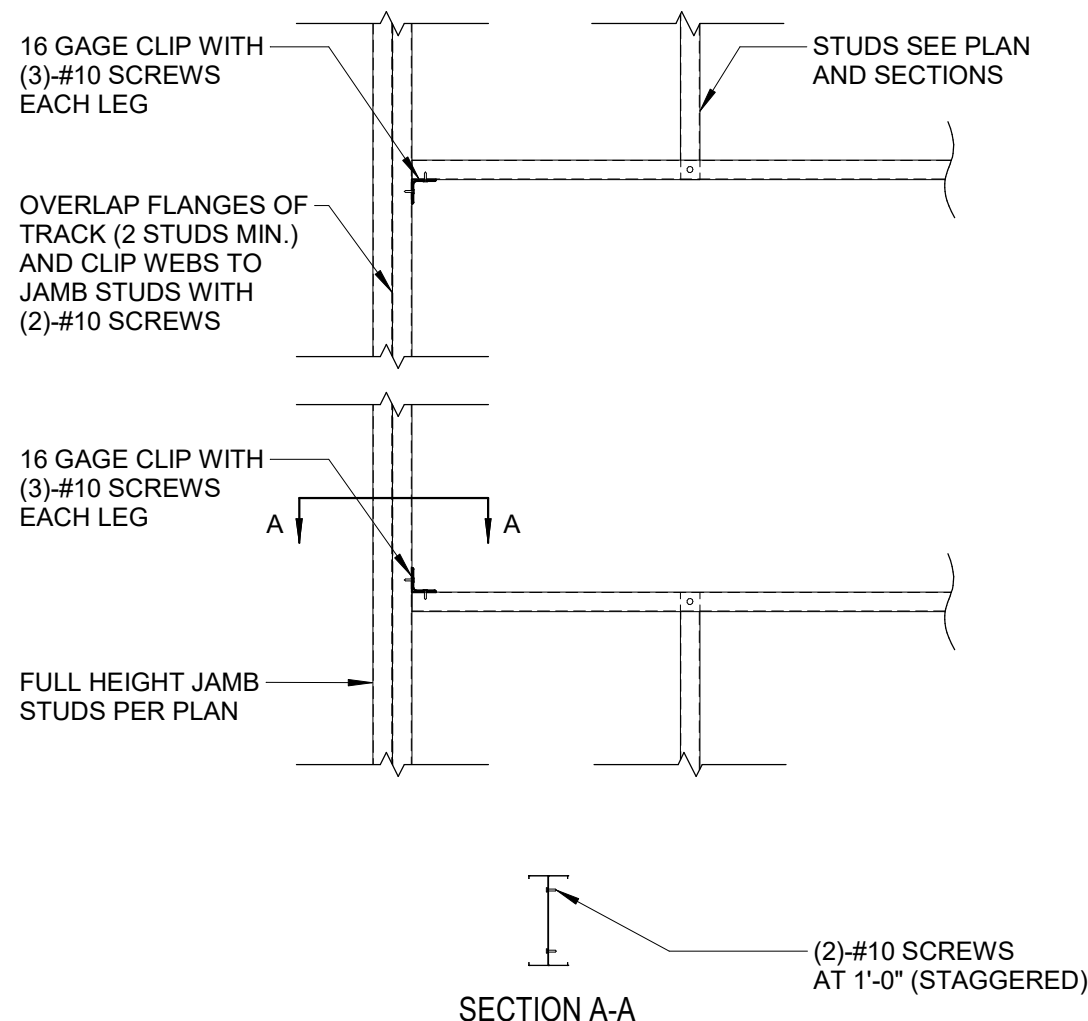
COLD FORMED METAL STUD DOOR HEADER ISO



TYPICAL COLD FORMED METAL STUD HEADER AT WINDOW OPTION 1

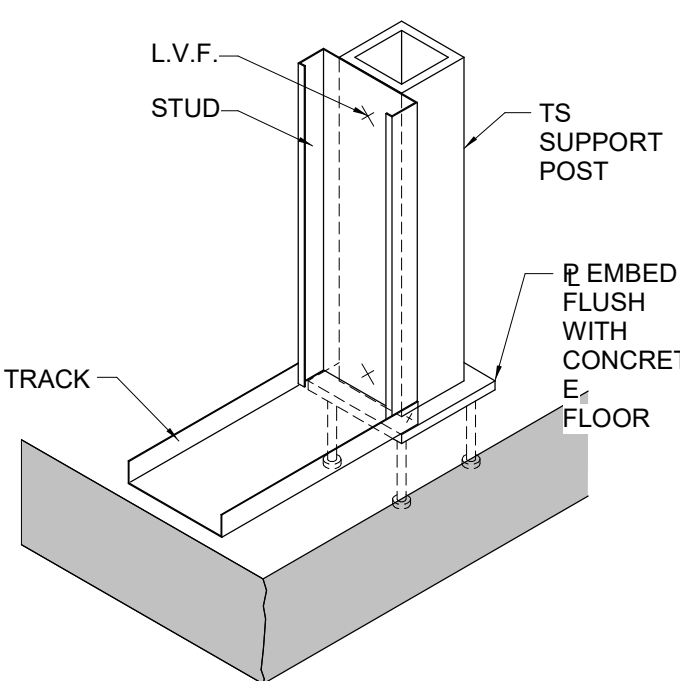


BACK TO BACK COLD FORMED METAL STUD HEADER AT WINDOW OPTION 2

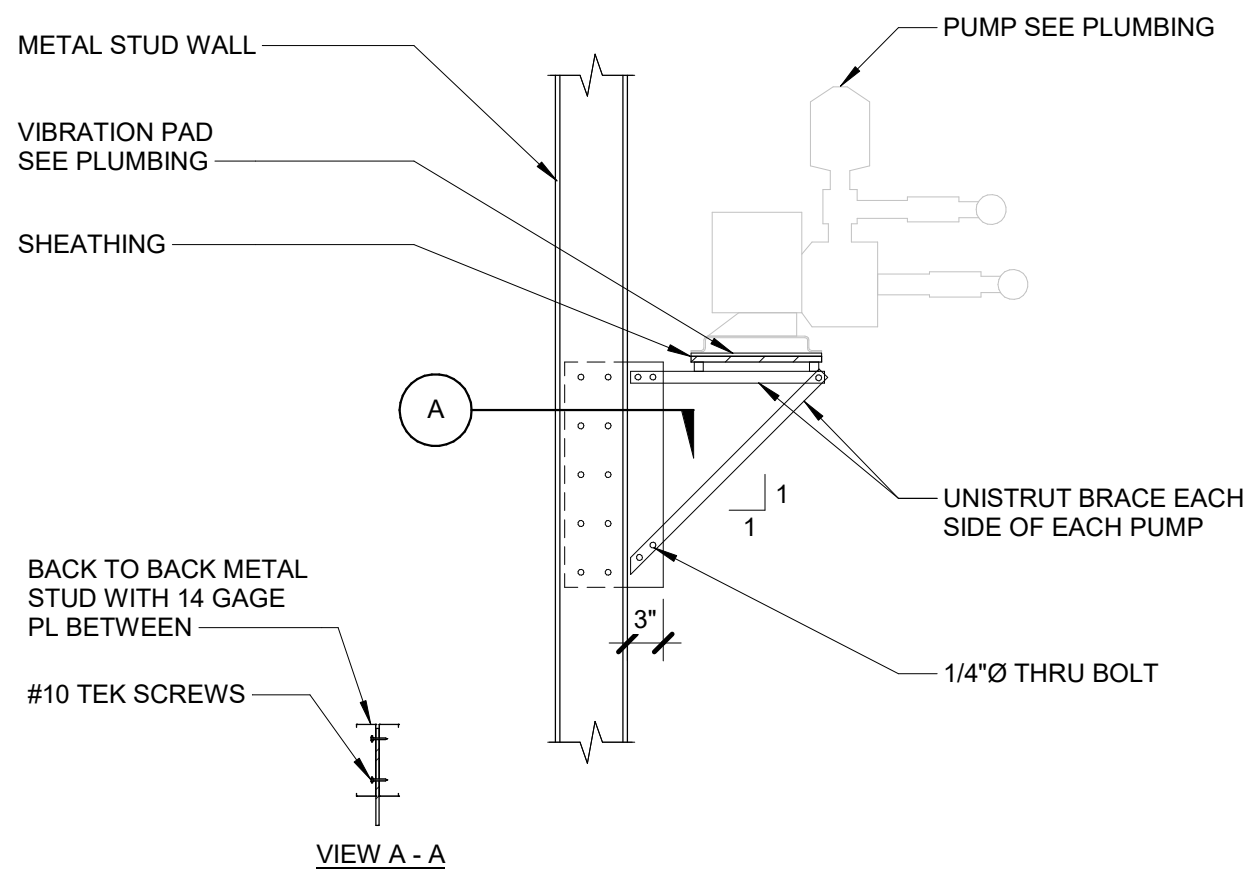


COLD FORMED METAL STUD TRACK HEADER AT WINDOW AT INTERIOR WALL

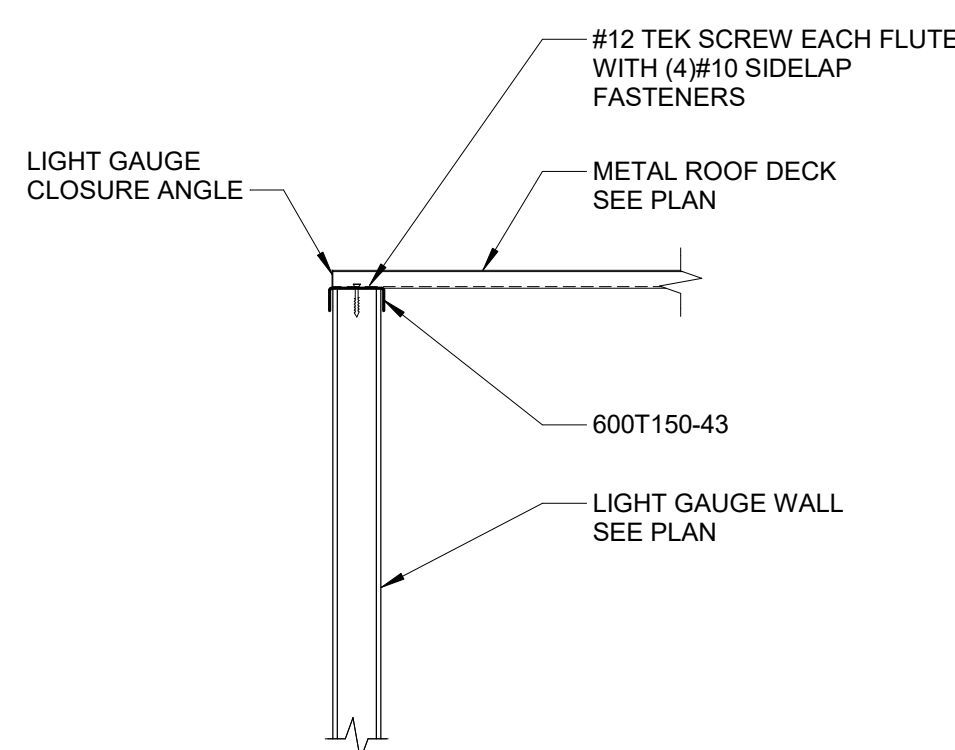
OPENING AT COLD FORMED METAL STUD TRACK HEADER SECTION



COLD FORMED METAL STUD CONNECTION TO STEEL COLUMN ISO



PUMP WALL SUPPORT



SECTION

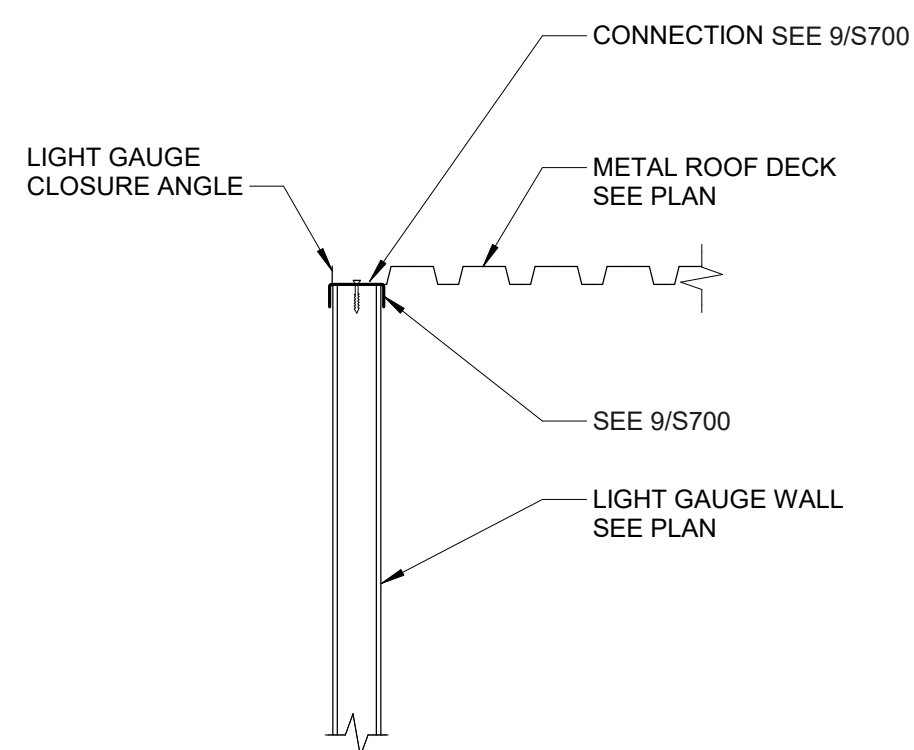
METAL STUD BOX HEADER SCHEDULE			
MARK	HEADER FRAMING	JAMBS STUDS	NOTES
MH1	(2)-S800S250-54 AND (2)-S600T250-54	(2)-S600S162-64	OPENINGS UP TO 10'-0"
MH2	(2)-S600S162-43 AND (2)-S600T150-43	(1)-S600S162-43	OPENINGS UP TO 10'-0"

NOTE:
 1. PROVIDE HEADERS AT ALL EXTERIOR WALLS PER SCHEDULE ABOVE AND PER NOTES IN SCHEDULE
 2. MH2 HEADER IS THE TYPICAL OPENING HEADER, UO/N.

LEVEL	SIZE	SPACING	NOTES
LEVEL 1	S600S162-54	1'-4"	TYPICAL EXTERIOR NON-LOAD BEARING WALLS

NOTE:

- WITHIN 6 FEET OF ALL BUILDING CORNERS, STUD SPACING TO BE 1/2" OF SPACING LISTED ABOVE
- INTERIOR NON-LOAD BEARING WALLS AND SLL EXTERIOR NON-LOAD BEARING WALLS BELOW TRANSFER SLAB TO ALLOW FOR 1" VERTICAL DEFLECTION OF CONCRETE SLAB(S), ALL VERTICAL STUDS TO BE SCREWED TO DEEP LEG TRACK OR PROVIDE MINIMUM OF (1)-ROW OF BRIDGING BELOW TRACK
- METAL STUD WALLS TO BE INSTALLED BELOW THE ABOVE INDICATED LEVEL.



SECTION

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[illegible]

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3101 EAST CAMELBACK ROAD
PHOENIX, ARIZONA 85016

PC/PM: LG
DRAWN BY: STAFF
JOB NO.: 24004991.01

DRAWING TITLE:
TYPICAL COLD
FORM SECTIONS

SHEET NO.:

S700

